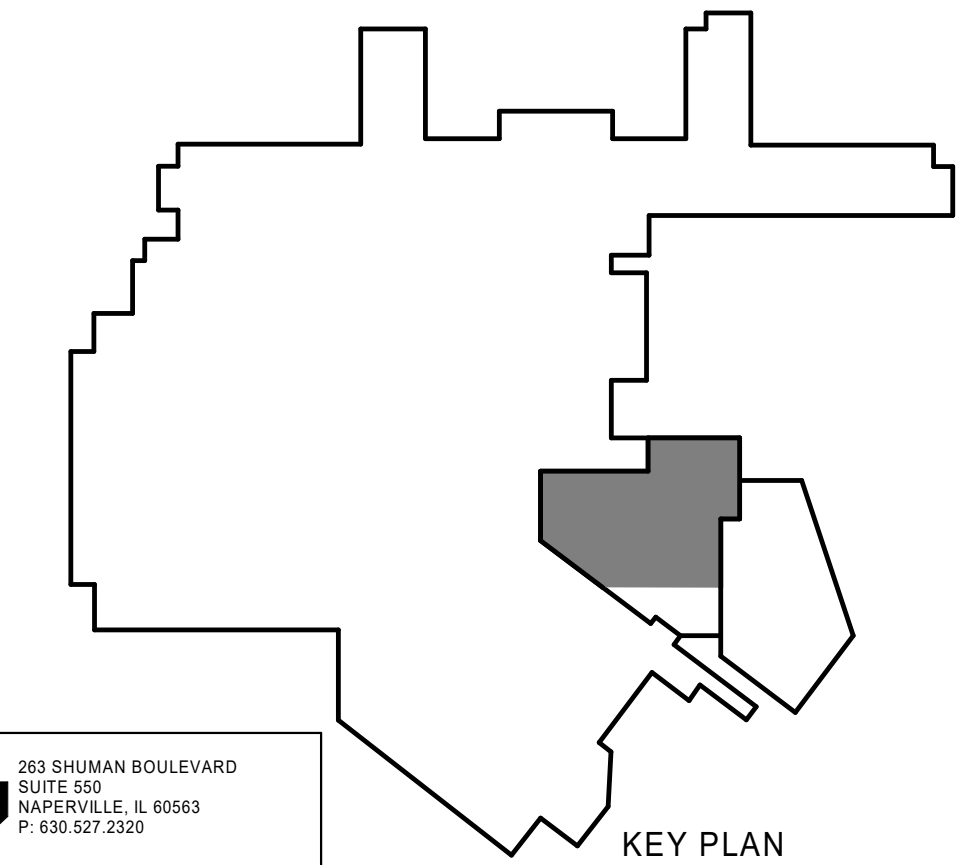


Project Number	556-402
Building Number	133CA
Drawing Number	M000



VA | U.S. Department
of Veterans Affairs



PLAN NORTH

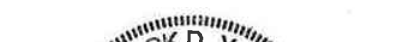
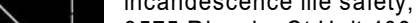
IMEG 263 SHUMAN BOULEVARD
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NAPERVILLE, IL 60563
P. 630.527.2320

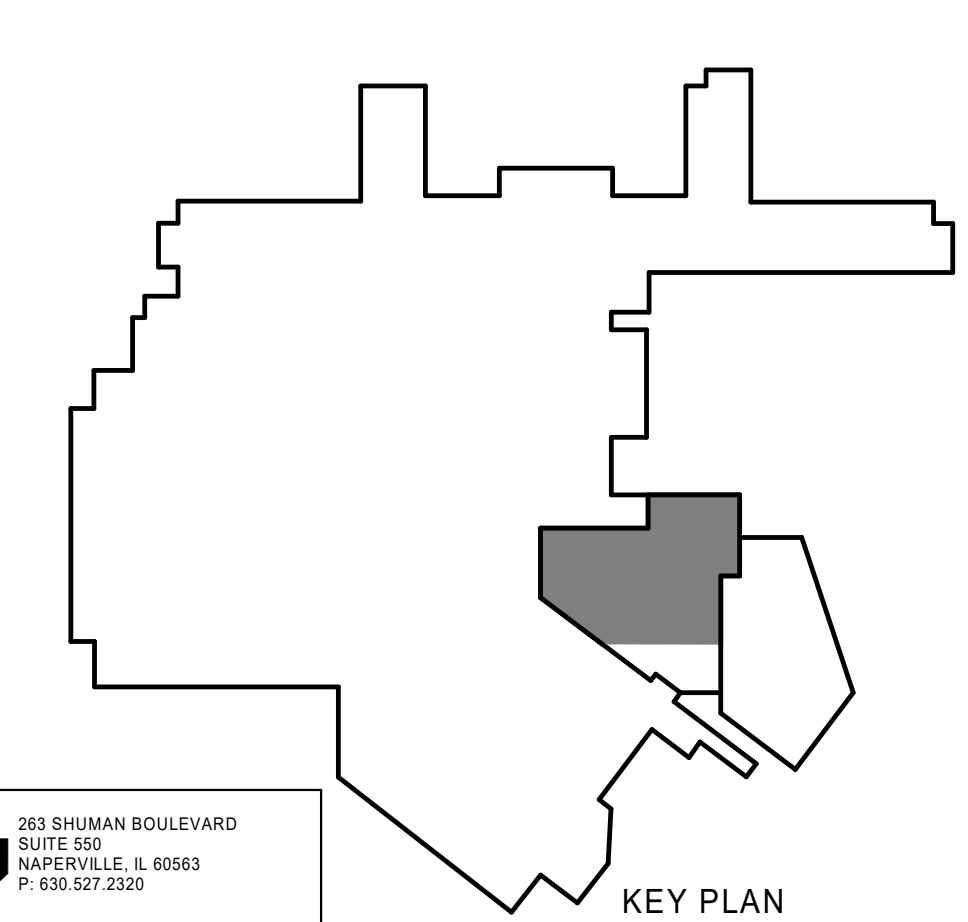
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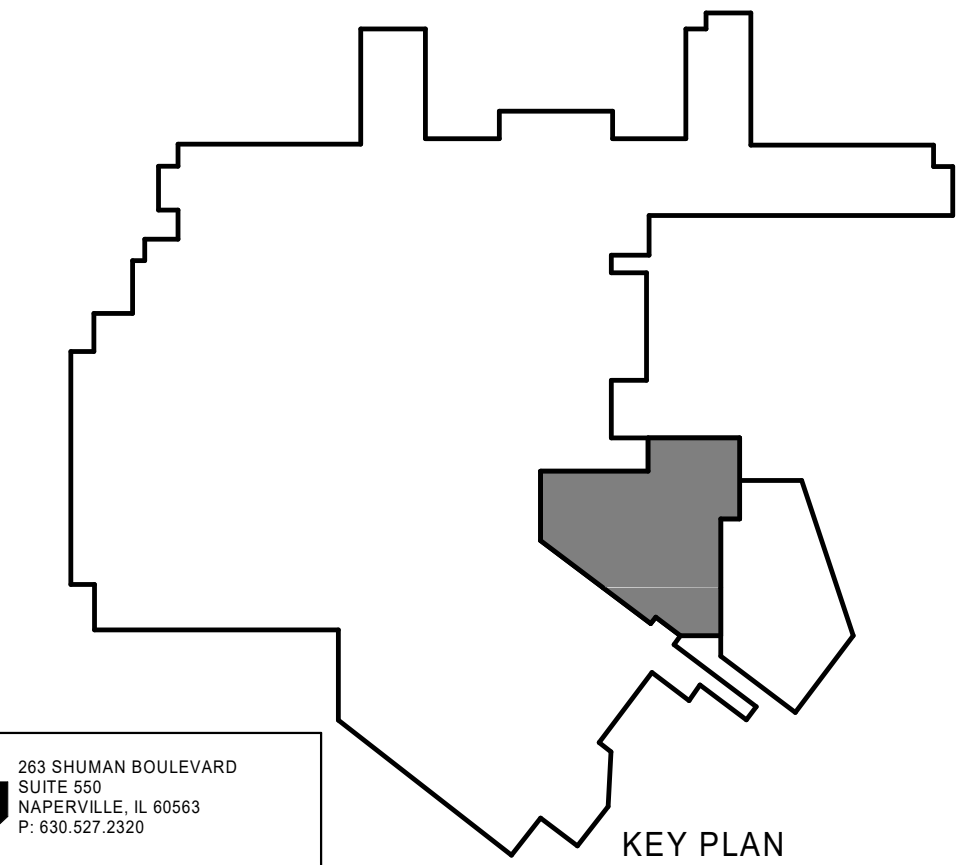
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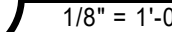
ARCHITECT/ENGINEER OF RECORD			STAMP	Office of Construction and Facilities Management	Drawing Title BASEMENT DEMOLITION - VENTILATION - PHASE 1	Phase 100% CONSTRUCTION DOCUMENTS	Project Title EXPAND STERILE PROCESSING SERVICE (SPS)	Project Number 556-402		
A3 Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-0941	 Incandescence Life Safety, Inc. 3575 Ringby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10801 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	 11/29/2022	VA U.S. Department of Veterans Affairs	Approved:  11/29/2022	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Drawing Number 133CA		
S3 Shaffer Stevens & Schapp P.C. 1070 Howard St. Omaha, Nebraska 68102 Phone: (402) 455-6998	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434					Issue Date 11/29/2022	Checked MRM	Drawn NVP	Drawing Number MD100.1



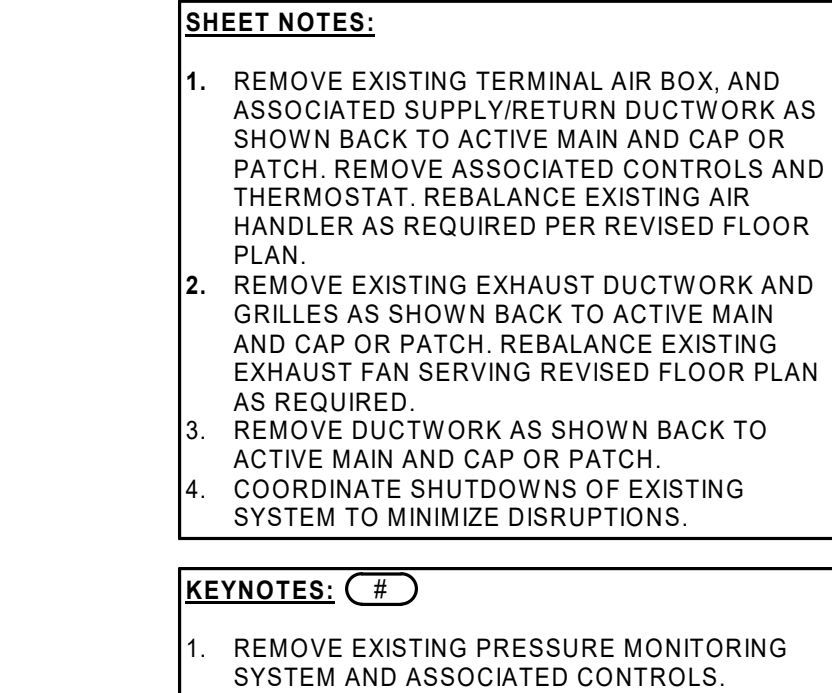
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PLAN
NORTH



ARCHITECT/ENGINEER OF RECORD			STAMP	Office of Construction and Facilities Management VA U.S. Department of Veterans Affairs	Drawing Title BASEMENT DEMOLITION - VENTILATION - PHASE 3	Phase 100% CONSTRUCTION DOCUMENTS	Project Title EXPAND STERILE PROCESSING SERVICE (SPS)	Project Number 556-402
 Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence life safety, inc. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	 11/29/2022 exp 11/29/2023		Approved: MD100.3	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Drawing Number 133CA



SHEET NOTES:	
1.	REMOVE EXISTING TERMINAL AIR BOX, AND ASSOCIATED SUPPLY/RETURN DUCTWORK AS SHOWN BACK TO ACTIVE MAIN AND CAP OR PATCH. REMOVE ASSOCIATED CONTROLS AND THERMOSTAT. REBALANCE EXISTING AIR HANDLER AS REQUIRED PER REVISED FLOOR PLAN.
2.	REMOVE EXISTING EXHAUST DUCTWORK AND GRILES AS SHOWN BACK TO ACTIVE MAIN AND CAP OR PATCH. REBALANCE EXISTING EXHAUST FAN SERVING REVISED FLOOR PLAN AS REQUIRED.
3.	REMOVE DUCTWORK AS SHOWN BACK TO ACTIVE MAIN AND CAP OR PATCH.
4.	COORDINATE SHUTDOWNS OF EXISTING SYSTEM TO MINIMIZE DISRUPTIONS.

KEYNOTES: (#)

1. REMOVE EXISTING PRESSURE MONITORING SYSTEM AND ASSOCIATED CONTROLS.


$$1/8^{\circ} = 1'-0"$$

ARCHITECT/ENGINEER OF RECORD			
	Alesia Architecture P.C. 3705 North 26th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence life safety, inc. 3575 Rinsley Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3031	 Force Protect Security Cons. 19901 Front Beach Road, Suite 1415, Panama City Beach, FL 32407
	Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68108 Phone: (402) 455-7698	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 703-0434

STAMP



Office of
Construction
and Facilities
Management

VA U.S. Department
of Veterans Affairs

Drawing Title

BASEMENT DEMOLITION -
VENTILATION - PHASE 4

Approved:

Phase

100% CONSTRUCTION
DOCUMENTS

FULLY SPRINKLERED

Project Title

EXPAND STERILE PROCESSING
SERVICE (SPS)

Location	Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064
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Issue Date	11/29/2022
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Checked	MRM
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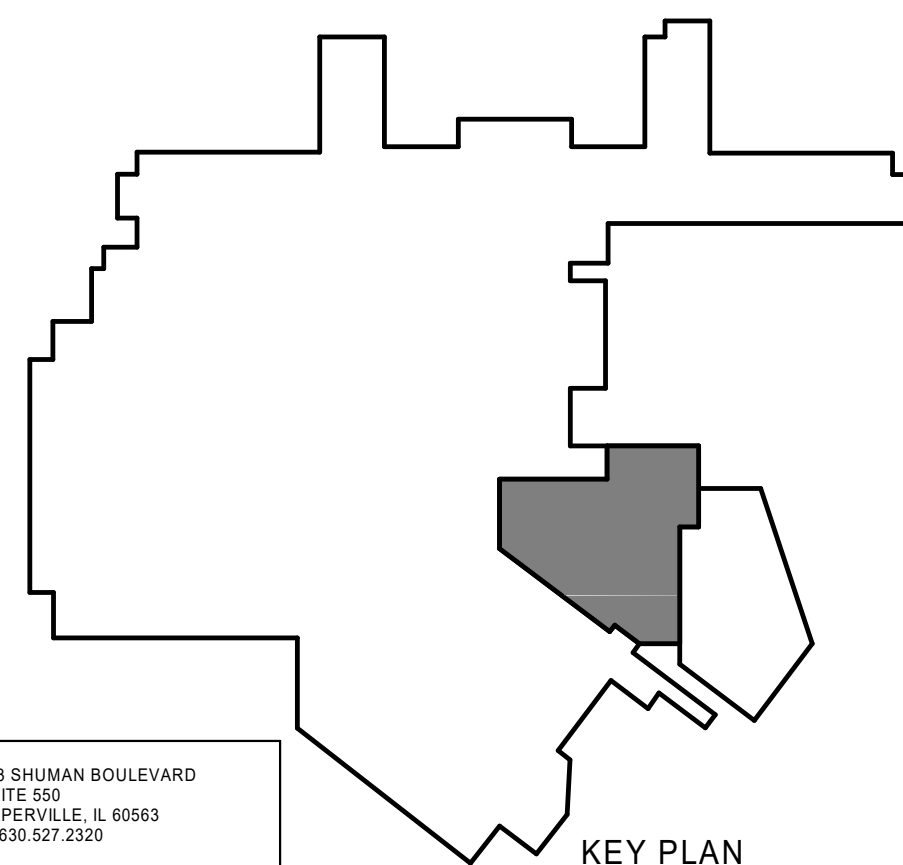
Drawn
NVP

Project Number

556-402
Building Number 133CA

Drawing Number	
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MD100.4



✓ KEY PLAIN



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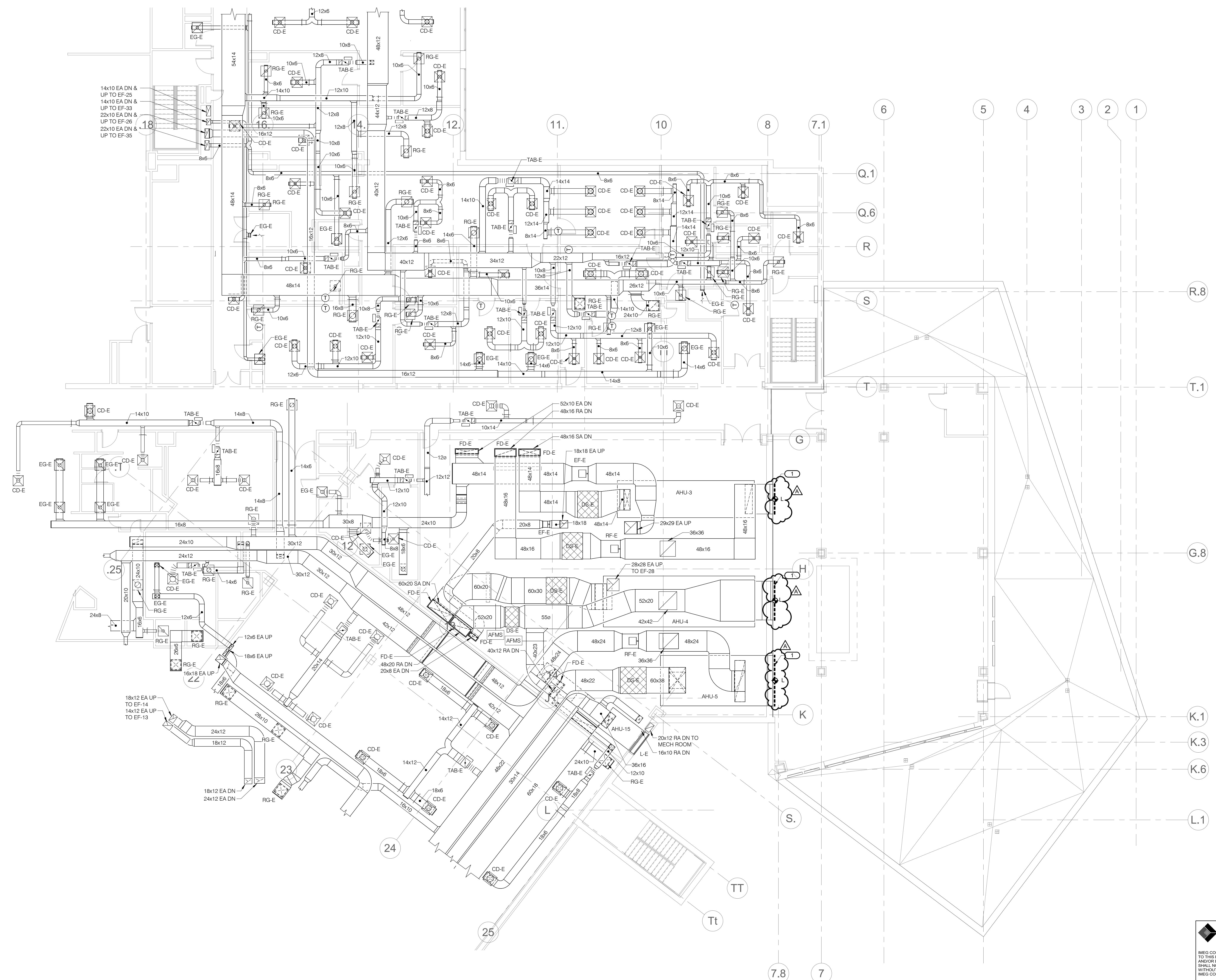
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REF. SCALE IN INCHES PROJECT #21008095.00

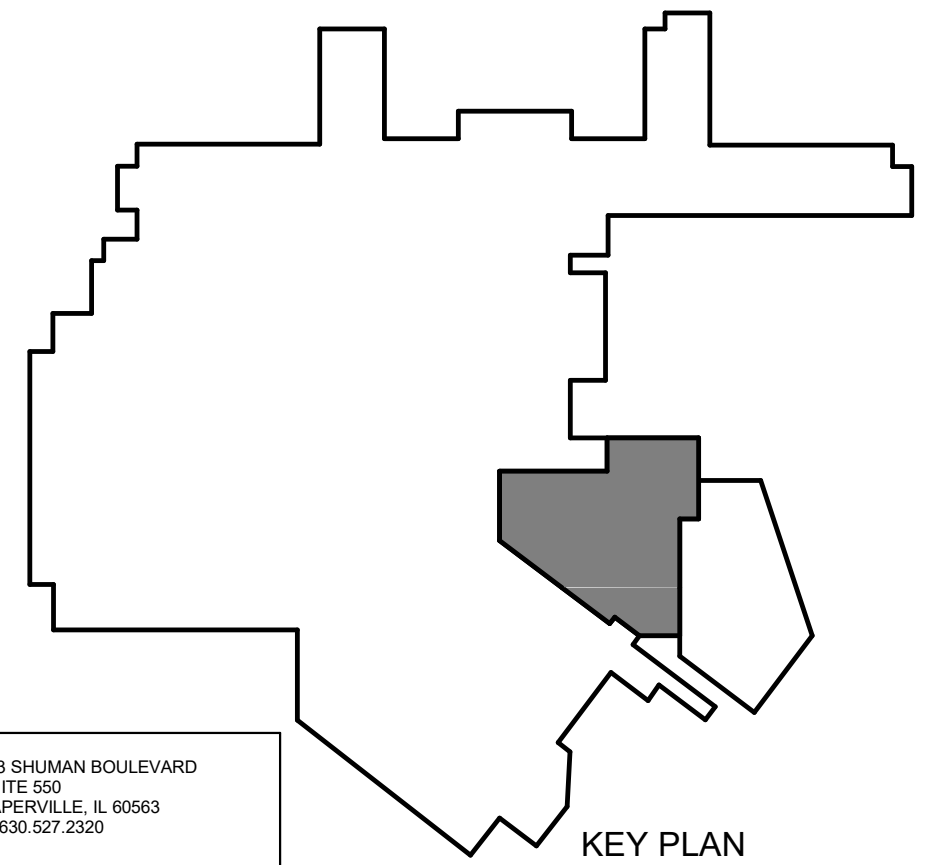
KEYNOTES

1. REMOVE EXISTING LOUVER AND PREPARE EXISTING PLENUM FOR NEW CONNECTIONS. CONTRACTOR TO FIELD VERIFY EXISTING LOUVERS AND PLENUM. DO NOT REMOVE EXISTING CONTROL DAMPERS AND AFMS ON OA DUCT. COORDINATE SHUTDOWNS OF EXISTING SYSTEM TO MINIMIZE DISRUPTIONS.



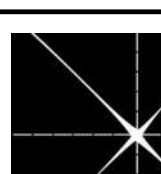
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Illinois Design Firm Registration #184.007637-0014
PROJECT #21008955.00

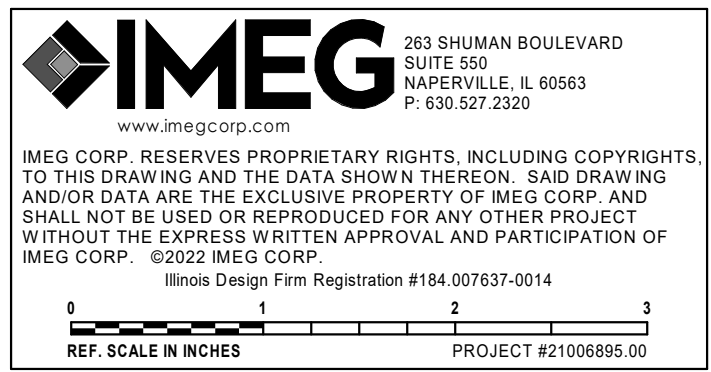
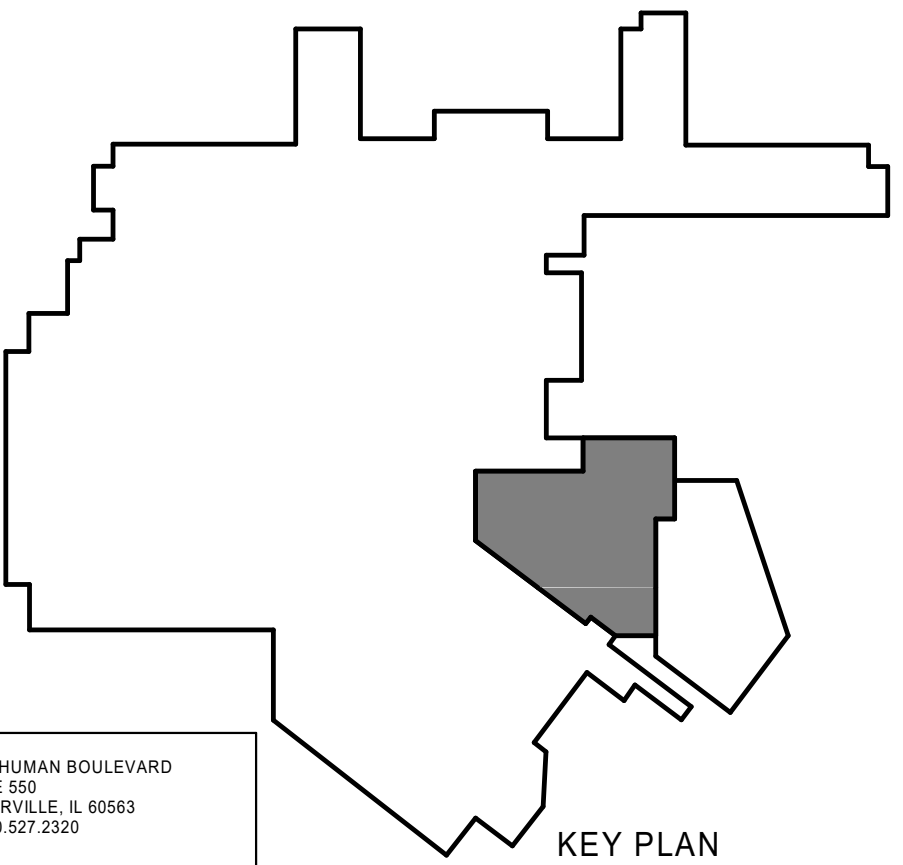
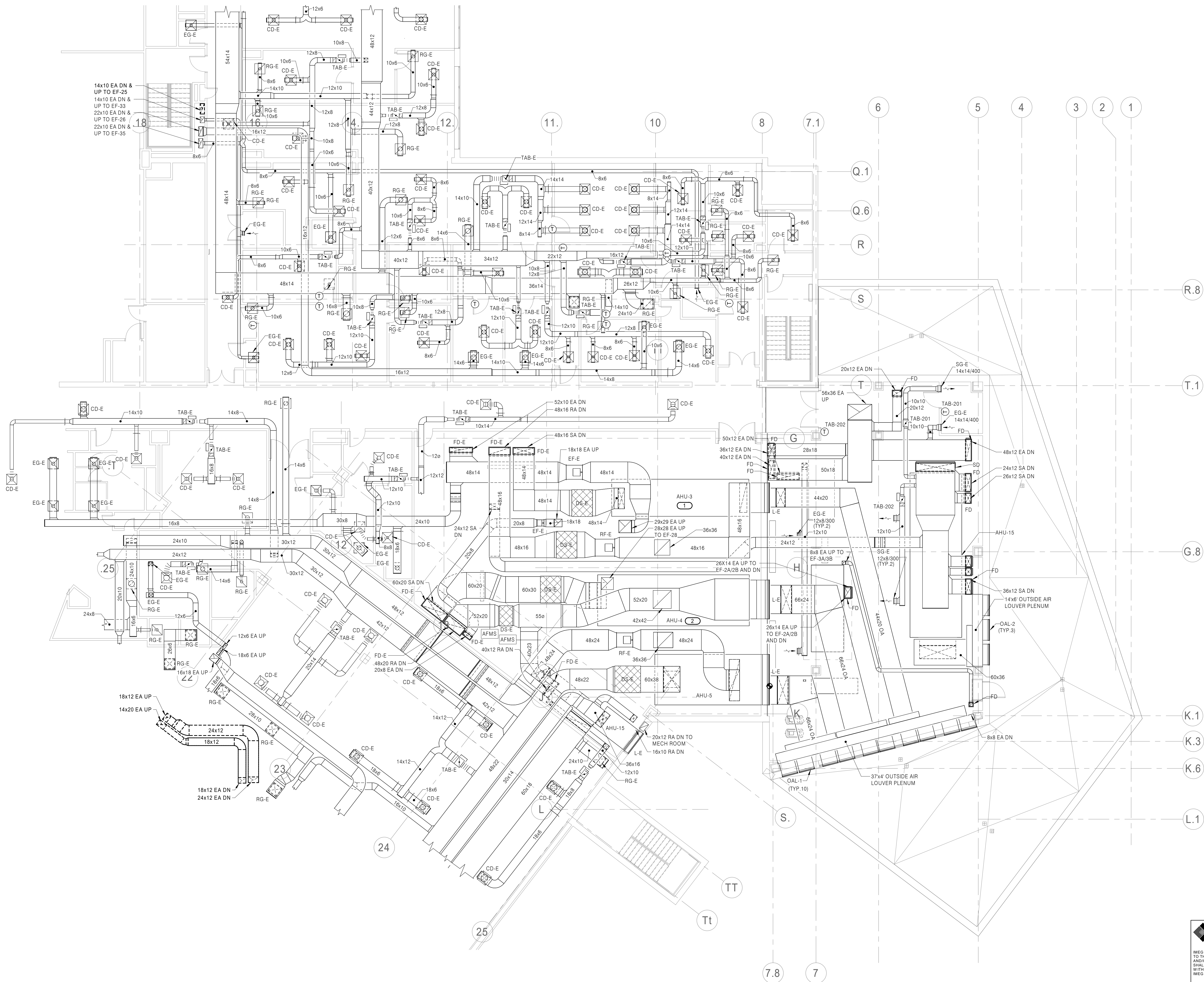


FIRST FLOOR DEMOLITION - VENTILATION - PHASE 2

1/8" = 1'-0"

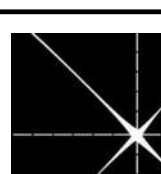
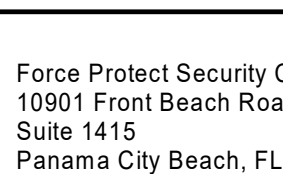
ARCHITECT/ENGINEER OF RECORD				STAMP	Office of Construction and Facilities Management	Drawing Title FIRST FLOOR DEMOLITION - VENTILATION - PHASE 2	Phase 100% CONSTRUCTION DOCUMENTS	Project Title EXPAND STERILE PROCESSING SERVICE (SPS)	Project Number 556-402
 Alesia Architecture P.C. 2705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434					
Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098					U.S. Department of Veterans Affairs	Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Drawing Number MD101.2
A VA Revisions Mark Revisions: 07/05/2023 Date:									

- KEYNOTES:**
1. REBALANCE EXISTING AHU-3 SUPPLY AIR FAN TO 21075 CFM AND RETURN AIR FAN TO 19125 CFM.
 2. REBALANCE EXISTING AHU-4 SUPPLY AIR FAN TO 10875 CFM AND RETURN AIR FAN TO 9350 CFM.



FIRST FLOOR DEMOLITION - VENTILATION - PHASE 4
1/8" = 1'-0"

Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD			
 Ayesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434
 Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434	



Office of
Construction
and Facilities
Management

VA U.S. Department
of Veterans Affairs

Drawing Title
**FIRST FLOOR DEMOLITION -
VENTILATION - PHASE 4**

Approved:

Phase
**100% CONSTRUCTION
DOCUMENTS**

FULLY SPRINKLERED

Project Title
**EXPAND STERILE PROCESSING
SERVICE (SPS)**

Location
Captain James A. Lovell Federal Healthcare Center
3001 Green Bay Road, North Chicago, IL 60064

Issue Date
11/29/2022

Checked
MRM

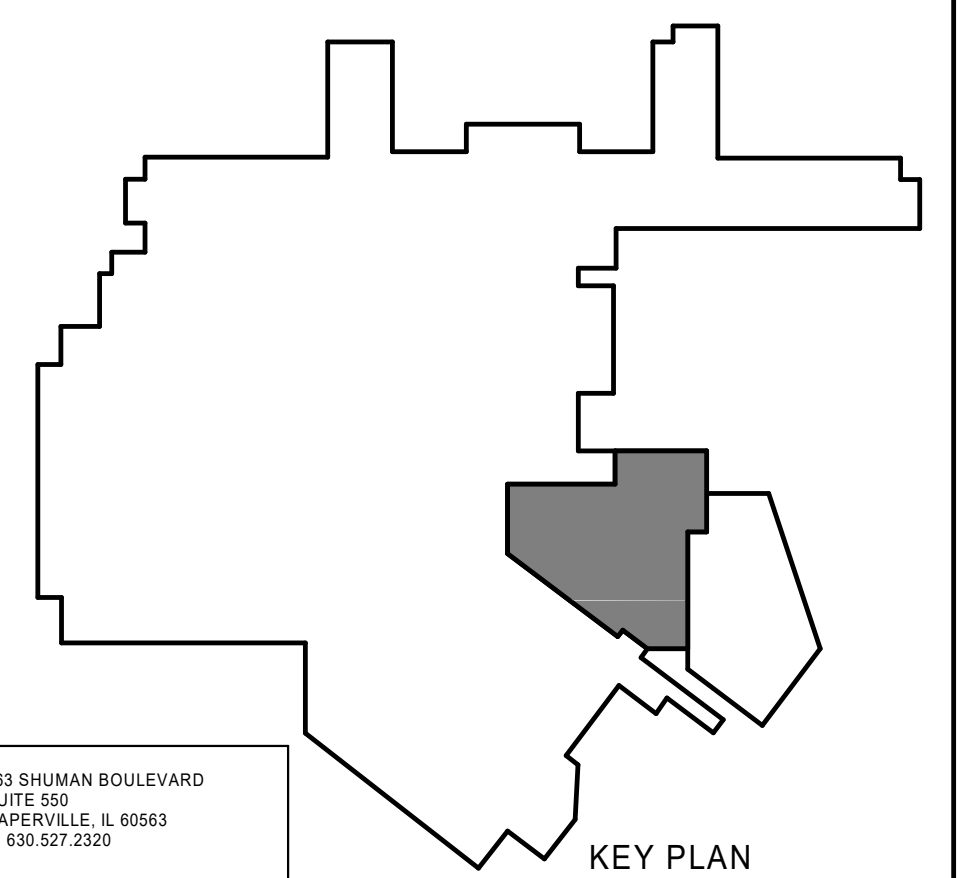
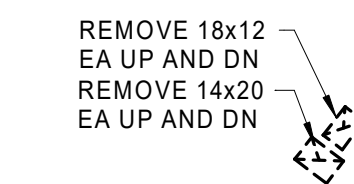
Drawn
NVP

Project Number
556-402

Building Number
133CA

Drawing Number
MD101.4

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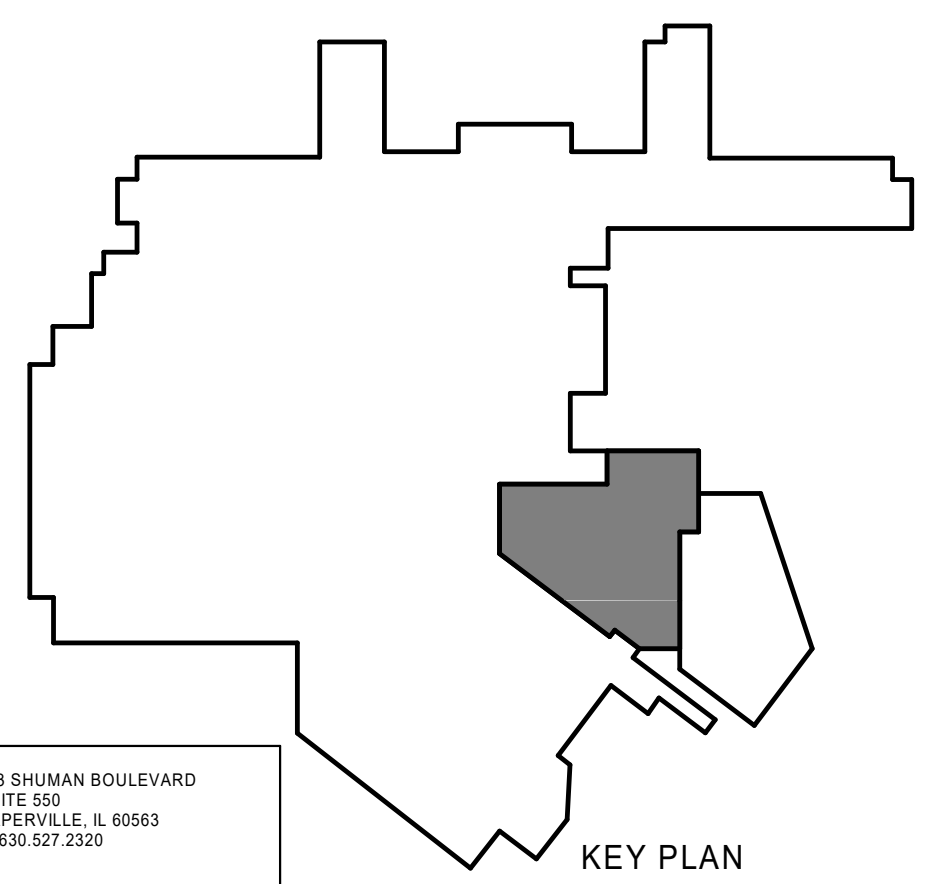
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SECOND FLOOR DEMOLITION – VENTILATION – PHASE 4

1/8" = 1'-0"

[illegible]

ARCHITECT/ENGINEER OF RECORD				STAMP	Office of Construction and Facilities Management VA U.S. Department of Veterans Affairs	Drawing Title	Phase	Project Title	Project Number		
	Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941		incandescence life safety, inc. 3575 Kingsley Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010			Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407		SECOND FLOOR DEMOLITION – VENTILATION – PHASE 4	100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402
	Building Number		133CA								
	Shaffer Stevens & Schaap P.C. 9070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7659		IMEG Corp. 9070 S Ridgeline Blvd, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320		Calibre Engineering, Inc. 9900 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434		Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3931 Green Bay Road, North Chicago, IL 60064	Drawing Number	
	MD102.4										


$$1/8'' = 1'-0''$$

Project Number	556-402
Building Number	133CA
Drawing Number	MD103.4



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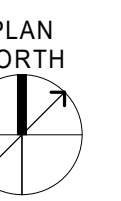
283 SHUMAN BOULEVARD
SUITE 550
NAPERVILLE, IL 60563
P: 630.527.2320

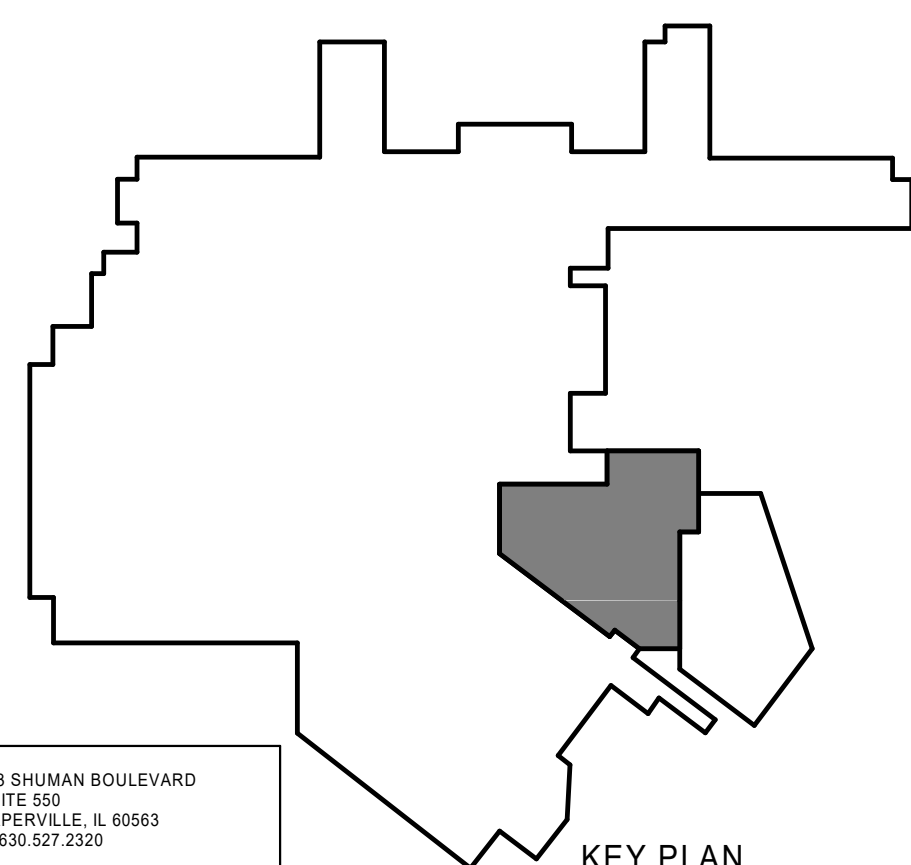
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Illinois Design Firm Registration #184-007637-0014



PROJECT #21006895.00





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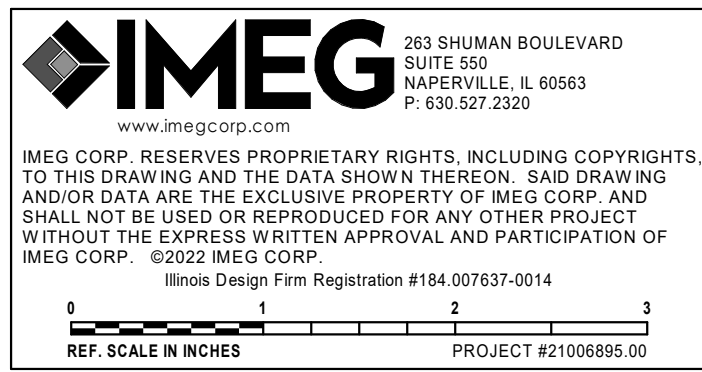
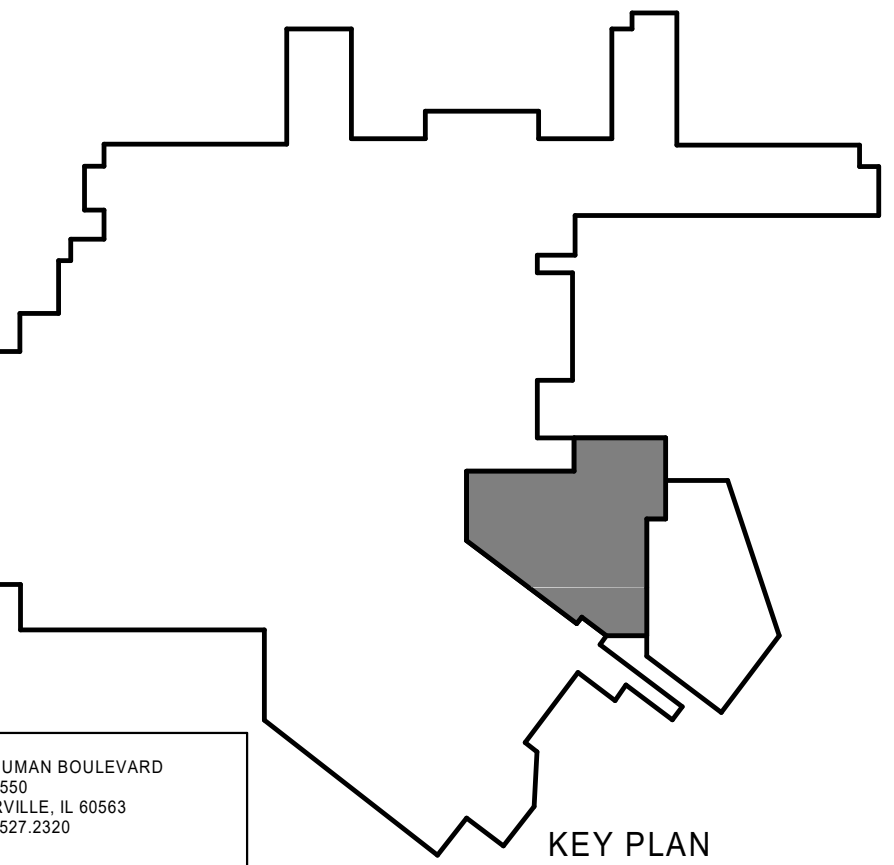
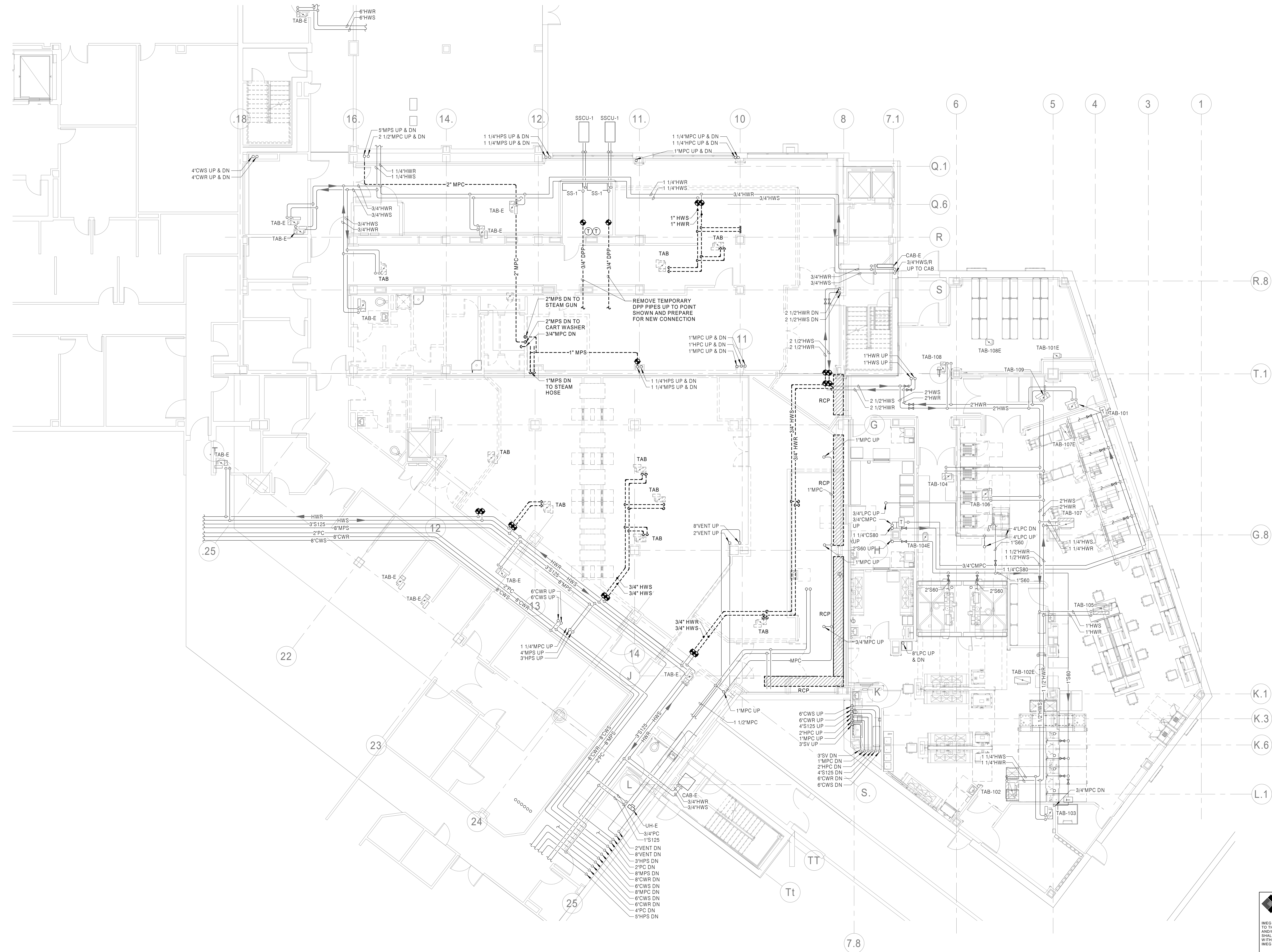
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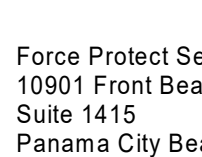
Project Number	556-402
Building Number	133CA
Drawing Number	MD104.4

- SHEET NOTES:**
1. REMOVE EXISTING HYDRONIC PIPING AS SHOWN BACK TO ACTIVE MAIN AND CAP.
 2. REMOVE EXISTING STEAM PIPING AND ASSOCIATED ACCESSORIES BACK TO ACTIVE MAIN AND CAP.
 3. COORDINATE SHUTDOWNS OF EXISTING SYSTEM TO MINIMIZE DISRUPTIONS.



1 BASEMENT DEMOLITION - PIPING - PHASE 4
1/8" = 1'-0"

Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD			
 Aesia Architecture P.C. 2705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence Life Safety, Inc. 3575 Ringold Ct Unit 408 Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434
 Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434	 BRADLEY D. YOUNG 062-060058 LICENSED PROFESSIONAL ENGINEER OF ILLINOIS 11/29/2022 03/03/2023



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and Facilities
Management

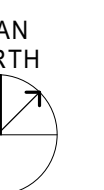
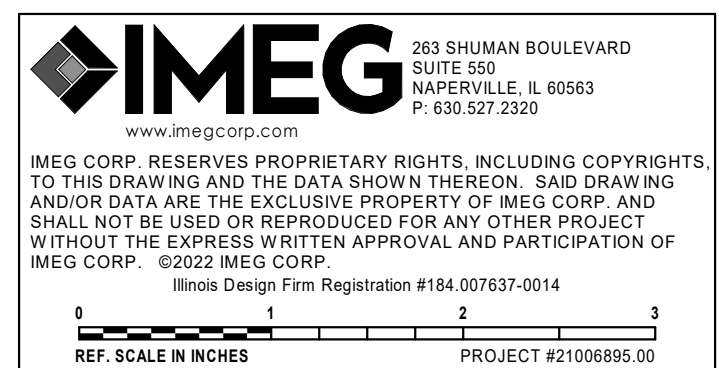
VA U.S. Department
of Veterans Affairs

Drawing Title	BASEMENT DEMOLITION - PIPING - PHASE 4
Approved:	

Phase	100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED	

Project Title	EXPAND STERILE PROCESSING SERVICE (SPS)
Location	Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064
Issue Date	11/29/2022
Checked	PGP
Drawn	NVP

Project Number	556-402
Building Number	133CA
Drawing Number	MD200.4

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ARCHITECT/ENGINEER OF RECORD			STAMP	Office of Construction and Facilities Management	Drawing Title	Project Title	Project Number				
	Alexia Architecture P.C. 3705 North 20th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941		incandescence fire safety, inc. 3575 Ringway Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3910			Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	PIPE BASEMENT - VENTILATION - PHASE 2	100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402	
	Shaffer Stovens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698		IMEG Corp. 1663 Sherman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320			Calibre Engineering, Inc. 2090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434	Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road North Chicago, IL 60064	Drawing Number 133CA	
 U.S. Department of Veterans Affairs								Issue Date 11/29/2022	Checked MRM	Drawn NVP	M1PB.2

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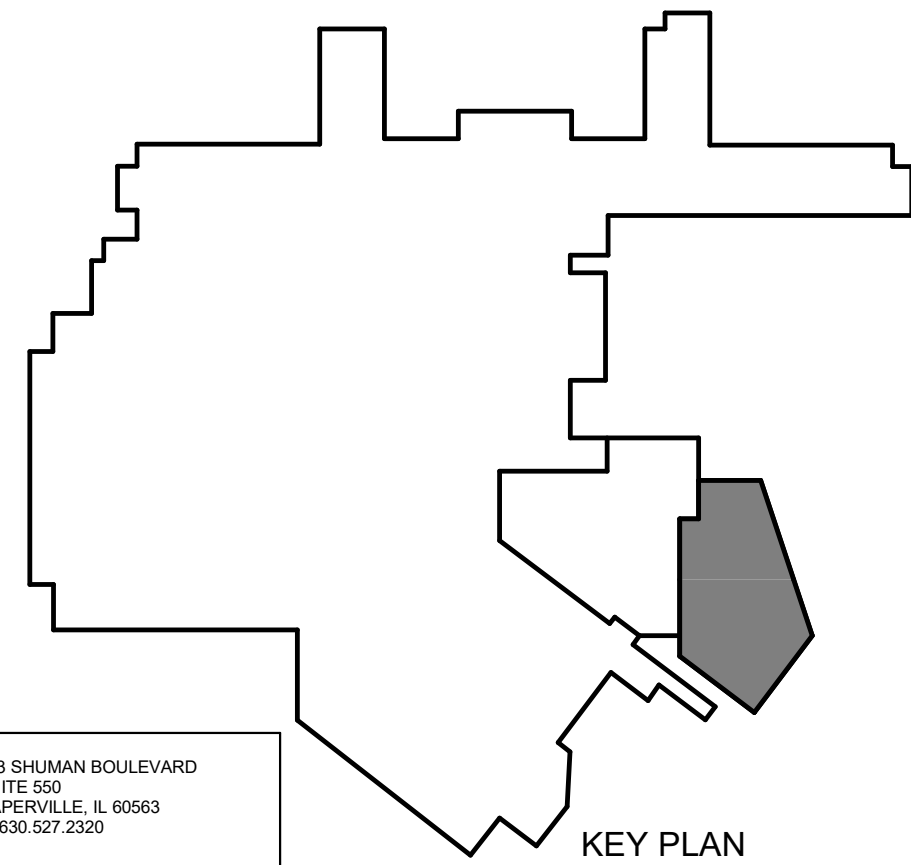
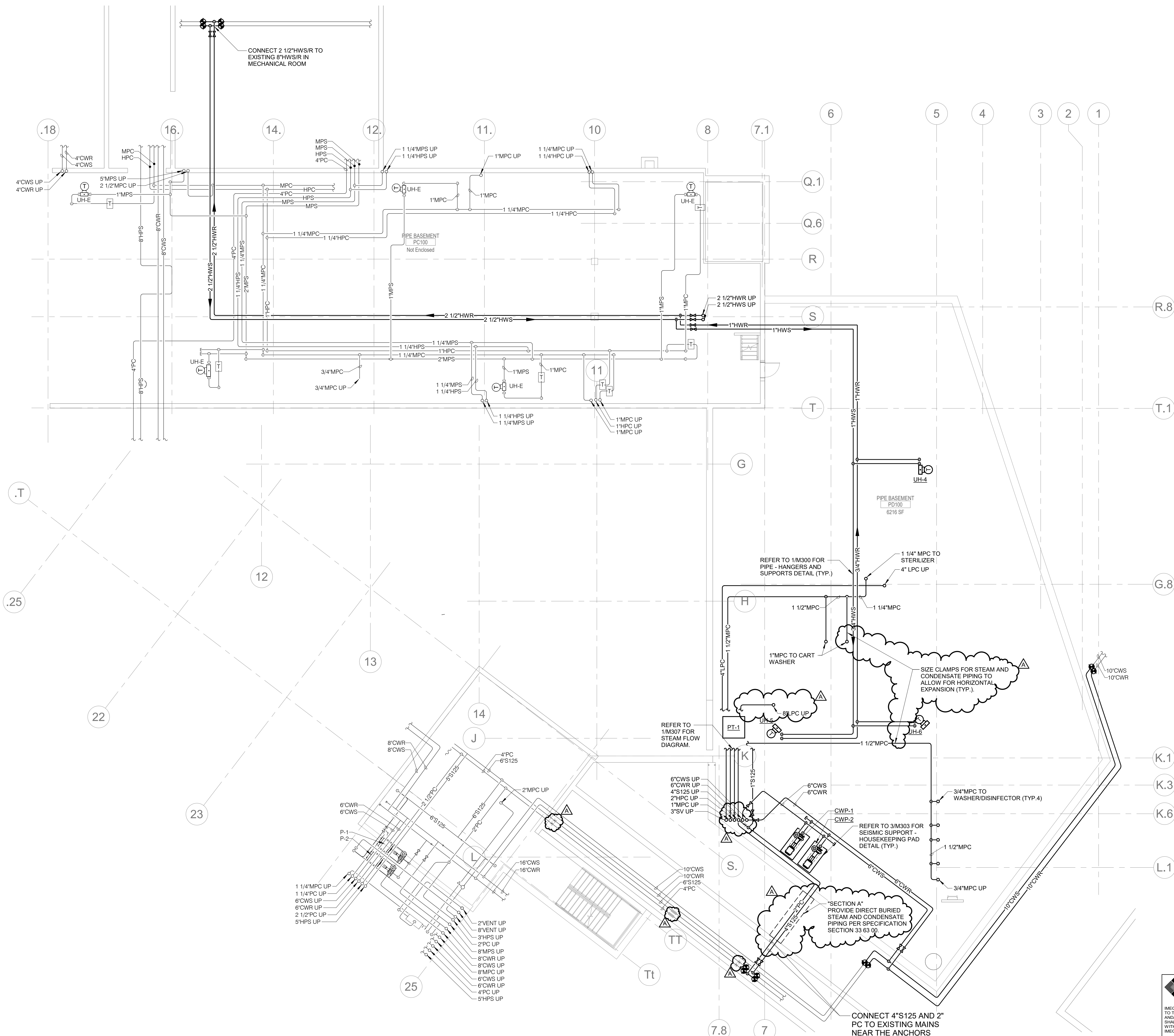
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VA

Autodesk Docs/2023-22-556-402 - VA Expand Sterile Processing Services (SPS) - Design/MEP/IT2_2100895.00_L-VA-North Chicago - L-SPS

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- SHEET NOTES:**
1. CONTRACTORS FINAL PIPE ROUTING SHALL ALLOW FULL ACCESS IN THE PIPE BASEMENT. PRIORITIZE ACCESS TO EQUIPMENT REQUIRING MAINTENANCE. PIPING SHALL NOT BLOCK ANY PATHWAYS.
 2. ALL STEAM SUPPLY PIPING AND ASSOCIATED FITTINGS SHALL BE OF STAINLESS STEEL MATERIAL. ALL CLEAN STEAM SUPPLY PIPING, CLEAN STEAM CONDENSATE PIPING, FITTINGS AND ASSOCIATED ACCESSORIES SHALL BE OF STAINLESS STEEL MATERIAL.



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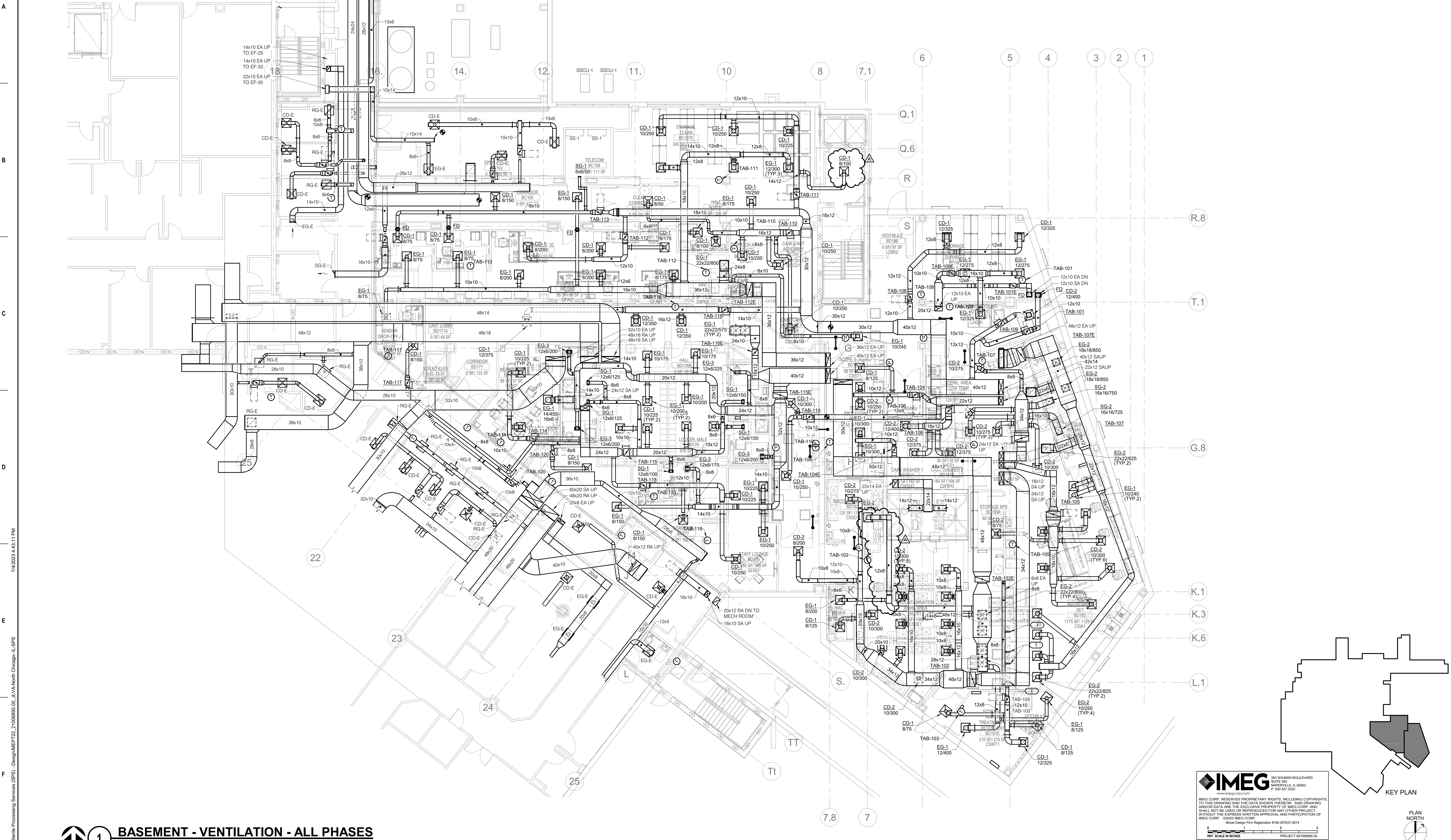
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1 PIPE BASEMENT - PIPING - PHASE 2
1/8" = 1'-0"

A		VA Revisions	07/05/2023
Mark	Revisions:		Date:

ARCHITECT/ENGINEER OF RECORD				STAMP	Office of Construction and Facilities Management	Drawing Title	Phase	Project Title	Project Number									
	Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941			incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010			Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407			11/29/2022	11/30/2023		U.S. Department of Veterans Affairs	Approved:	PIPE BASEMENT - PIPING - PHASE 2	100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402
		Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098			IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320			Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434										
Issue Date 11/29/2022		Checked PGP			Drawn NVP			Drawing Number M2PB.2										

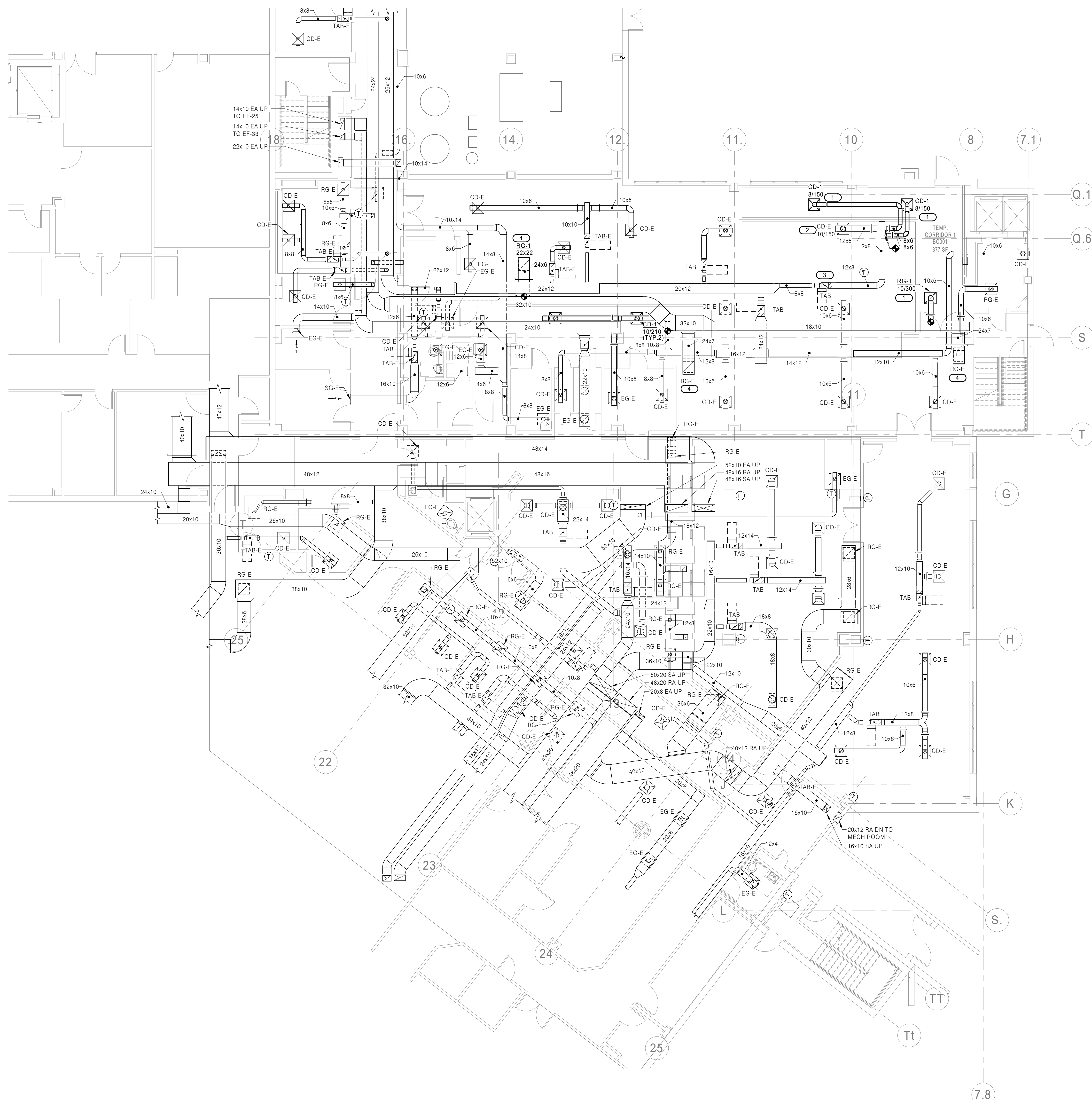
SHEET NOTES:
1. THIS DRAWING REPRESENTS OVERALL AREA OF WORK. REFER TO INDIVIDUAL PHASE SHEETS FOR DETAILED INFORMATION OF SCOPE OF WORK.



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VA FORM 08 - 6231

1 BASEMENT - VENTILATION - ALL PHASES
1/8" = 1'-0"

ARCHITECT/ENGINEER OF RECORD			STAMP	Office of Construction and Facilities Management	Drawing Title	Phase	Project Title	Project Number	
 Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence life safety, inc. 3575 Ringcity Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	 11/29/2022 11/30/2023		BASEMENT - VENTILATION - ALL PHASES	100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402	
					Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Building Number 133CA	
 Shaffer Stevens & Schapp P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434	 U.S. Department of Veterans Affairs			Issue Date 11/29/2022 07/05/2023 REVISIONS	Checked MRM	Drawn NVP	Drawing Number M100.0



SHEET NOTES:

1. FLEX DUCTWORK IS NOT ALLOWED IN SPS DEPARTMENT PER VA STANDARDS.

KEYNOTES: (#)

1. PROVIDE TEMPORARY DIFFUSERS/GRILLES TO SERVE TEMPORARY CORRIDOR AS SHOWN.

2. REBALANCE EXISTING GRILLE/DIFFUSER TO MEET MAX VALUE FLOW.

3. REBALANCE EXISTING TERMINAL AIR BOX TO 450 CFM COOLING/HEATING MAX AND BOX MIN VALUE.

4. PROVIDE NEW RETURN GRILLE AS SHOWN.

5. REBALANCE EXISTING RETURN GRILLES SERVING EXISTING AREA ALONGWITH NEW RETURN GRILLE AS REQUIRED TO MEET REVISED FLOOR PLAN.

BASEMENT - VENTILATION - PHASE 1

$$1/8^{\circ} = 1'-0''$$
[illegible]

ARCHITECT/ENGINEER OF RECORD



Alesia Architecture P.C.
3705 North 200th Street
Elkhorn, Nebraska 68022
Phone: (402) 291-6941



incandescence life safety, inc.
3575 Ringsby Ct Unit 408,
Denver, CO 80216
Phone: (720) 722-3010



Force Protect Security Cons.
10901 Front Beach Road,
Suite 1415
Panama City Beach, FL 32407



Shaffer Stevens & Schaap P.C.
1070 Howard St
Omaha, Nebraska 68102
Phone: (402) 455-7698



IMEG Corp.
263 Shuman Boulevard
Naperville, IL 60563
Phone: (630) 527-2320



Calibre Engineering, Inc.
9090 S Ridgeline Blvd #105,
Highlands Ranch, CO 80129
Phone: (303) 730-0434

STAMP



Office of
Construction
and Facilities
Management

VA U.S. Department
of Veterans Affairs

Drawing Title
BASEMENT - VENTILATION - PHASE 1

Approved:

Phase
100% CONSTRUCTION
DOCUMENTS

FULLY SPRINKLERED

Project Title	EXPAND STERILE PROCESSING SERVICE (SPS)
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Location	
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Issue Date	11/29/2022
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Checked
MRM

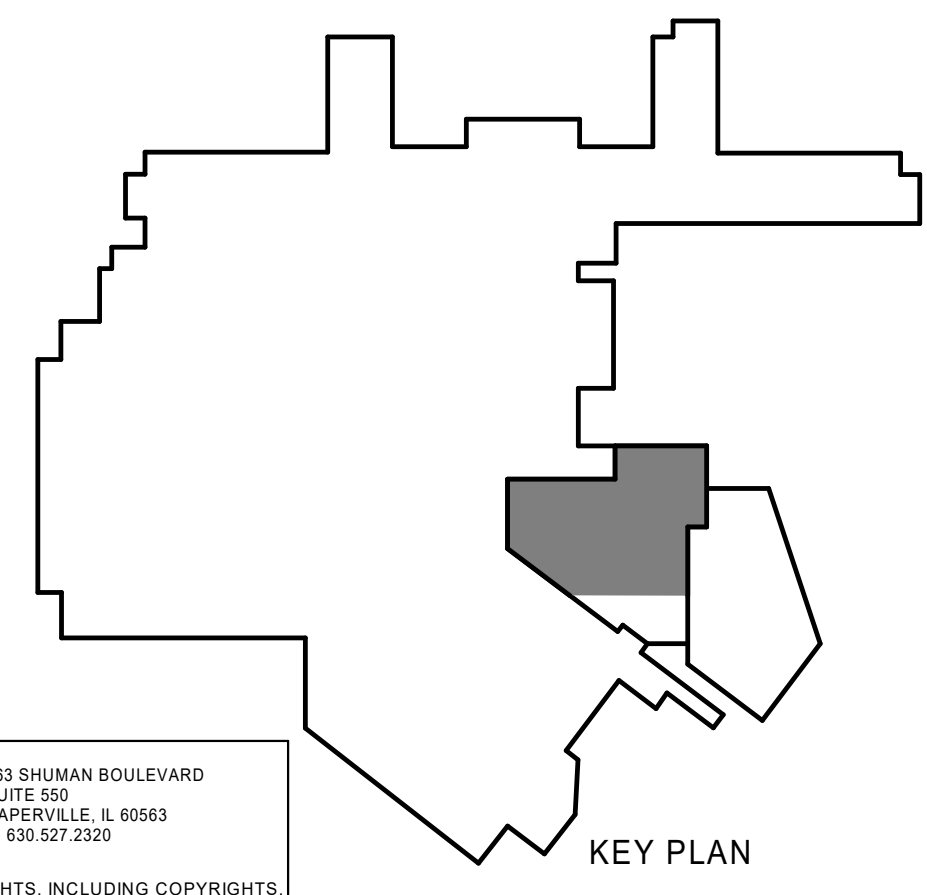
Drawn
NVP

Project Number
556-402

Building Number
133CA

Drawing Number

M100.1



KEY PLAN



IMEG

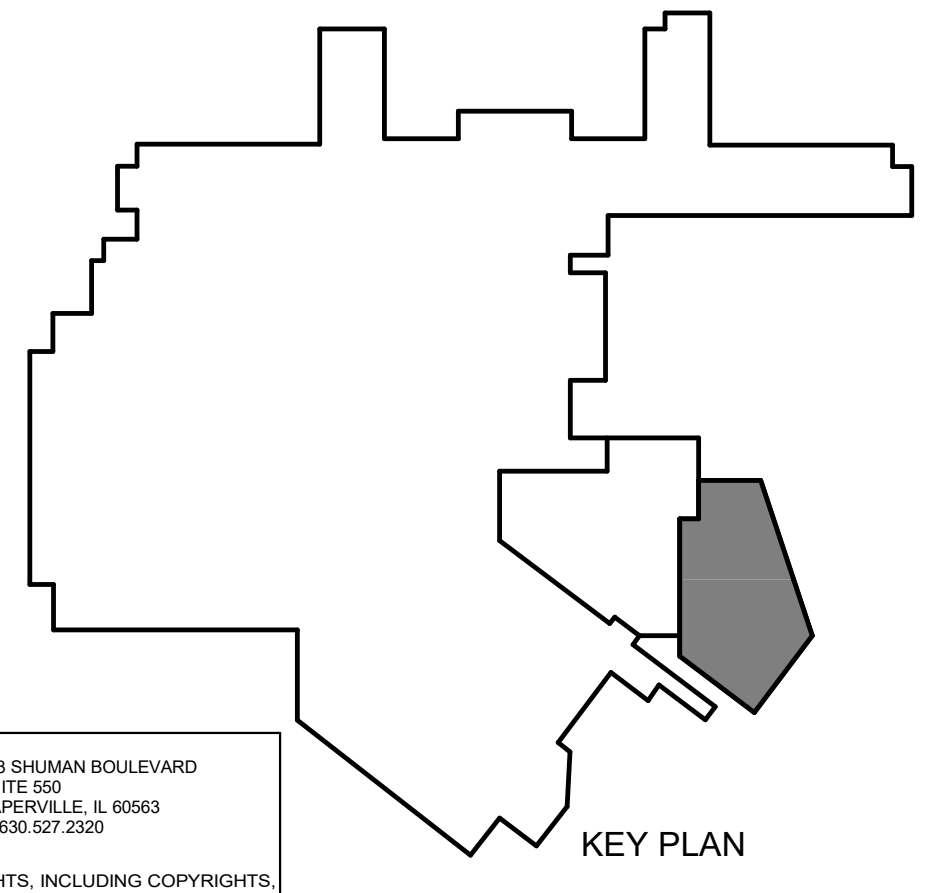
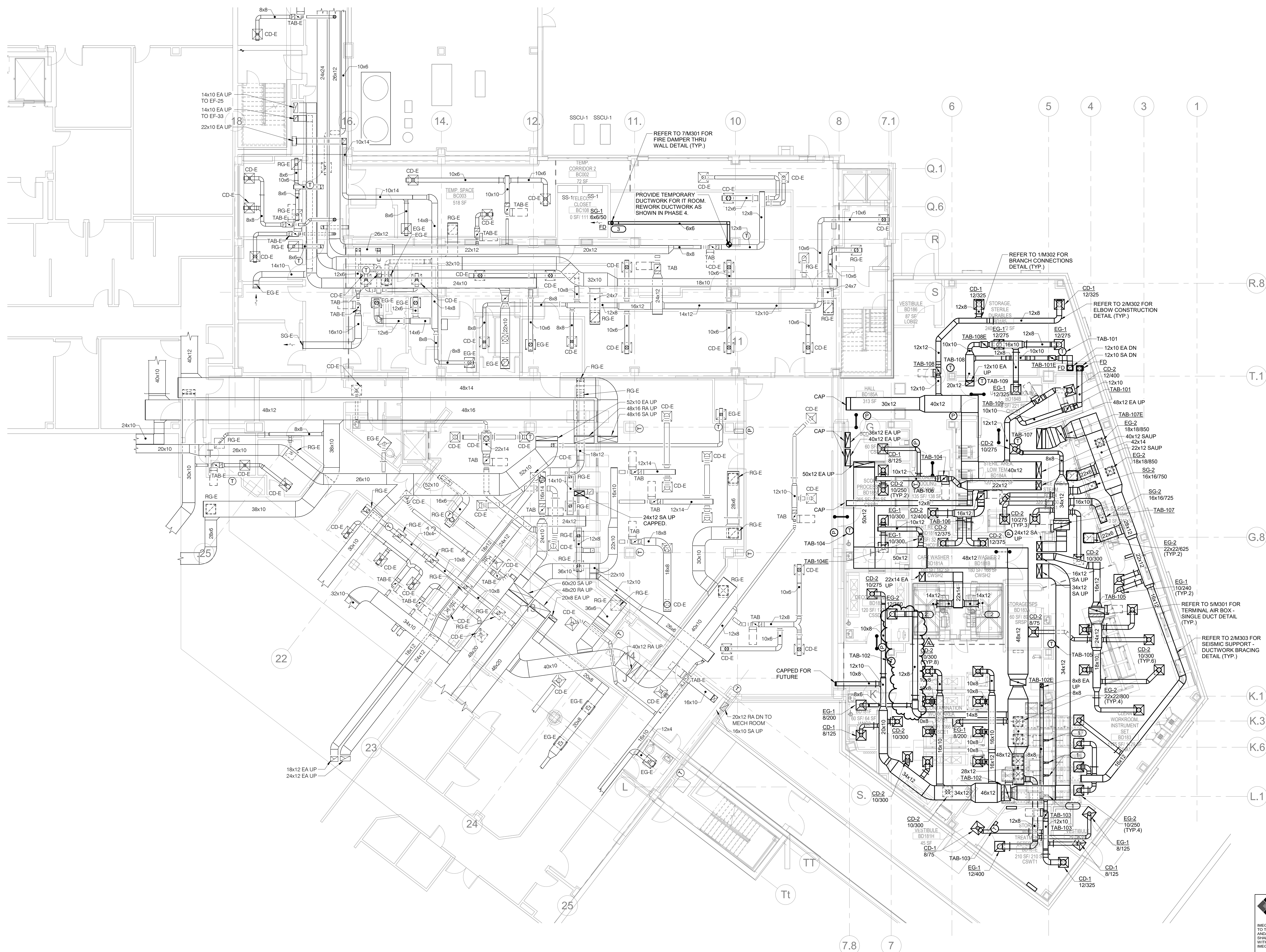
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Illinois Design Firm Registration #184-007637-0014

0 1 2 3
REF. SCALE IN INCHES PROJECT #21008695.00

- SHEET NOTES:**
1. FLEX DUCTWORK IS NOT ALLOWED IN SPS DEPARTMENT PER VA STANDARDS.
- KEYNOTES:** #
1. 3"Ø EA DOWN TO WASHER/DISINFECTOR. COORDINATE WITH MANUFACTURER FOR EXACT CONNECTION REQUIREMENTS.
 2. CONNECT EXHAUST DUCT TO CART WASHER. VERIFY FINAL CONNECTION SIZE AND REQUIREMENTS WITH EQUIPMENT MANUFACTURER RECOMMENDATIONS.
 3. PROVIDE ACCESS PANEL FOR FIRE DAMPER. COORDINATE WITH ARCHITECT PLANS FOR CEILING DETAILS.



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Basic Design Firm Registration #184.007637-0014

1 BASEMENT - VENTILATION - PHASE 2
1/8" = 1'-0"

Revision Day			
	A	VA Revisions	07/05/2023
	Mark	Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD			
A Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	Force Protect Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	Calibre Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434
S Shaffer Stevens & Schapp P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098	IMEG IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320		

STAMP

BRADY D. YOUNG
062-060058
LICENSED
PROFESSIONAL
ENGINEER
OF
ILLINOIS
11/29/2022
10/30/2023

Office of
Construction
and Facilities
Management

VA U.S. Department
of Veterans Affairs

Drawing Title
BASEMENT - VENTILATION - PHASE 2

Approved:

Phase
**100% CONSTRUCTION
DOCUMENTS**

FULLY SPRINKLERED

Project Title
**EXPAND STERILE PROCESSING
SERVICE (SPS)**

Location
Captain James A. Lovell Federal Healthcare Center
3001 Green Bay Road, North Chicago, IL 60064

Issue Date
11/29/2022
07/05/2023 REVISIONS

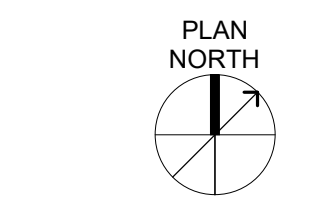
Checked
MRM

Drawn
NVP

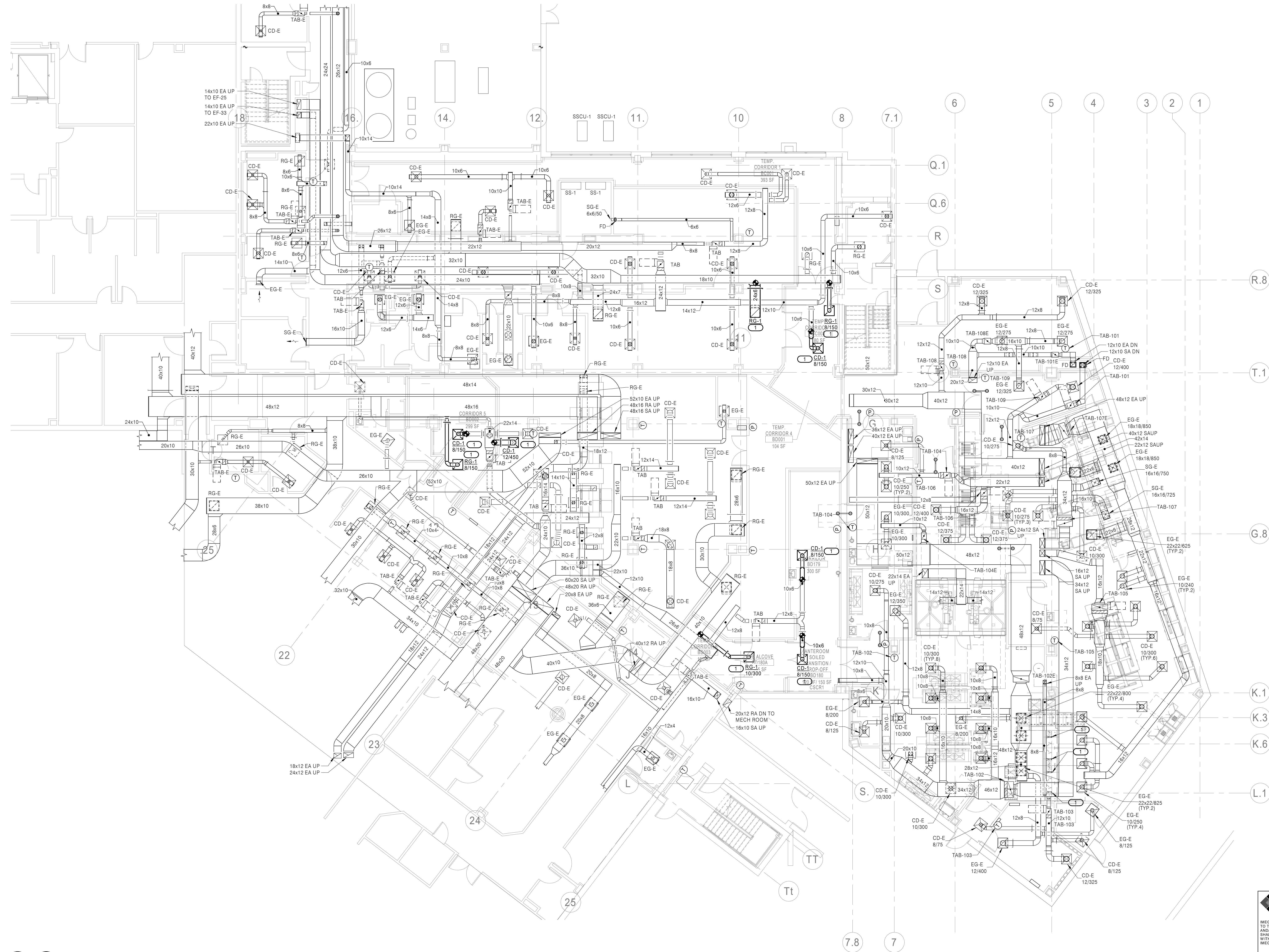
Project Number
556-402

Building Number
133CA

Drawing Number
M100.2



11/29/2022 1:28:24 AM
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Revision: 001

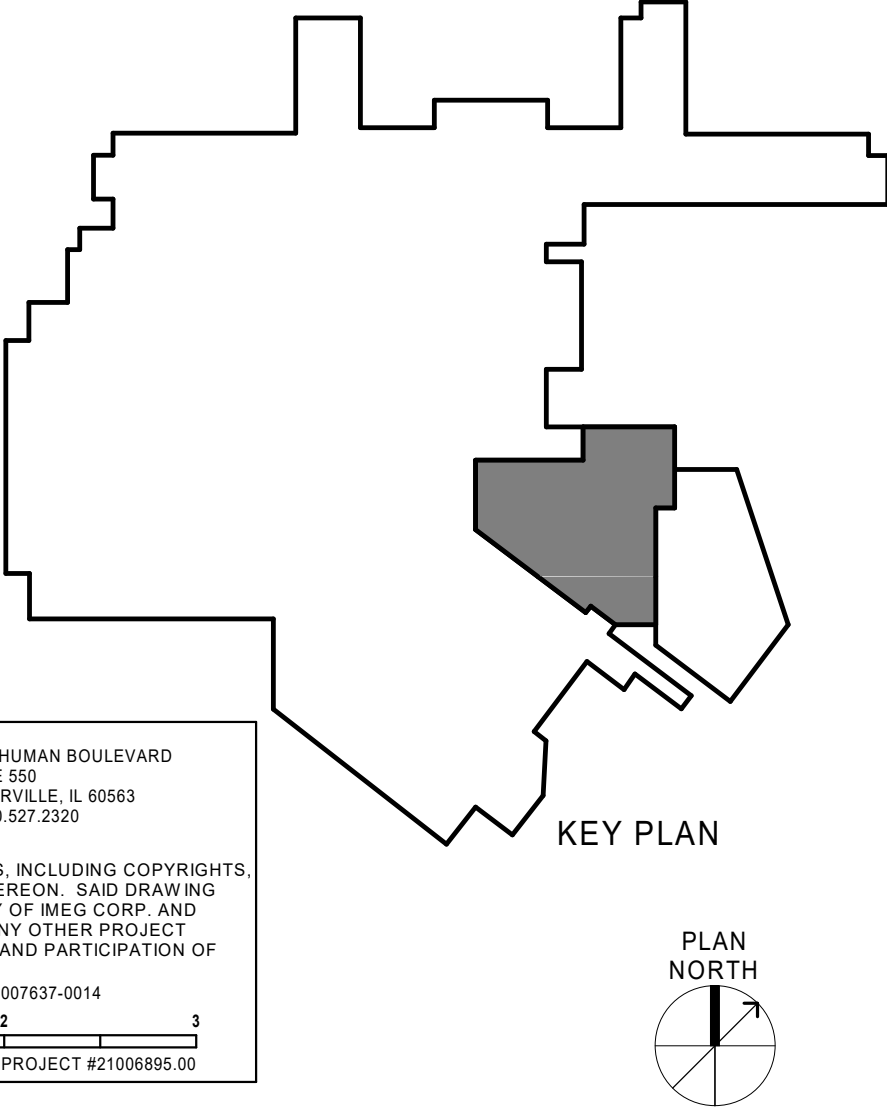


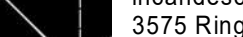
- SHEET NOTES:**
1. FLEX DUCTWORK IS NOT ALLOWED IN SPS DEPARTMENT PER VA STANDARDS.
- KEYNOTES:** (C) #
1. PROVIDE TEMPORARY DIFFUSERS/GRILLES TO SERVE TEMPORARY CORRIDORS AS SHOWN AND BALANCE THE TERMINAL AIR BOX AS REQUIRED.

1 BASEMENT - VENTILATION - PHASE 3
1/8" = 1'-0"

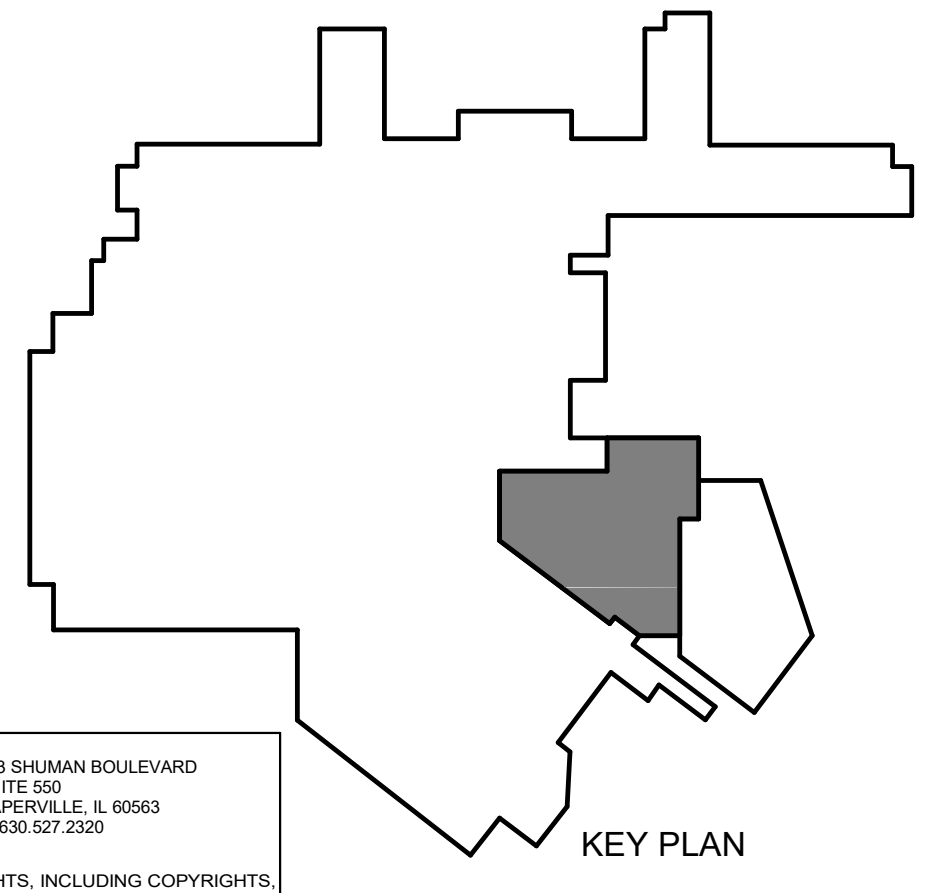
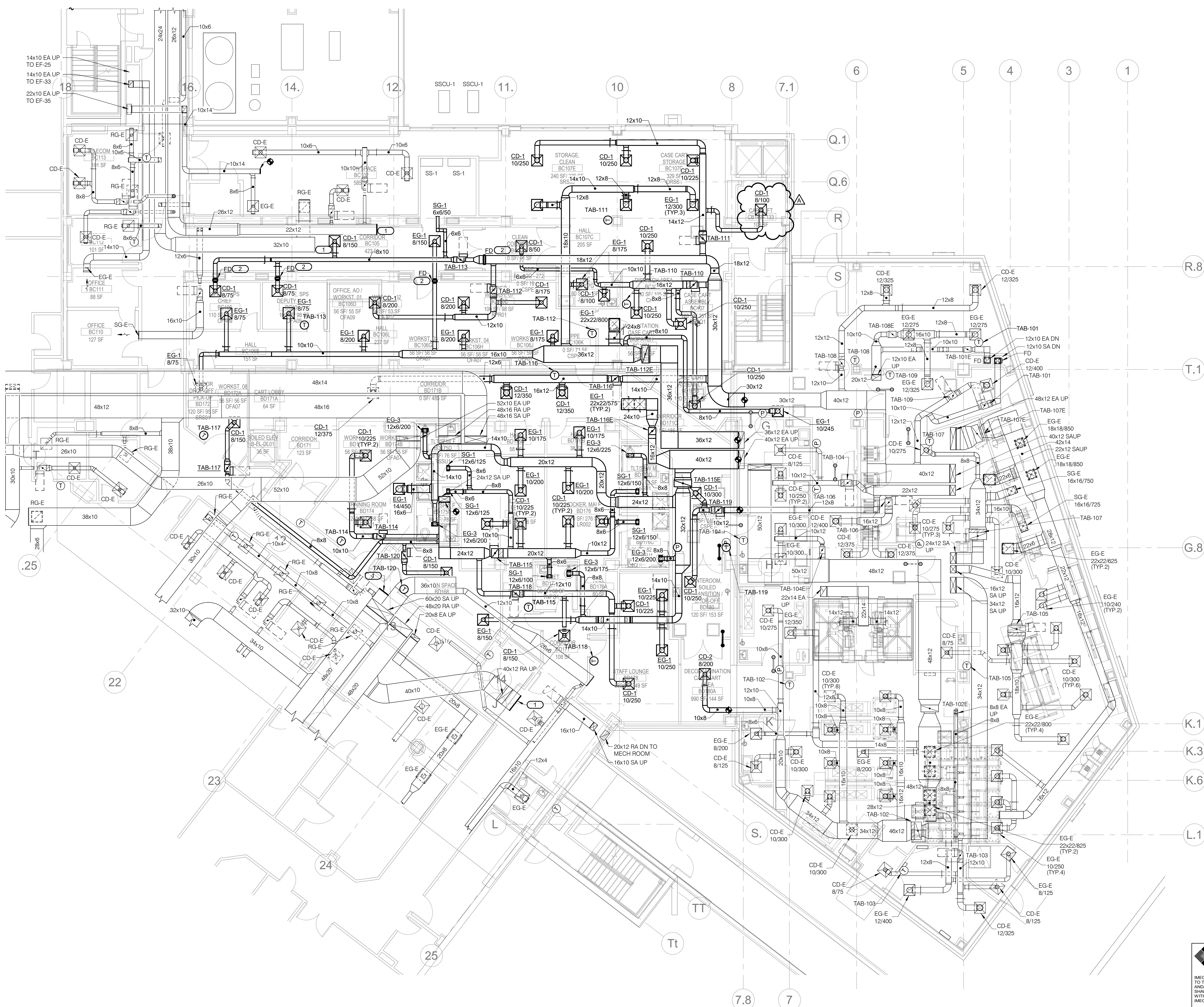
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ARCHITECT/ENGINEER OF RECORD				STAMP	Office of Construction and Facilities Management	Drawing Title	Phase	Project Title	Project Number
 Aesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road Suite 1415 Panama City Beach, FL 32407		BASEMENT - VENTILATION - PHASE 3		100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402	
				Approved:				Building Number	
								133CA	
								Drawing Number	
 Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698	 IMEG Corp. 263 Shuman Boulevard, Suite 500 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434		FULLY SPRINKLERED		Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	M100.3		
				Issue Date	Checked	Drawn			
				11/29/2022	MRM	NVP			

- SHEET NOTES:**
1. FLEX DUCTWORK IS NOT ALLOWED IN SPS DEPARTMENT PER VA STANDARDS.
- KEYNOTES:** **C**
1. CAP DUCT, REBALANCE EXISTING SUPPLY/RETURN FAN AS REQUIRED TO MEET REVISED FLOOR PLANS.
 2. PROVIDE ACCESS PANEL FOR FIRE DAMPER. COORDINATE WITH ARCHITECT PLANS FOR CEILING DETAILS.
- BID ALTERNATES:**
- BASE BID: PROVIDE AND INSTALL ALL DUCTWORK, ASSOCIATED ACCESSORIES AND HANGER/SUPPORTS AS SHOWN.
- BID ALTERNATES: REFER TO INDIVIDUAL BID ALTERNATE SHEETS FOR DETAILS OF WORK.

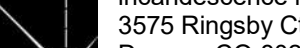


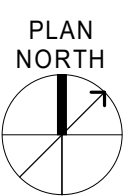
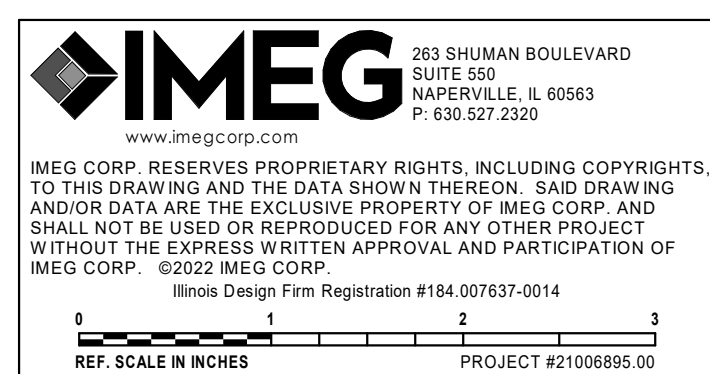
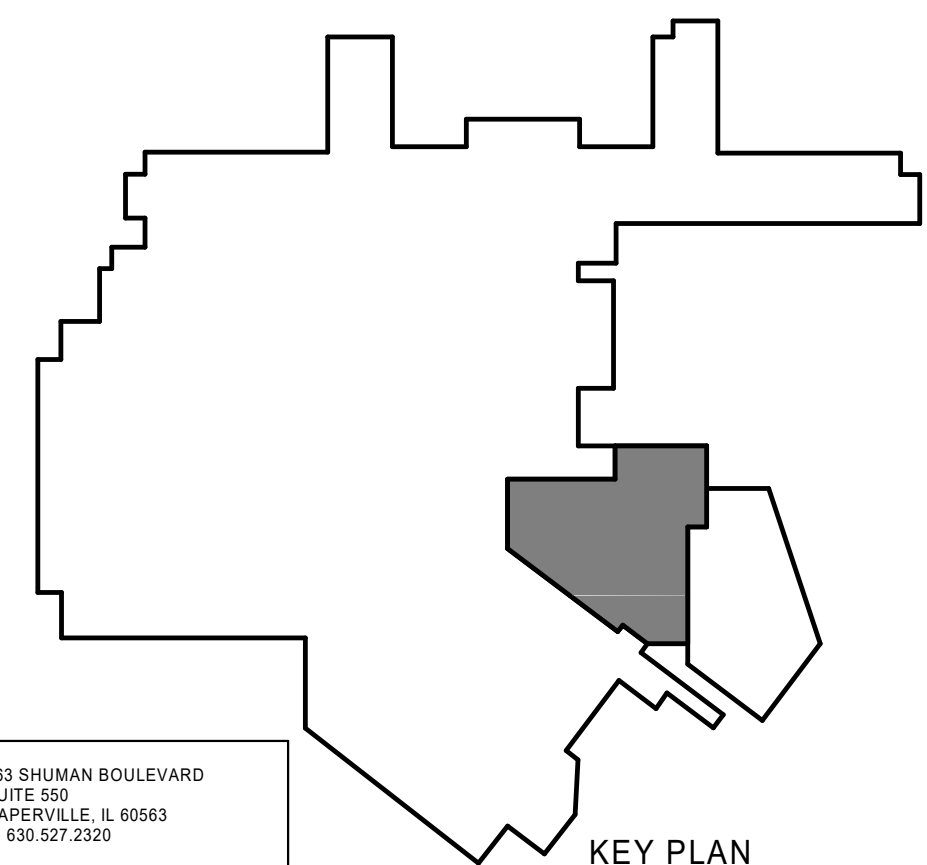
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Broad Design Firm Registration #184.007637-0014

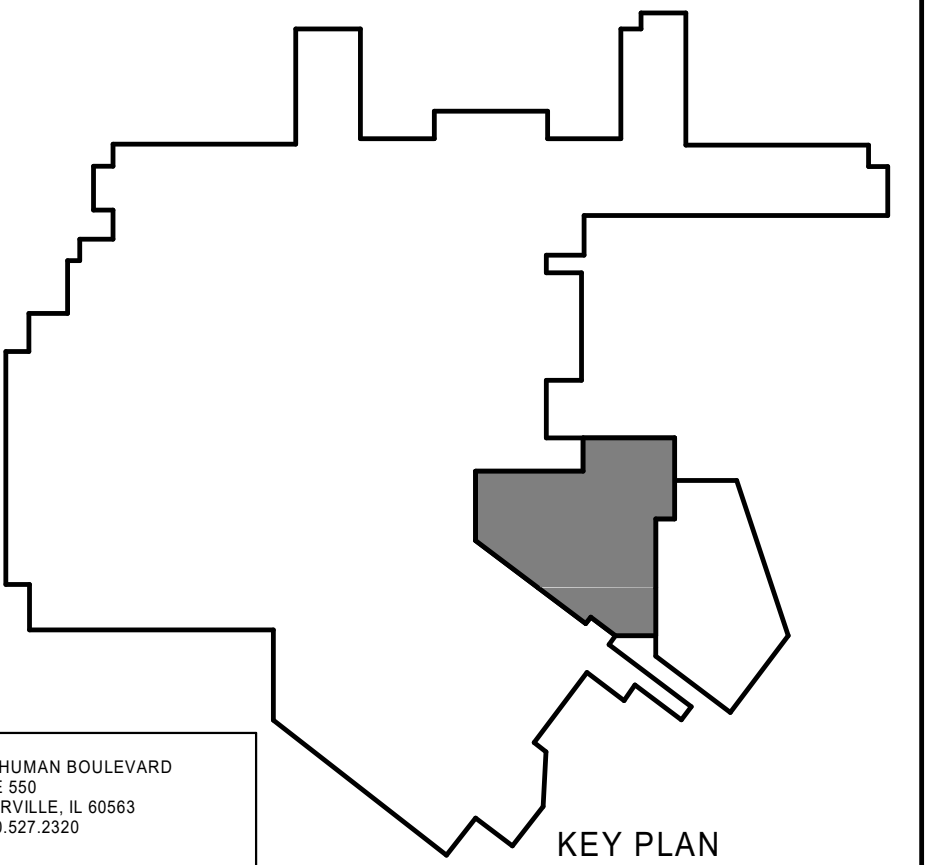
REF. SCALE IN INCHES PROJECT #21008895.00

1 BASEMENT - VENTILATION - PHASE 4
1/8" = 1'-0"

ARCHITECT/ENGINEER OF RECORD				STAMP	Office of Construction and Facilities Management	Drawing Title	Phase	Project Title	Project Number	
	Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941		Incandescence Life Safety, Inc. 3575 Ringby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010			Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	BASEMENT - VENTILATION - PHASE 4	100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402
	Shaffer Stevens & Schaap P.C. 1070 Howard St. Omaha, Nebraska 68102 Phone: (402) 455-7098		IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320			Calibre Engineering Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434	Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Building Number 133CA
									Issue Date 11/29/2022 07/05/2023 REVISIONS	Drawing Number M100.4
									Checked MRM	Drawn NVP
						U.S. Department of Veterans Affairs				

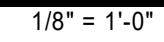

$$1/8'' = 1'-0''$$

Project Number	556-402
Building Number	133CA
Drawing Number	M100.42



0 1 2 3
REF. SCALE IN INCHES PROJECT #21006895.00

A diagram of a circle representing Earth. A vertical line segment from the top center to the top edge is labeled 'NORTH'. A horizontal line segment from the center to the right edge is labeled 'EQUATOR'. A point on the equator is marked with a dot and labeled 'A'. An arrow points from the center to point A.

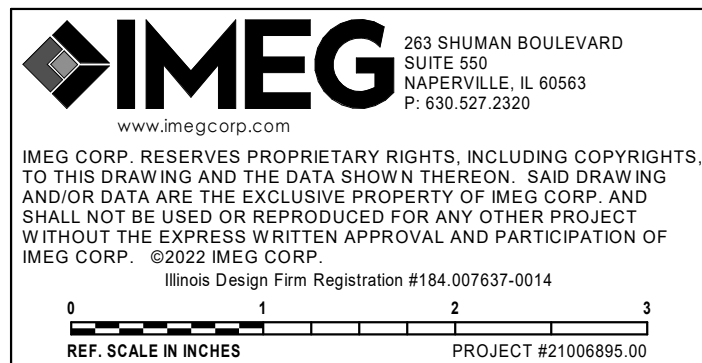
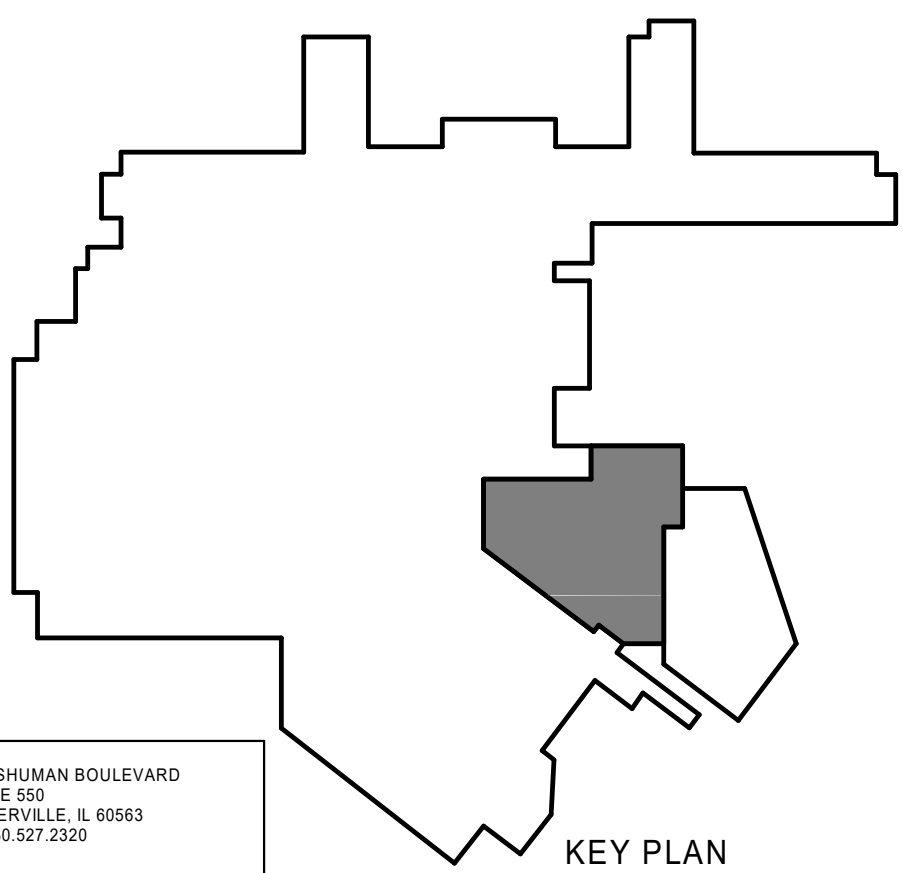
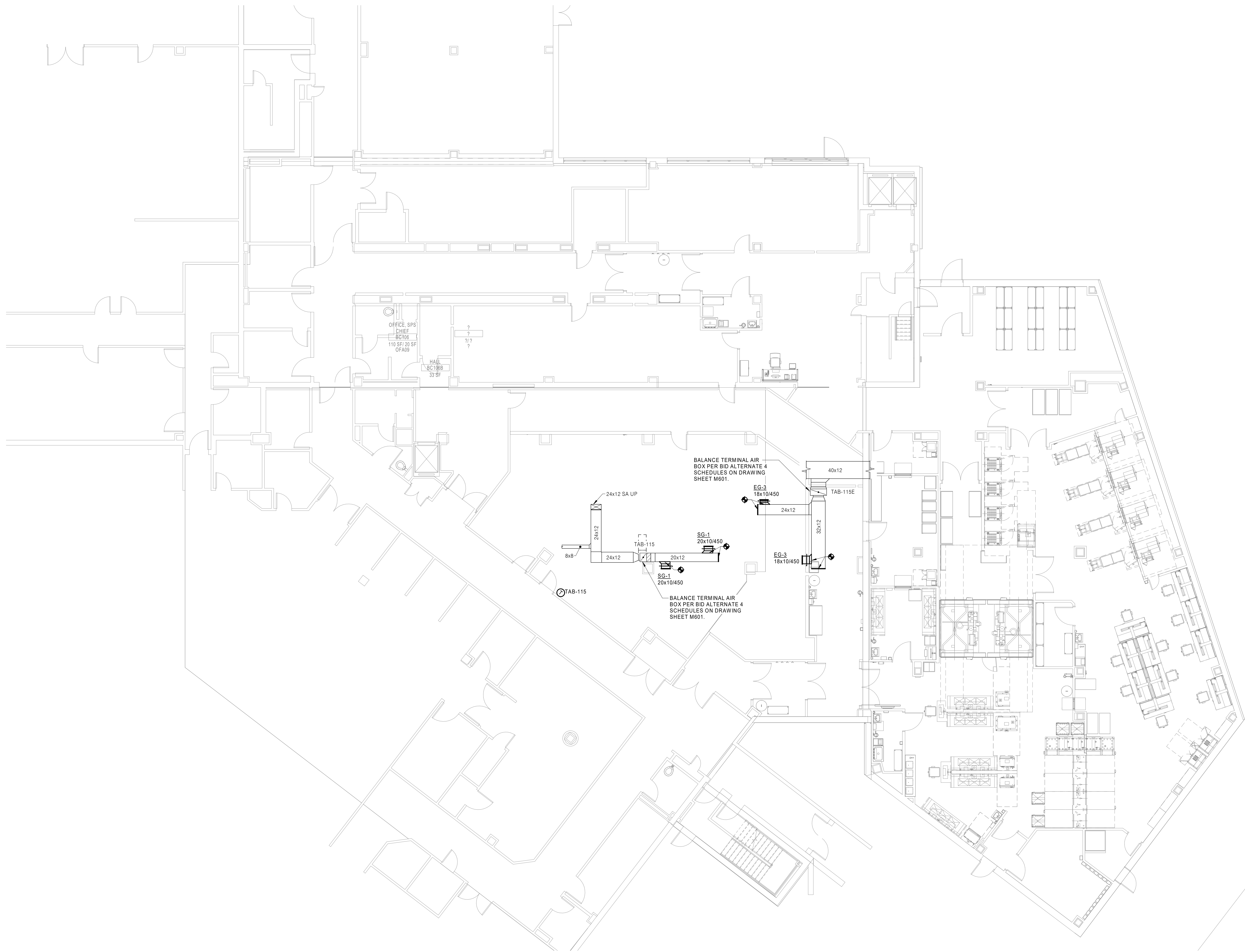


ARCHITECT/ENGINEER OF RECORD			STAMP	Office of Construction and Facilities Management	Drawing Title	Phase	Project Title	Project Number
 Alexia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence life safety, inc. 3705 North 200th Street Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407			BASEMENT – VENTILATION – PHASE 4 - BID ALTERNATE 3	100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402 Building Number 133CA
 Shaffer, Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434		VA U.S. Department of Veterans Affairs	Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3901 Green Bay Road, North Chicago, IL 60064 Issue Date 11/29/2022	Drawing Number M100.43 Checked MRM Drawn NVP

1. FLEX DUCTWORK IS NOT ALLOWED IN SPS DEPARTMENT PER VA STANDARDS.

BASE BID: PROVIDE AND INSTALL ALL DUCTWORK,
ASSOCIATED ACCESSORIES AND
HANGER/SUPPORTS AS SHOWN IN BASE BID
SHEET M100.4 .

BID ALTERNATE 4: PROVIDE AND INSTALL ALL DUCTWORK, FITTINGS, ACCESSORIES AND HANGER/SUPPORTS WITHIN BID ALTERNATE 4 SCOPE AS SHOWN. REFER TO ARCHITECT DRAWINGS FOR EXTENT OF BID ALTERNATE SCOPE AREA AND RELATED DETAILS.



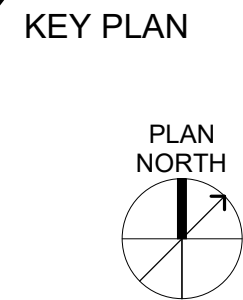
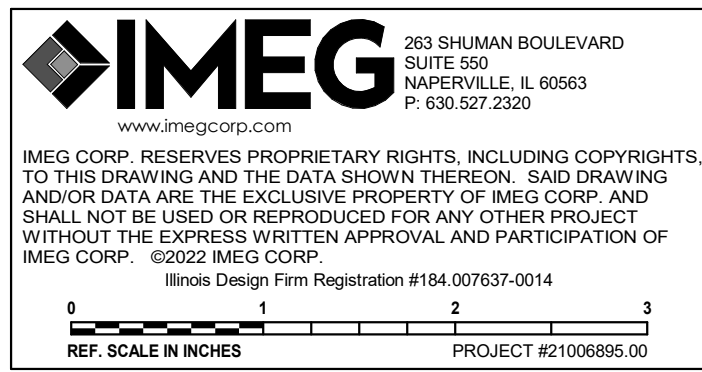
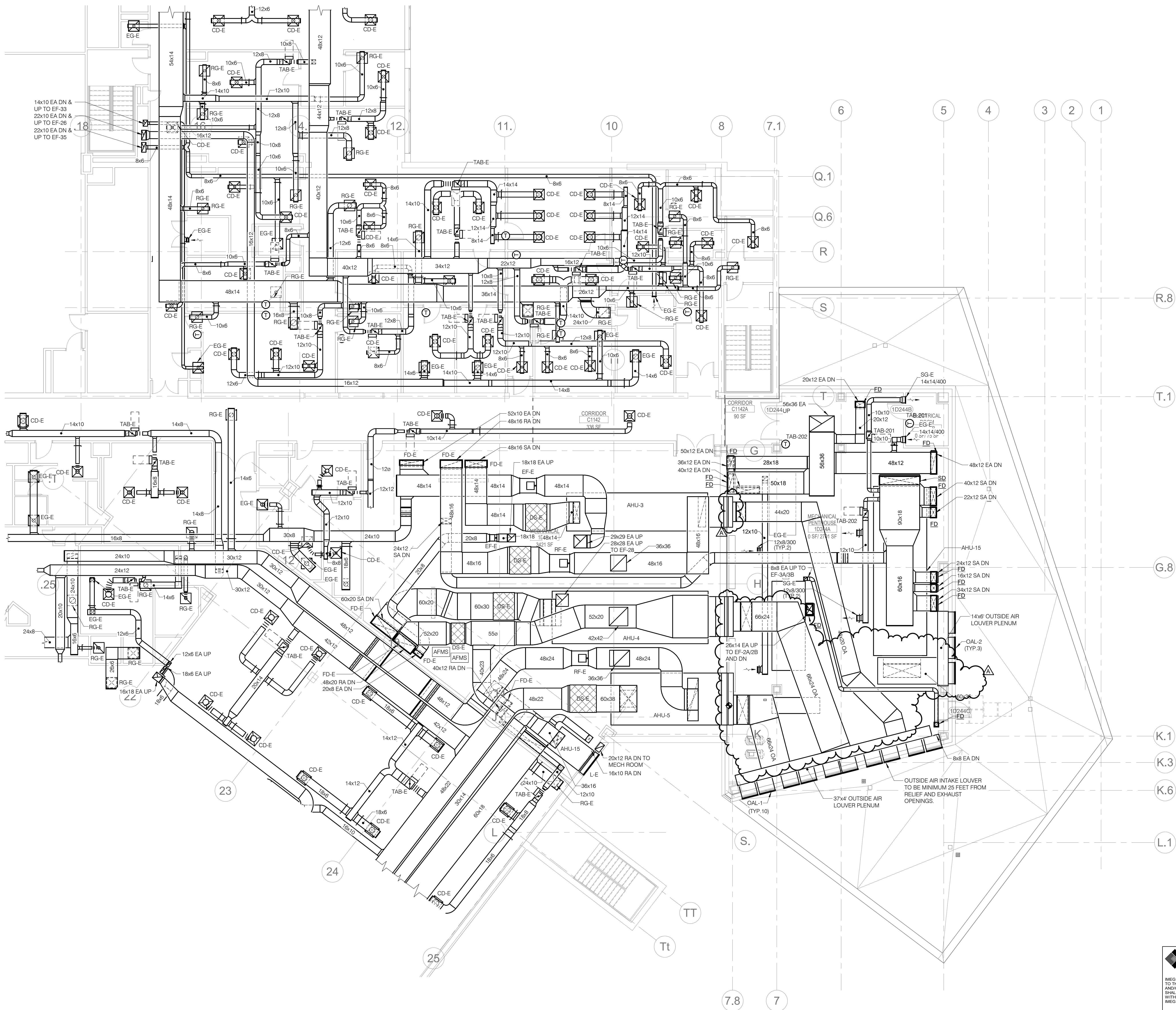
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BASEMENT – VENTILATION – PHASE 4 - BID ALTERNATE 4

$$1/8'' = 1'-0'$$
[illegible]

ARCHITECT/ENGINEER OF RECORD				STAMP	Office of Construction and Facilities Management VA U.S. Department of Veterans Affairs	Drawing Title	Phase	Project Title	Project Number		
	Alexia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941		incandescence life safety, inc. 3575 North 200th Street Denver, CO 80216 Phone: (720) 722-3010			Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407		BASEMENT – VENTILATION – PHASE 4 - BID ALTERNATE 4	100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402
						Building Number 133CA					
	Shaffer, Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698		IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320		Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434		Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Drawing Number M100.44	

SHEET NOTES:
1. THIS DRAWING REPRESENTS OVERALL AREA OF WORK. REFER TO INDIVIDUAL PHASE SHEETS FOR DETAILED INFORMATION OF SCOPE OF WORK.



FIRST FLOOR - VENTILATION - ALL PHASES

A		VA Revisions	07/05/2023
Mark		Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD			
A Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	incandescence life safety, inc. 3575 Ringway Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	Force Protect Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	IMEG IMEG Corp. 263 Shuman Boulevard, Suite 500 Naperville, IL 60563 Phone: (630) 527-2320
S Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098	IMEG IMEG Corp. 263 Shuman Boulevard, Suite 500 Naperville, IL 60563 Phone: (630) 527-2320	Calibre Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434	

STAMP
BRADY D. YOUNG 062-060058 LICENSED PROFESSIONAL ENGINEER OF ILLINOIS 11/29/2022

Office of Construction and Facilities Management
VA U.S. Department of Veterans Affairs

Drawing Title FIRST FLOOR - VENTILATION - ALL PHASES
Approved:

Phase 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED

Project Title EXPAND STERILE PROCESSING SERVICE (SPS)
Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064
Issue Date 11/29/2022 07/05/2023 REVISIONS
Checked MRM
Drawn NVP

Project Number 556-402
Building Number 133CA
Drawing Number M101.0

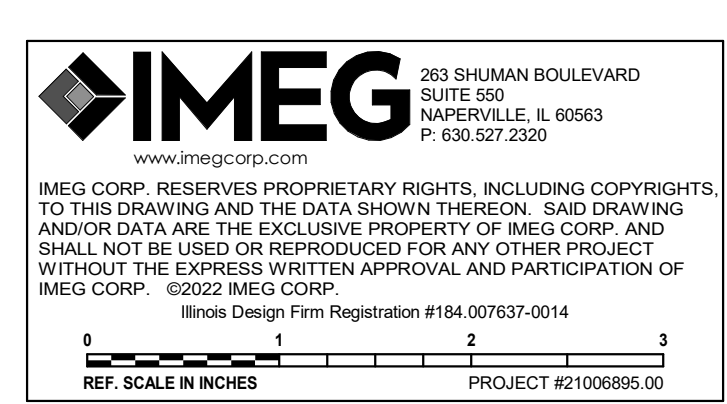
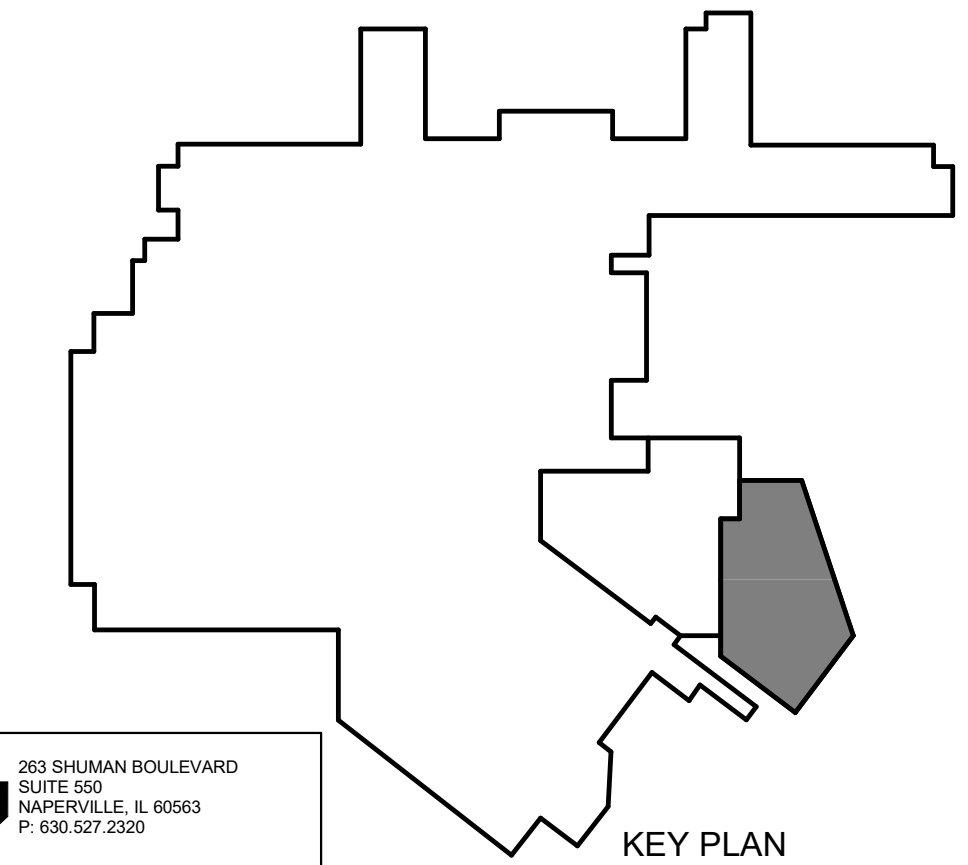
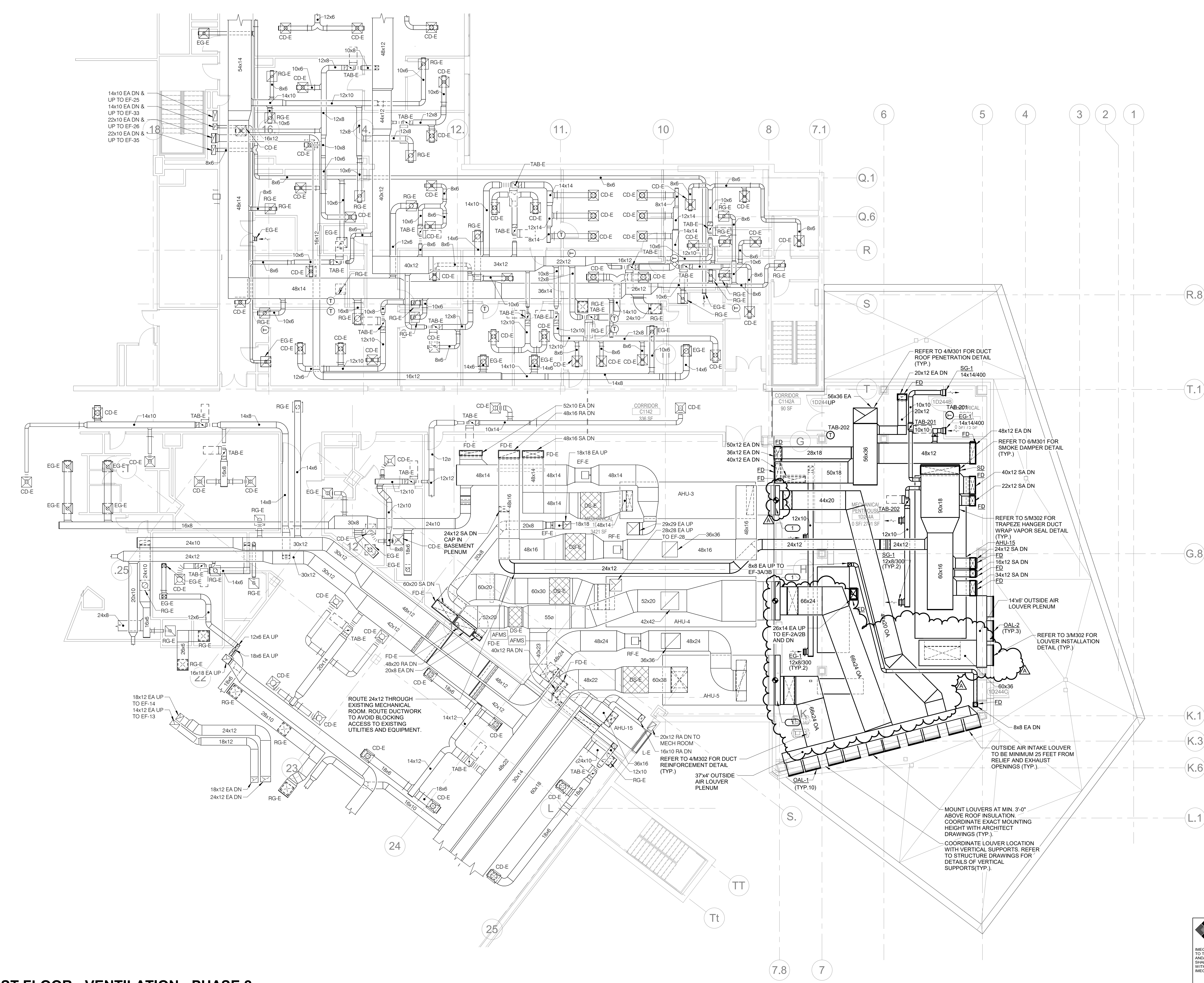
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SHEET NOTES:

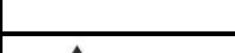
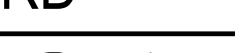
1. FLEX DUCTWORK IS NOT ALLOWED IN SPS DEPARTMENT PER VA STANDARDS.

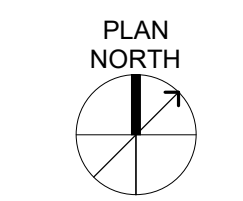
KEYNOTES:

1. EXTEND EXISTING OUTSIDE AIR PLENUM. CONNECT NEW OUTSIDE AIR DUCTS TO EXISTING OUTSIDE AIR PLENUM. CONTRACTOR TO FIELD VERIFY EXISTING PLENUM SIZES. DO NOT REMOVE EXISTING CONTROL DAMPERS AND AFMS ON OA DUCT. CONNECT DUCTWORK TO EXISTING AHU PRIOR TO PHASE 2. PROVIDE TEMPORARY DUCTWORK IN FIRST PART OF PHASE 2 TO KEEP EXISTING AHU IN SERVICE. EXTEND TEMPORARY DUCTWORK UP TO EXISTING ROOF AND PROVIDE TEMPORARY MEDIA FILTER OVER INTAKE DURING CONSTRUCTION OF PHASE 2. TEMPORARY OA DUCTWORK SIZES:
AHU-3 : 44x20
AHU-4 : 66x24
AHU-5 : 66x24

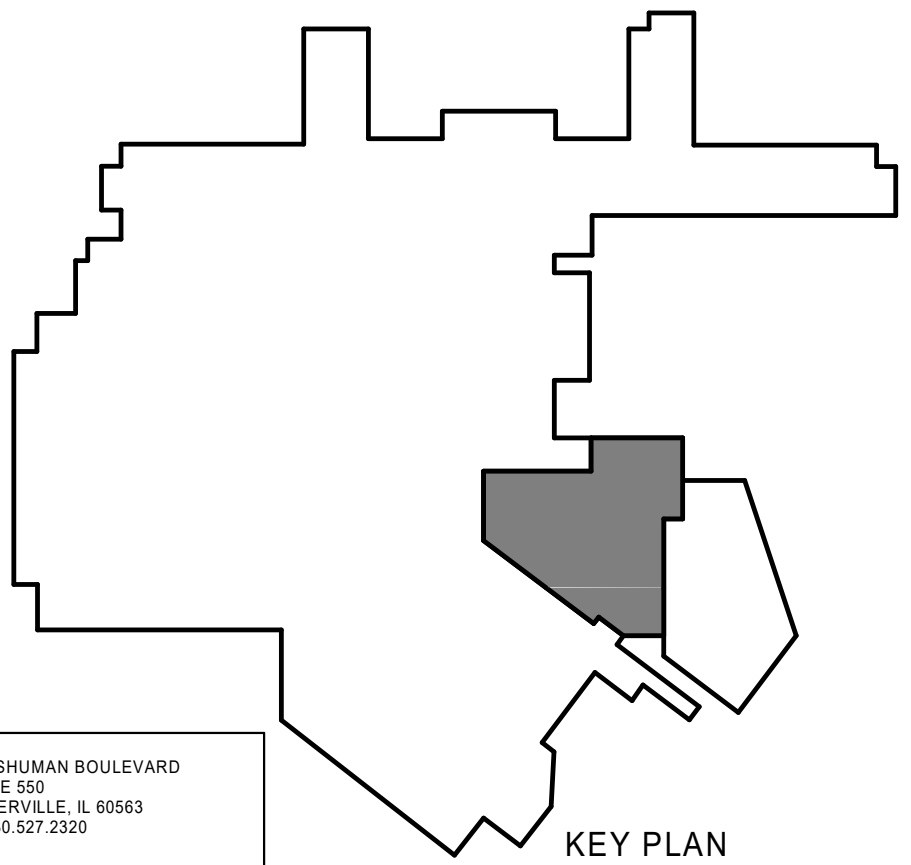
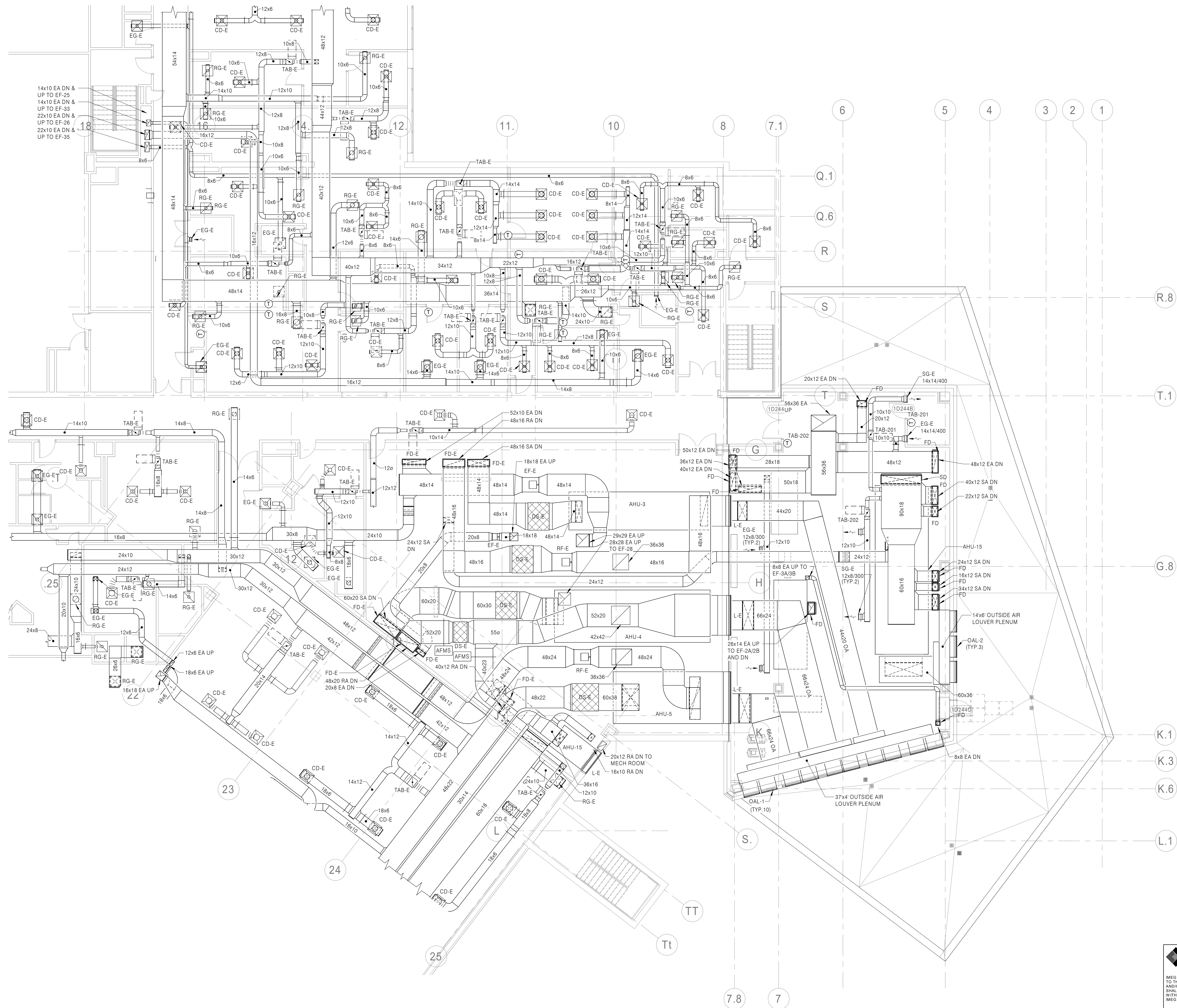


1 FIRST FLOOR - VENTILATION - PHASE 2
1/8" = 1'-0"

ARCHITECT/ENGINEER OF RECORD			STAMP	Office of Construction and Facilities Management	Drawing Title	Phase	Project Title	Project Number
 Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	2		FIRST FLOOR - VENTILATION - PHASE	100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402
			Approved:					
 Shaffer Stevens & Schapp P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434	 11/29/2022 11/30/2023	VA U.S. Department of Veterans Affairs	FULLY SPRINKLERED		Location	Drawing Number
					Issue Date	Checked	Drawn	M101.2
							Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	
							07/05/2023	
							07/05/2023 REVISIONS	



SHEET NOTES:
1. FLEX DUCTWORK IS NOT ALLOWED IN SPS
DEPARTMENT PER VA STANDARDS.



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Illinois Design Firm Registration #184.007637-0014
PROJECT #2100895.05

1 FIRST FLOOR - VENTILATION - PHASE 4
1/8" = 1'-0"

Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD			
A Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	Force Protect Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	Calibre Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434
S Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098	IMEG IMEG Corp. 263 Shuman Boulevard, Suite 500 Naperville, IL 60563 Phone: (630) 527-2320		

STAMP

BRADICK D. YOUNG
062-060058
LICENSED
PROFESSIONAL
ENGINEER
OF
ILLINOIS
11/29/2022
10/30/2023

**Office of
Construction
and Facilities
Management**

VA U.S. Department
of Veterans Affairs

Drawing Title
FIRST FLOOR - VENTILATION - PHASE 4

Approved: _____

Phase
**100% CONSTRUCTION
DOCUMENTS**

FULLY SPRINKLERED

Project Title
**EXPAND STERILE PROCESSING
SERVICE (SPS)**

Location
Captain James A. Lovell Federal Healthcare Center
3001 Green Bay Road, North Chicago, IL 60064

Issue Date
11/29/2022

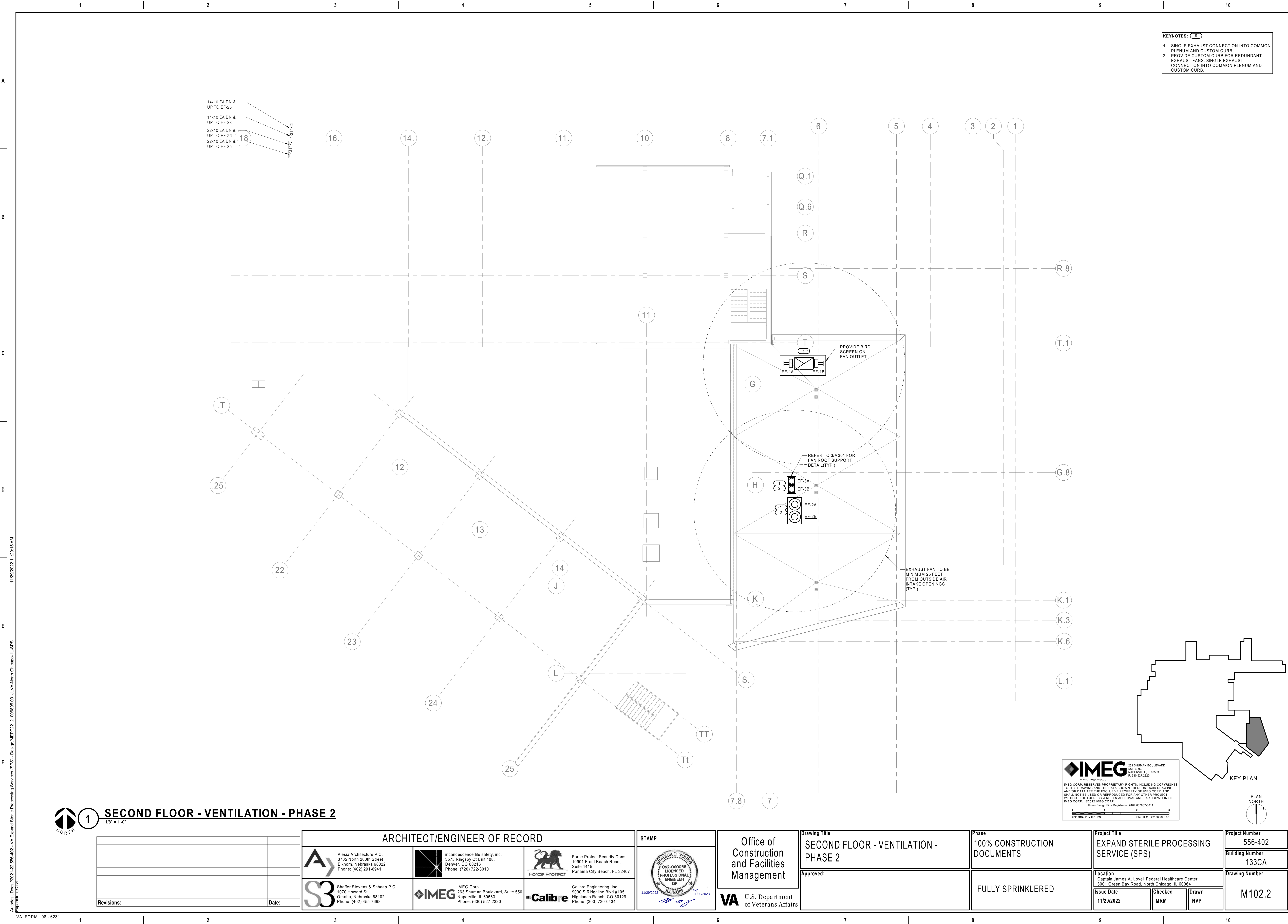
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MRM

Drawn
NVP

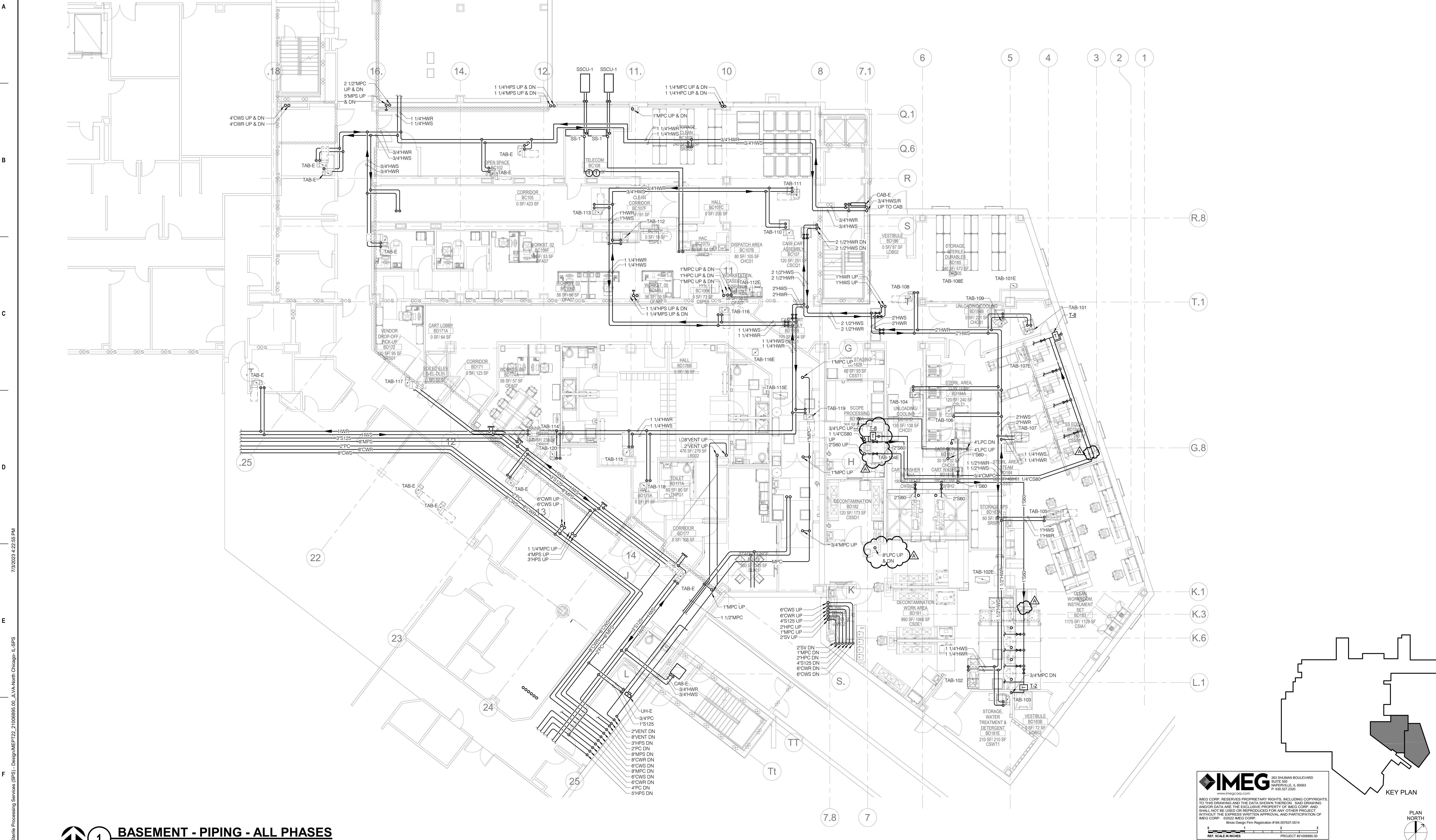
Project Number
556-402

Building Number
133CA

Drawing Number
M101.4



SHEET NOTES:
1. THIS DRAWING REPRESENTS OVERALL AREA OF WORK. REFER TO INDIVIDUAL PHASE SHEETS FOR DETAILED INFORMATION OF SCOPE OF WORK.



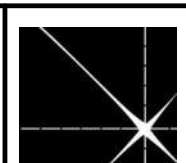
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AutoSave: Docs\2020-22-554-402 - VA Expand Sterile Processing Services (SPS) - Design\MEP\T22_2100895.00_LVA-North Chicago - LSPS
VA FORM 08 - 6231

1 BASEMENT - PIPING - ALL PHASES
1/8" = 1'-0"

ARCHITECT/ENGINEER OF RECORD



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Phone: (402) 291-6941



incandescence life safety, inc.
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Denver, CO 80216
Phone: (720) 722-3010



Force Protect Security Cons.
10901 Front Beach Road,
Suite 1415
Panama City Beach, FL 32407

STAMP



Office of
Construction
and Facilities
Management

VA U.S. Department
of Veterans Affairs

Drawing Title
BASEMENT - PIPING - ALL PHASES

Approved:

Phase
**100% CONSTRUCTION
DOCUMENTS**

FULLY SPRINKLERED

Project Title
**EXPAND STERILE PROCESSING
SERVICE (SPS)**

Location
Captain James A. Lovell Federal Healthcare Center
3001 Green Bay Road, North Chicago, IL 60064

Issue Date
11/29/2022
07/05/2023 REVISIONS

Checked
PGP

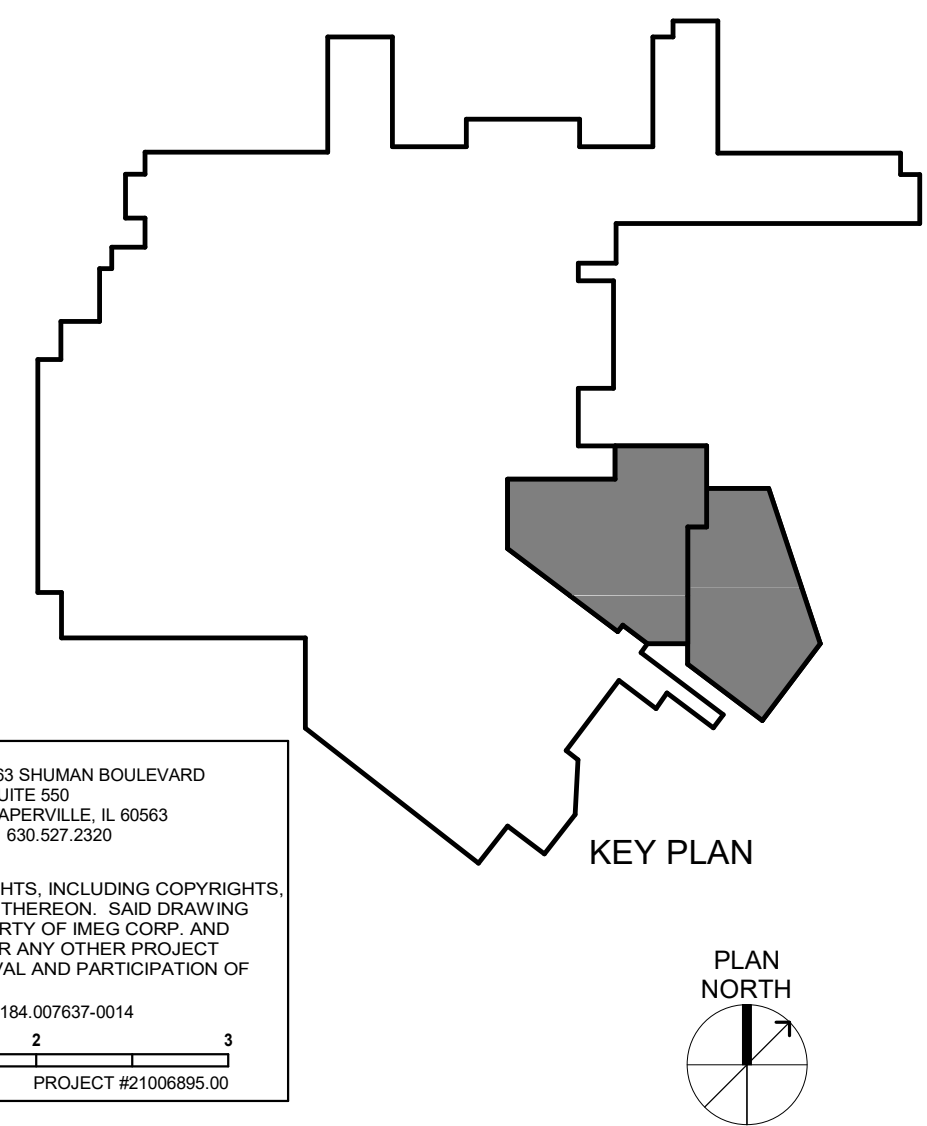
Drawn
NVP

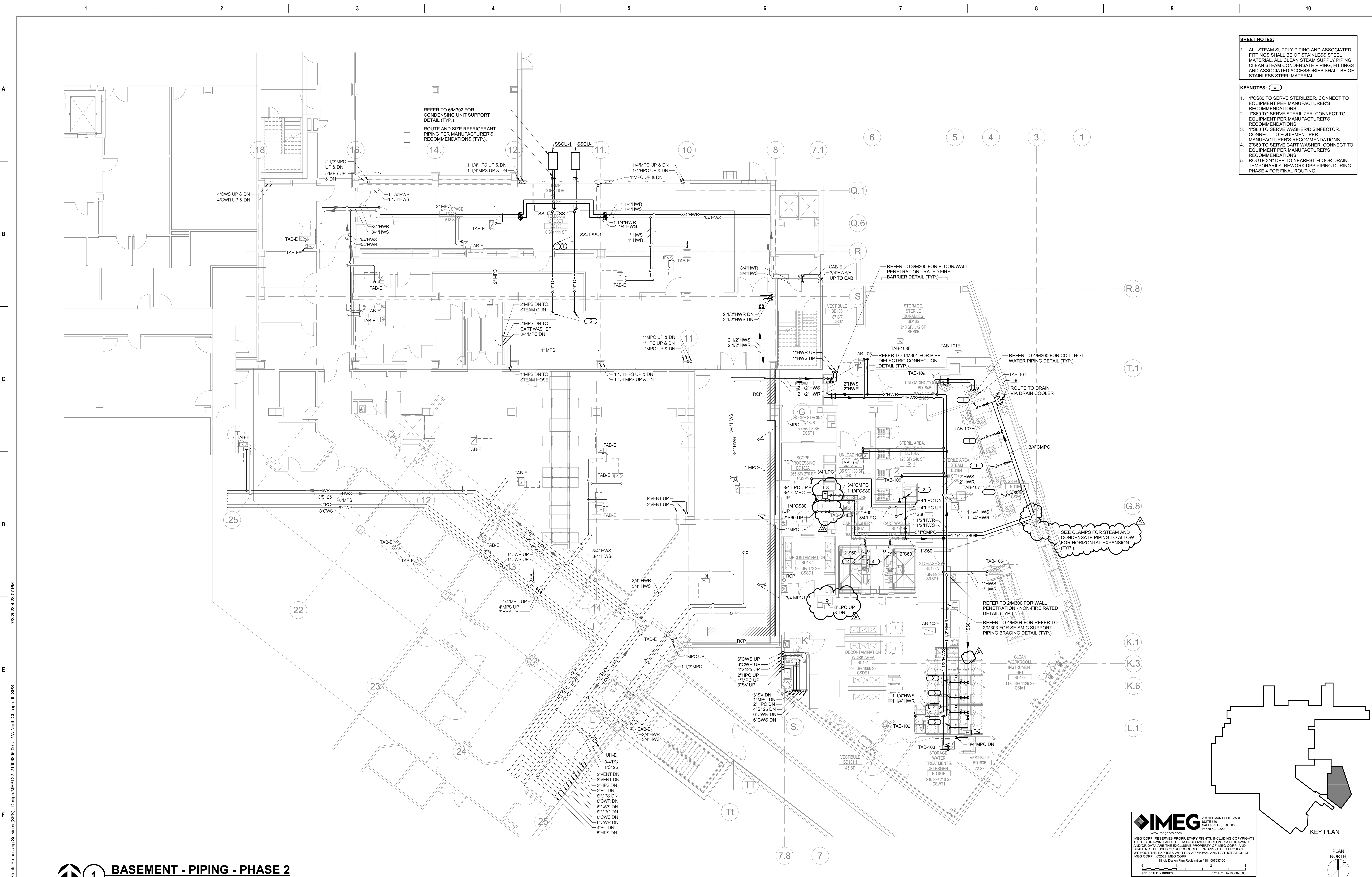
Project Number
556-402

Building Number
133CA

Drawing Number
M200.0

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Illinois Design Firm Registration #184.007637-0014
PROJECT #2100895.00
REF. SCALE IN INCHES 0 1 2 3





- SHEET NOTES:**
1. ALL STEAM SUPPLY PIPING AND ASSOCIATED FITTINGS SHALL BE OF STAINLESS STEEL. MATERIAL: ALL CLEAN STEAM SUPPLY PIPING, CLEAN STEAM CONDENSATE PIPING, FITTINGS AND ASSOCIATED ACCESSORIES SHALL BE OF STAINLESS STEEL MATERIAL.
- KEYNOTES: (E)**
1. 1"CS80 TO SERVE STERILIZER. CONNECT TO EQUIPMENT PER MANUFACTURER'S RECOMMENDATIONS.
 2. 1"SS60 TO SERVE STERILIZER. CONNECT TO EQUIPMENT PER MANUFACTURER'S RECOMMENDATIONS.
 3. 1"SS60 TO SERVE WASHER/DISINFECTOR. CONNECT TO EQUIPMENT PER MANUFACTURER'S RECOMMENDATIONS.
 4. 2"SS60 TO SERVE CART WASHER. CONNECT TO EQUIPMENT PER MANUFACTURER'S RECOMMENDATIONS.
 5. ROUTE 3/4" DPP TO NEAREST FLOOR DRAIN TEMPORARILY. REWORK DPP PIPING DURING PHASE 4 FOR FINAL ROUTING.

1

1/8" = 1'-0"

NORTH

ARCHITECT/ENGINEER OF RECORD

A

Alesia Architecture P.C.
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Phone: (720) 722-3010

S

Shaffer Stevens & Schapp P.C.
1070 Howard St
Omaha, Nebraska 68102
Phone: (402) 291-6941

IMEG

IMEG Corp.
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Naperville, IL 60563
Phone: (630) 527-2320

Calibre

Calibre Engineering, Inc.
9090 S Ridgeline Blvd #105,
Highlands Ranch, CO 80129
Phone: (303) 730-0434

STAMP

BRADY D. YOUNG

062-060058

PROFESSIONAL ENGINEER

OF

ILLINOIS

11/29/2022

03/03/2023

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

Drawing Title

BASEMENT - PIPING - PHASE 2

Approved:

Phase

100% CONSTRUCTION DOCUMENTS

Project Title

EXPAND STERILE PROCESSING SERVICE (SPS)

Location

Captain James A. Lovell Federal Healthcare Center
3001 Green Bay Road, North Chicago, IL 60064

Issue Date

11/29/2022

Checked

PGP

Drawn

NVP

Project Number

556-402

Building Number

133CA

Drawing Number

M200.2

Revisions:

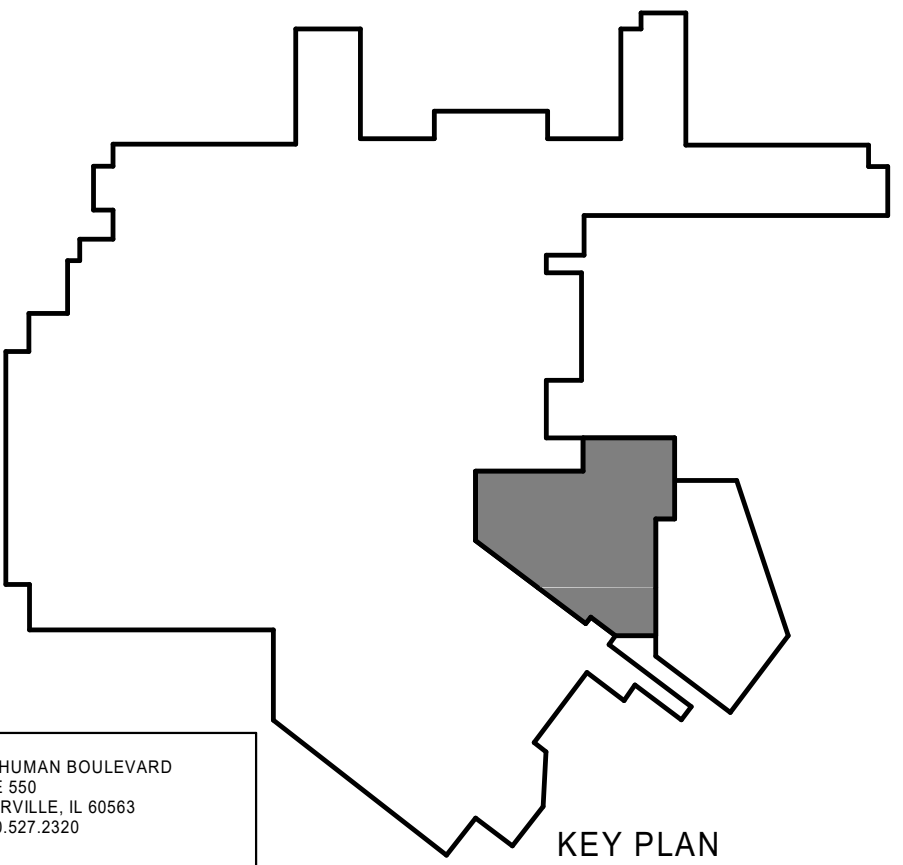
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VA Revisions

07/05/2023

Date:

VA FORM 08 - 6231 1 2 3 4 5 6 7 8 9 10



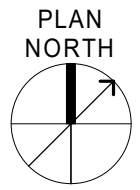
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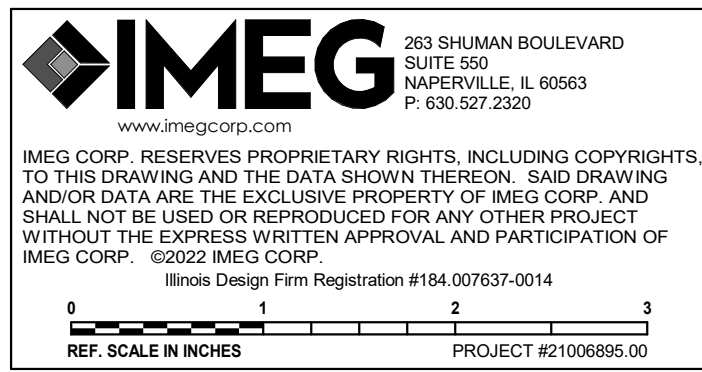
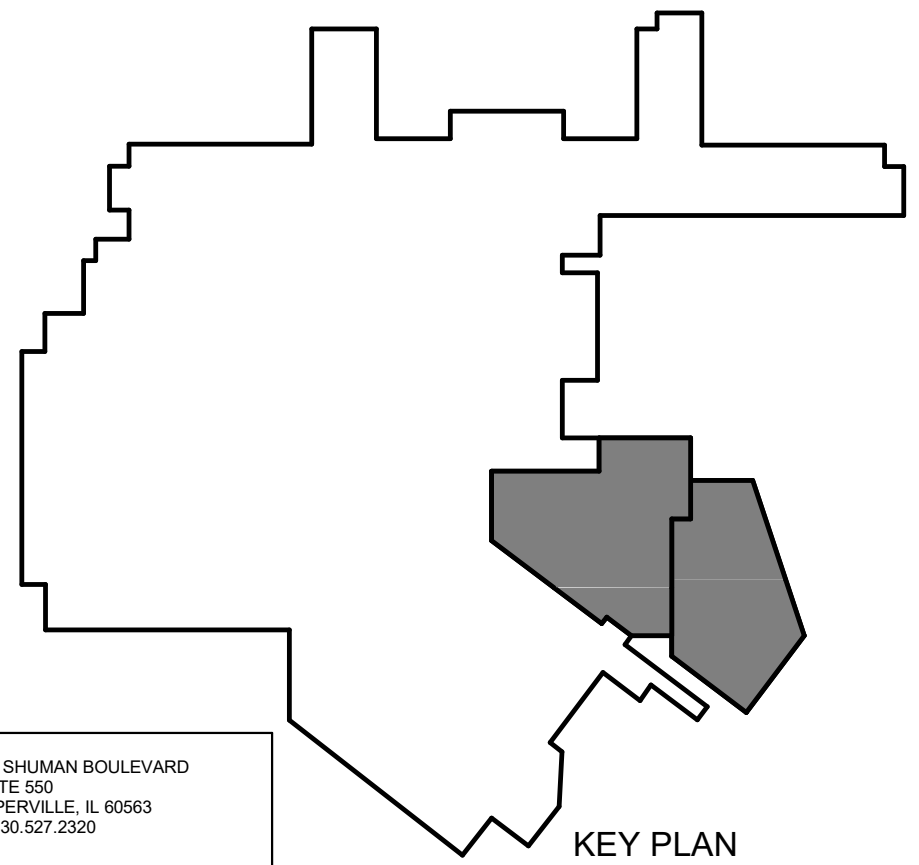
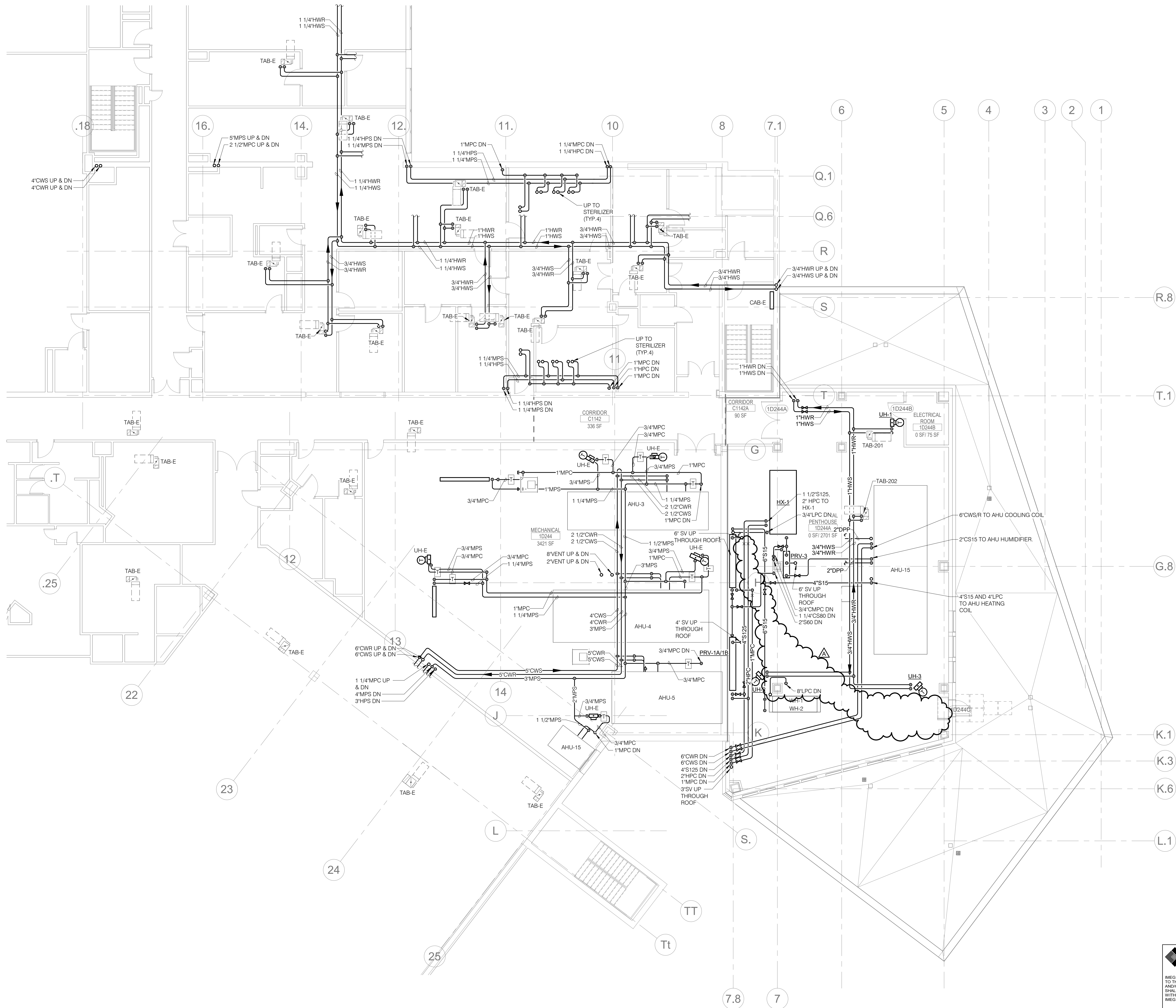
Illinois Design Firm Registration #A04 007637-0014

REF. SCALE IN INCHES PROJECT #2100089500

[illegible]

ARCHITECT/ENGINEER OF RECORD			STAMP	Office of Construction and Facilities Management VA U.S. Department of Veterans Affairs	Drawing Title BASEMENT - PIPING - PHASE 4	Phase 100% CONSTRUCTION DOCUMENTS	Project Title EXPAND STERILE PROCESSING SERVICE (SPS)	Project Number 556-402
 Alexia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 incandescence fire safety, inc. 3575 Ringway Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407			Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Drawing Number M200.4
 Shaffer, Stovens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698	 IMEG Corp. 1063 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 6090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434		Approved:	FULLY SPRINKLERED	Issue Date 11/29/2022	Checked PGP	Drawn NVP

SHEET NOTES:
1. THIS DRAWING REPRESENTS OVERALL AREA OF WORK. REFER TO INDIVIDUAL PHASE SHEETS FOR DETAILED INFORMATION OF SCOPE OF WORK.



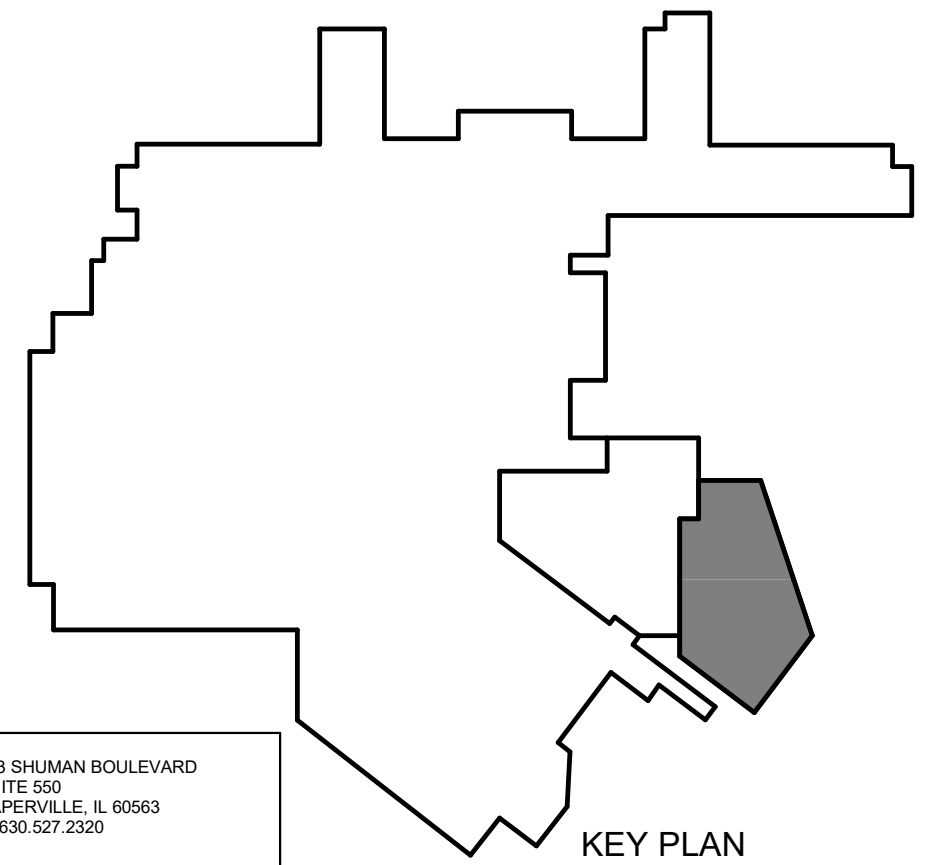
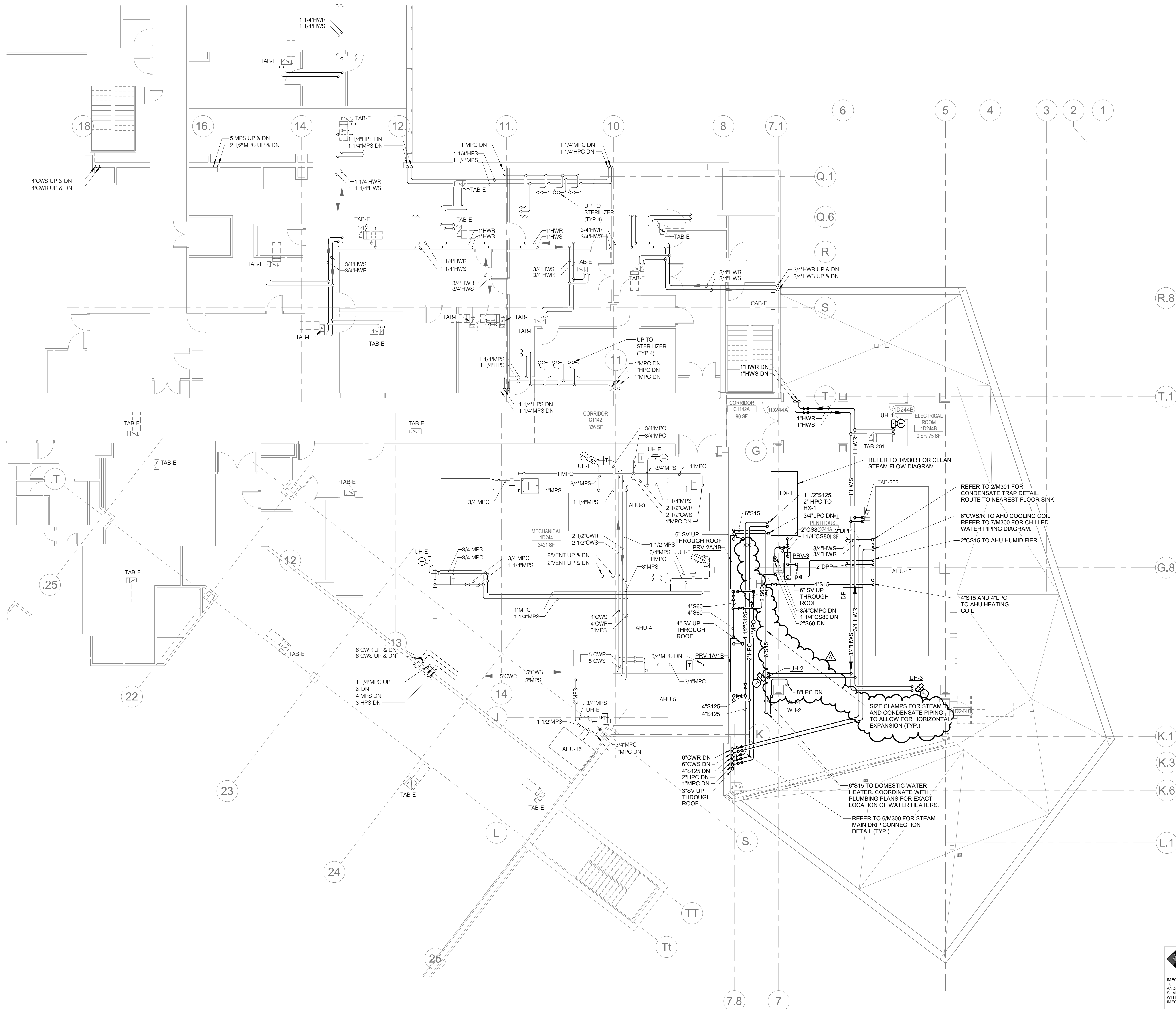
1 FIRST FLOOR - PIPING - ALL PHASES
1/8" = 1'-0"

A		VA Revisions	07/05/2023
Mark	Revisions:	Date:	

ARCHITECT/ENGINEER OF RECORD				STAMP	Office of Construction and Facilities Management VA U.S. Department of Veterans Affairs	Drawing Title	Phase	Project Title	Project Number
 Alesia Architecture P.C. 3575 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407		FIRST FLOOR - PIPING - ALL PHASES		100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402	
				Approved:		FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Building Number 133CA	
							Issue Date 11/29/2022 07/05/2023 REVISIONS	Drawing Number M201.0	
							Checked PGP	Drawn NVP	

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VA FORM 08 - 6231

SHEET NOTES:
1. ALL STEAM SUPPLY PIPING AND ASSOCIATED FITTINGS SHALL BE OF STAINLESS STEEL.
MATERIAL ALL CLEAN STEAM SUPPLY PIPING, CLEAN STEAM CONDENSATE PIPING, FITTINGS AND ASSOCIATED ACCESSORIES SHALL BE OF STAINLESS STEEL MATERIAL.



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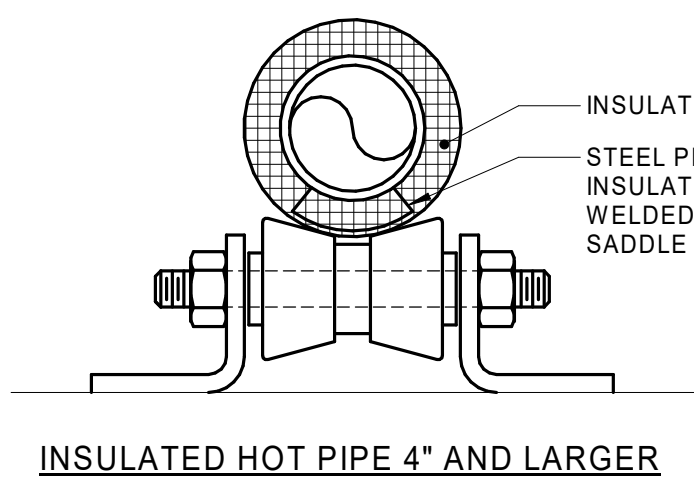
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Illinois Design Firm Registration #184.007637-0014
PROJECT #21008955.00

1 FIRST FLOOR - PIPING - PHASE 2
1/8" = 1'-0"

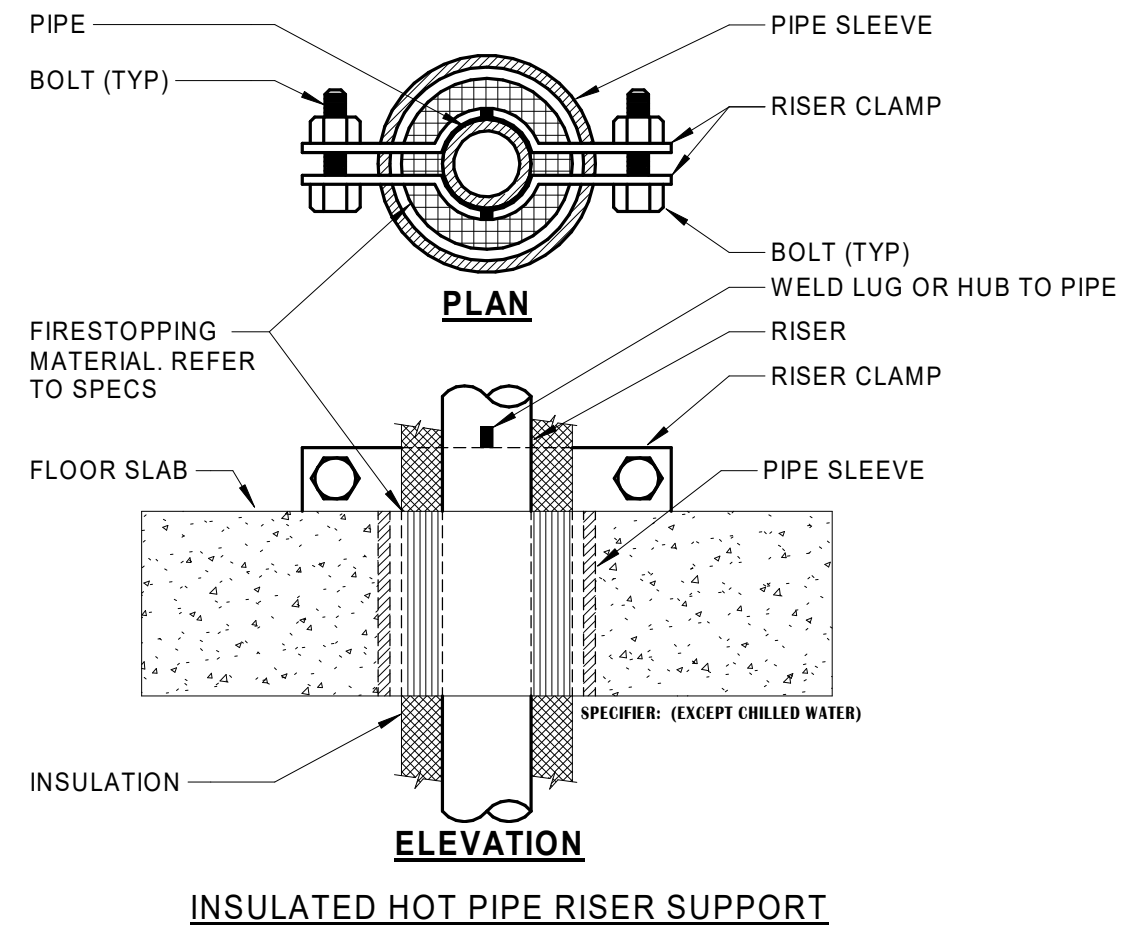
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Mark	Revisions:		Date:

ARCHITECT/ENGINEER OF RECORD			STAMP	Office of Construction and Facilities Management	Drawing Title	Phase	Project Title	Project Number
 Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence Life Safety, Inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	 11/29/2022 11/30/2023		FIRST FLOOR - PIPING - PHASE 2	100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402
 Shaffer Stevens & Schaap P.C. 1070 Howard St. Omaha, Nebraska 68102 Phone: (402) 455-7098	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434	 11/29/2022 11/30/2023		Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Drawing Number
 U.S. Department of Veterans Affairs					Issue Date 11/29/2022	Checked PGP	Drawn NVP	M201.2
					07/05/2023 REVISIONS			

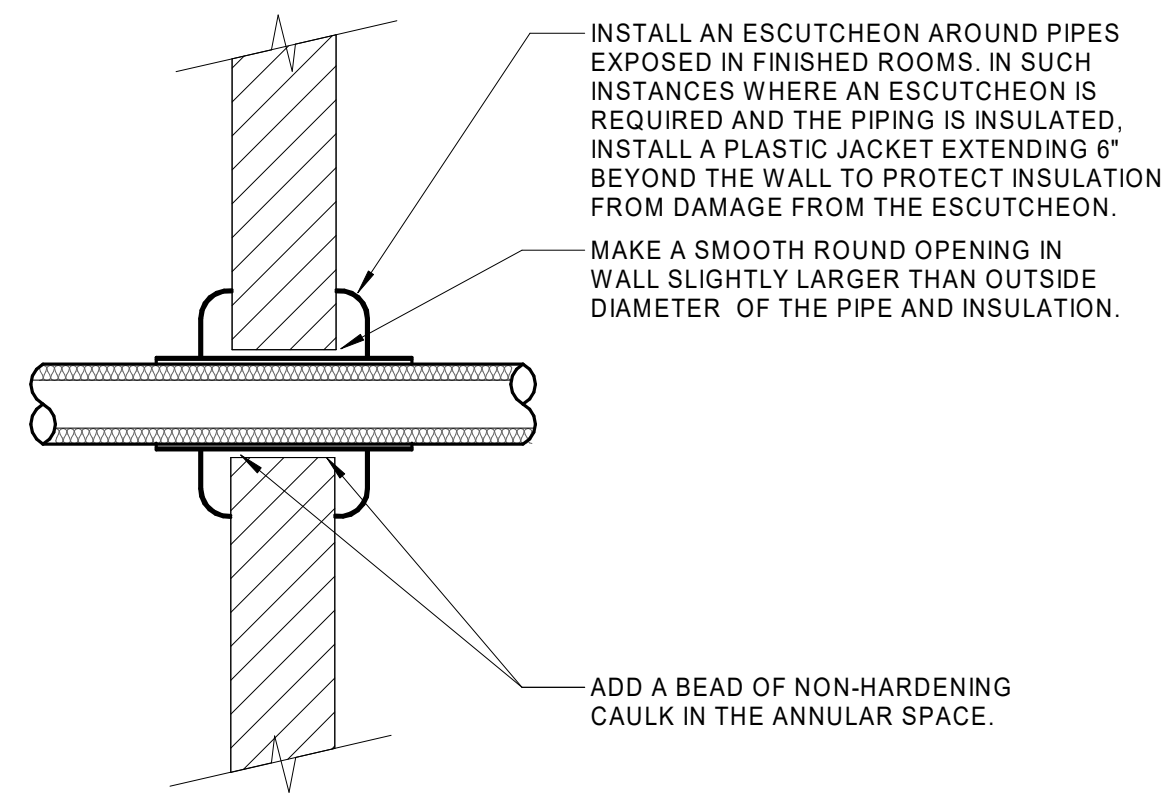
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VA FORM 08 - 6231



INSULATED HOT PIPE 4" AND LARGER

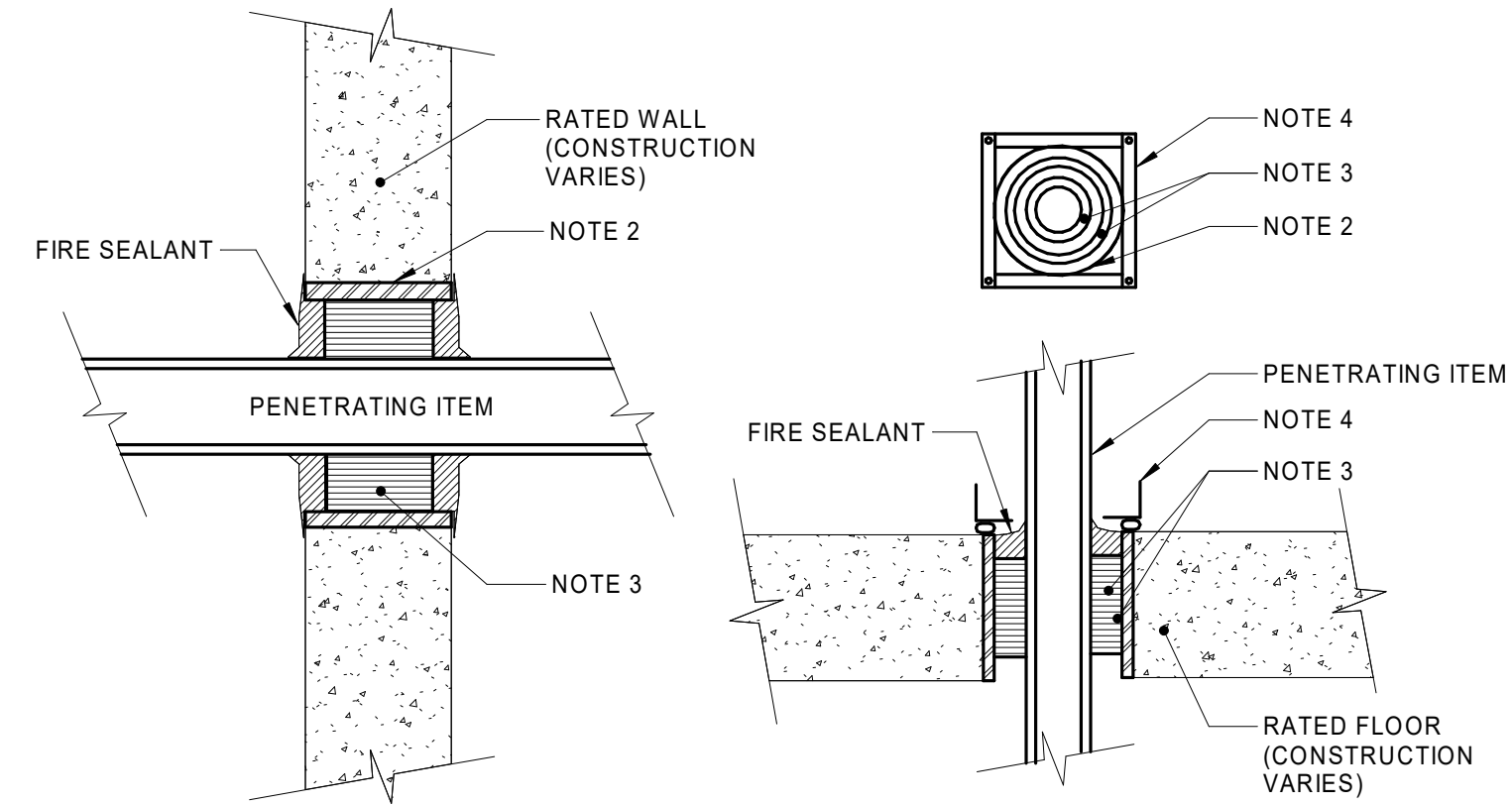


INSULATED HOT PIPE RISER SUPPORT



2 WALL PENETRATION - NON-FIRE RATED

- NOTES:
- THIS DETAIL APPLIES TO ALL PIPES. THE INTENTION IS TO CONTINUE THE INSULATION AND VAPOR BARRIER THROUGH ALL PENETRATIONS. PERMIT THERMAL EXPANSION WITHOUT DAMAGING INSULATION, AND TO SEAL AIRTIGHT AROUND INSULATED AND UNINSULATED PIPES FOR NOISE TRANSMISSION CONTROL.
 - FLOOR OPENINGS ARE SIMILAR SEE SPECIFICATIONS FOR DIFFERENCES BETWEEN FLOOR AND WALL PENETRATIONS.
 - SEE SPECIFICATIONS FOR ADDITIONAL INFORMATION.

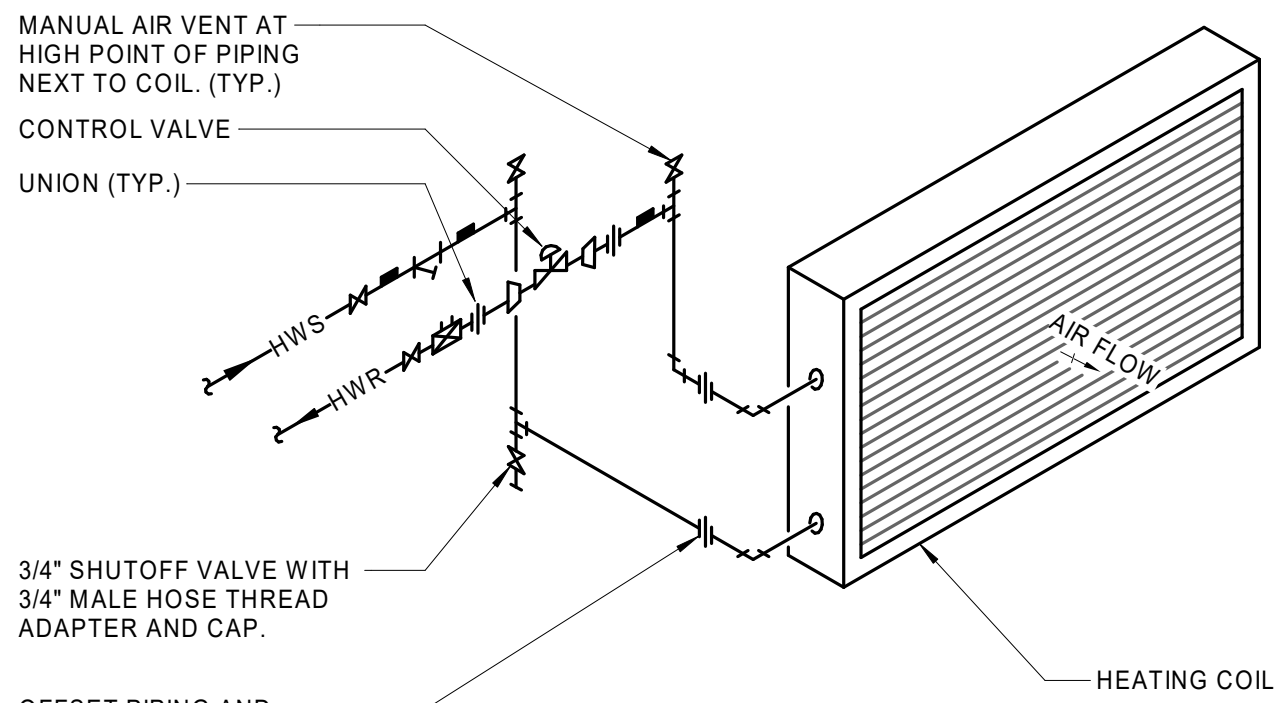


3 FLOOR/WALL PENETRATION - RATED FIRE BARRIER

- NOTES:
- THIS GENERAL DETAIL APPLIES TO ALL ITEMS PENETRATING FIRE RATED WALLS OR FLOORS. THE INTENT IS TO MAINTAIN THE FIRE RATING AND TO ALLOW LONGITUDINAL MOVEMENT. REFER TO SPECIFICATIONS FOR SELECTION OF THROUGH PENETRATION FIRE STOPPING.
 - SCHEDULE 80 PIPE SLEEVE EMBEDDED IN WALL OR FLOOR, OR SMOOTH CORE DRILL. EACH CONTRACTOR FURNISHES SLEEVE TO G.C. COORDINATES SLEEVE LOCATIONS AND DEBURS SLEEVE. G.C. BUILDS SLEEVE INTO WALL OR FLOOR ALLOWING NO GAP AROUND SLEEVE. IF SLEEVE IS NOT PROVIDED WHEN WALL OR FLOOR IS BUILT, CONTRACTOR SHALL INSTALL SLEEVE. SLEEVE SIZE SHALL ALLOW ANNUAL SPACE REQUIRED BY THE SELECTED FIRE STOP SYSTEM.
 - INSTALL BACKING MATERIAL, SUCH AS MINERAL WOOL SAFING, AS REQUIRED FOR FIRE STOP SYSTEM. INSTALL IN ACCORDANCE WITH FIRE STOP SYSTEM APPLICATION LISTING. SECURE TO WALL OR FLOOR TO ALLOW LONGITUDINAL MOVEMENT OF PENETRATING ITEM WITHOUT MOVEMENT OF FIRE BARRIER.
 - WATERTIGHT WELDED 1"x1" 20 GAUGE MINIMUM GALVANIZED SHEET METAL ANGLE FRAME. BY CONTRACTOR IN EQUIPMENT ROOMS FOR WATER STOP. PLACE A BEAD OF WATERPROOF SEALANT BETWEEN FLOOR AND BOTTOM OF ANGLE FRAME. SECURE TO FLOOR WITH MASONRY ANCHORS IN CORNERS AND ON 12" MAXIMUM CENTERS. MULTIPLE PENETRATING ITEMS MAY BE ENCLOSED IN ONE FRAME.

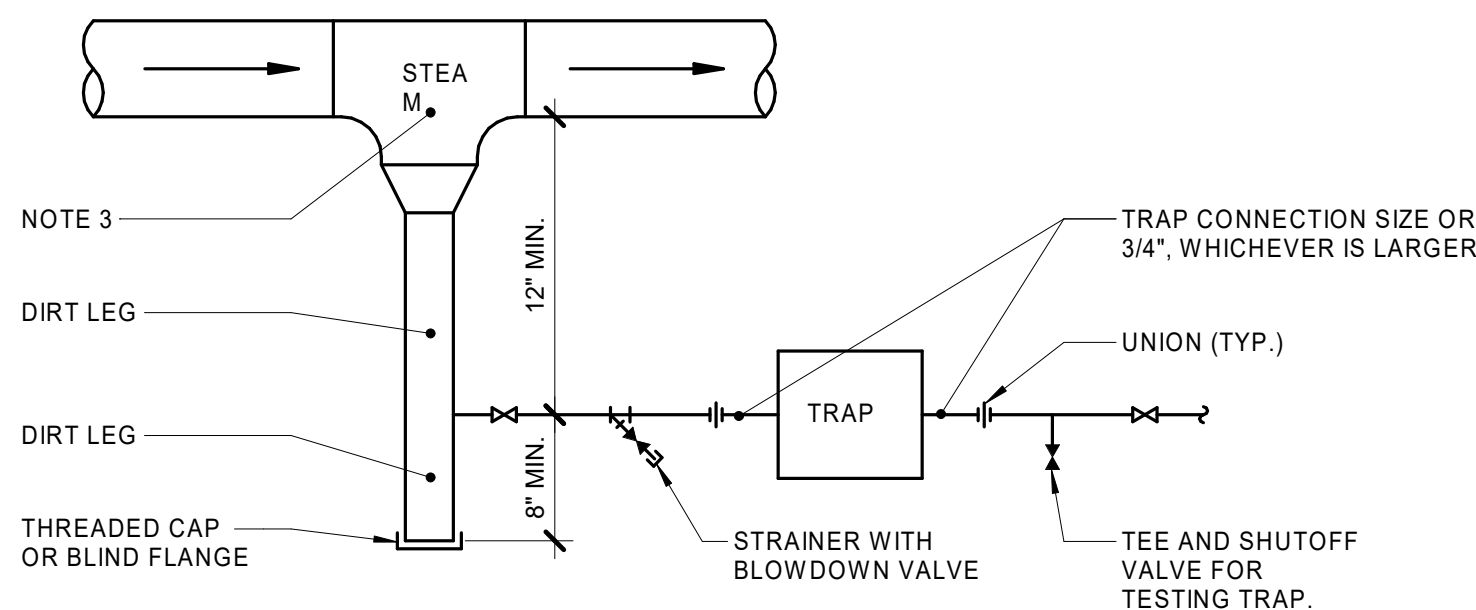
1 PIPE - HANGERS AND SUPPORTS

- NO SCALE
- NOTES:
- REFER TO SPECIFICATION SECTION 23 05 10 AND 23 07 11.



4 COIL - HOT WATER PIPING

- NO SCALE
- NOTES:
- SEE SPECIFICATION SECTION 23 21 00 - HYDRONIC PIPING FOR BALANCE VALVE SIZING REQUIREMENTS.

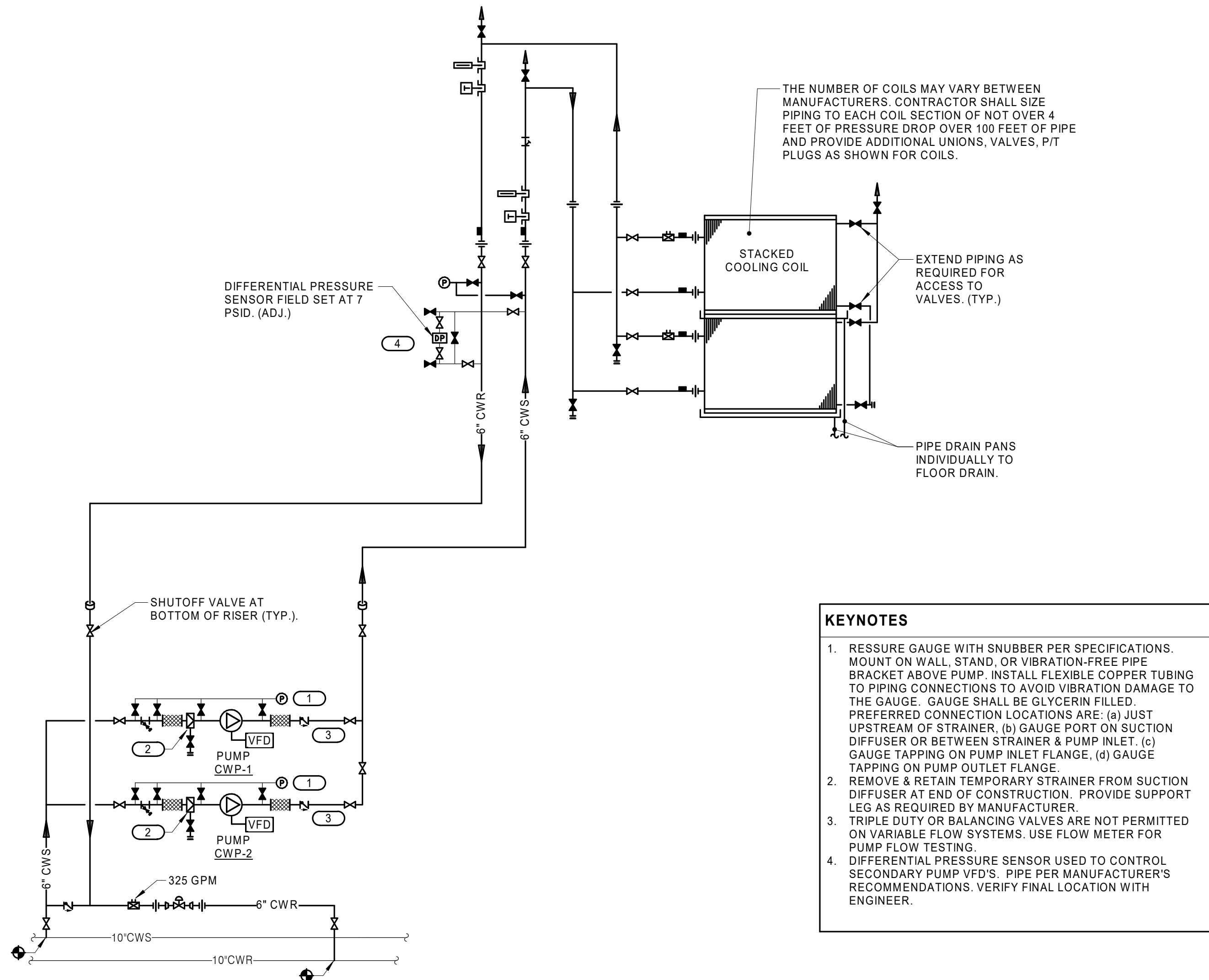
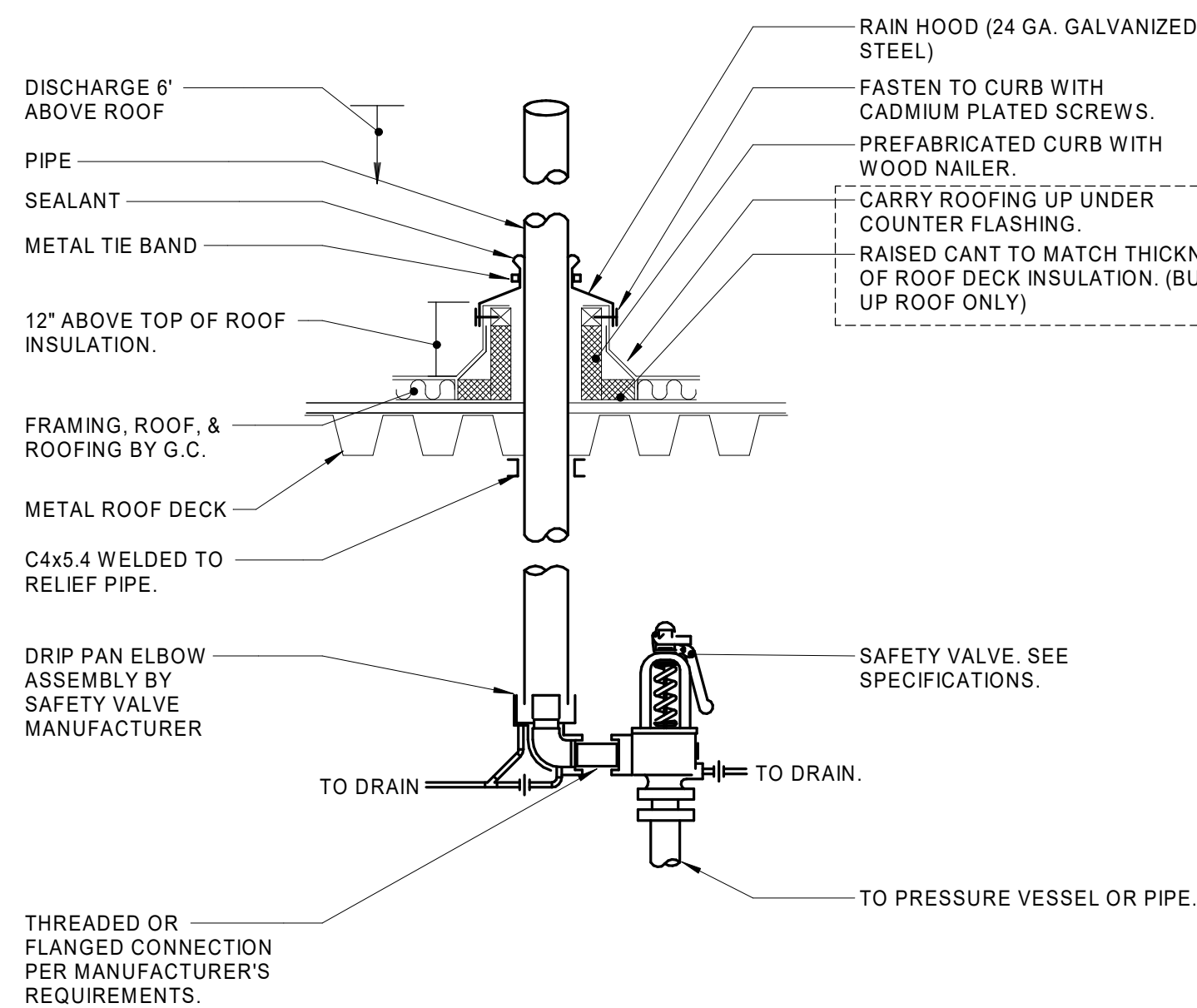


6 STEAM MAIN DRIP CONNECTION

- NOTES:
- DRIP AND DIRT LEGS SHALL BE AT LEAST TWICE THE DIAMETER OF THE TRAP INLET.
 - INSTALL LEGS OF STRAINERS IN HORIZONTAL POSITION TO MINIMIZE CONDENSATE HOLDING.
 - TEE SHALL BE FILL SIZE FOR 4" AND SMALLER MAINS. 4" FOR 5" AND 6" MAINS AND 1/2 OF MAIN DIAMETER FOR LARGER MAINS.
 - LOCATE DRIP TRAPS AT 300 FOOT MAXIMUM INTERVALS AND UPSTREAM OF ALL EXPANSION DEVICES, BRANCH CONNECTIONS OR CONTROL VALVES.

5 SAFETY VALVE DISCHARGE PIPING

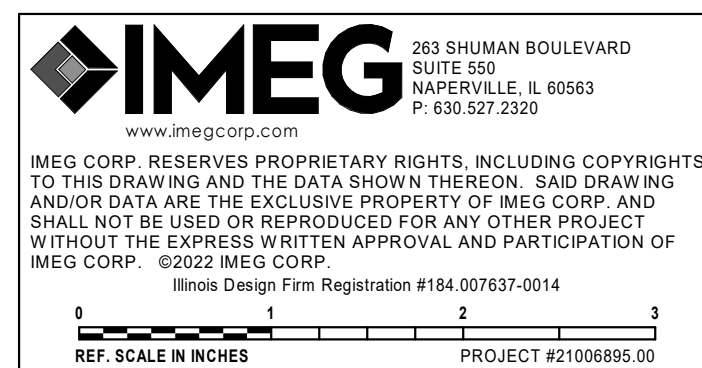
NO SCALE



7 CHILLED WATER PIPING

NO SCALE

- KEYNOTES
- RESSURE GAUGE WITH SNUBBER PER SPECIFICATIONS. MOUNT ON WALL, STAND, OR VIBRATION-FREE PIPE BRACKET ABOVE PUMP. INSTALL FLEXIBLE COPPER TUBING TO PIPING CONNECTIONS TO AVOID VIBRATION DAMAGE TO THE GAUGE. GAUGE SHALL BE GLYCERIN FILLED. PREFERRED CONNECTION LOCATIONS ARE: (a) JUST UPSTREAM OF STRAINER, (b) GAUGE PORT ON SUCTION DIFFUSER OR BETWEEN STRAINER & PUMP INLET, (c) GAUGE TAPPING ON PUMP INLET FLANGE, (d) GAUGE TAPPING ON PUMP OUTLET FLANGE.
 - REMOVE & RETAIN TEMPORARY STRAINER FROM SUCTION DIFFUSER AT END OF CONSTRUCTION. PROVIDE SUPPORT LEG AS REQUIRED BY MANUFACTURER.
 - TRIPLE DUTY OR BALANCING VALVES ARE NOT PERMITTED ON VARIABLE FLOW SYSTEMS. USE FLOW METER FOR PUMP FLOW TESTING.
 - DIFFERENTIAL PRESSURE SENSOR USED TO CONTROL SECONDARY PUMP VFD'S. PIPE PER MANUFACTURER'S RECOMMENDATIONS. VERIFY FINAL LOCATION WITH ENGINEER.



ARCHITECT/ENGINEER OF RECORD

A Alesia Architecture P.C. 2705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	incandescence life safety, inc. 3575 Ringway Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	Force Protect Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407
S Shaffer Stevens & Schapp P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098	IMEG IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	Calibre Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434

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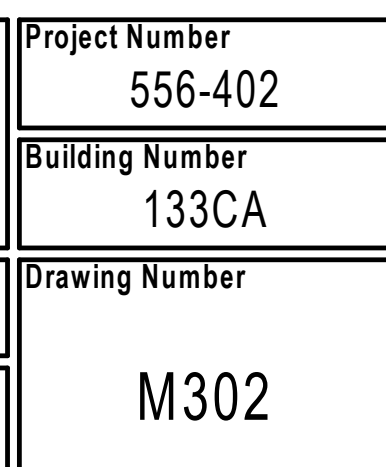
Office of Construction and Facilities Management VA U.S. Department of Veterans Affairs
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Drawing Title MECHANICAL DETAILS
Approved:

Phase 100% CONSTRUCTION DOCUMENTS
FULLY SPRINKLERED

Project Title EXPAND STERILE PROCESSING SERVICE (SPS)
Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064
Issue Date 11/29/2022
Checked MRM
Drawn NVP

Project Number 556-402
Building Number 133CA
Drawing Number M300



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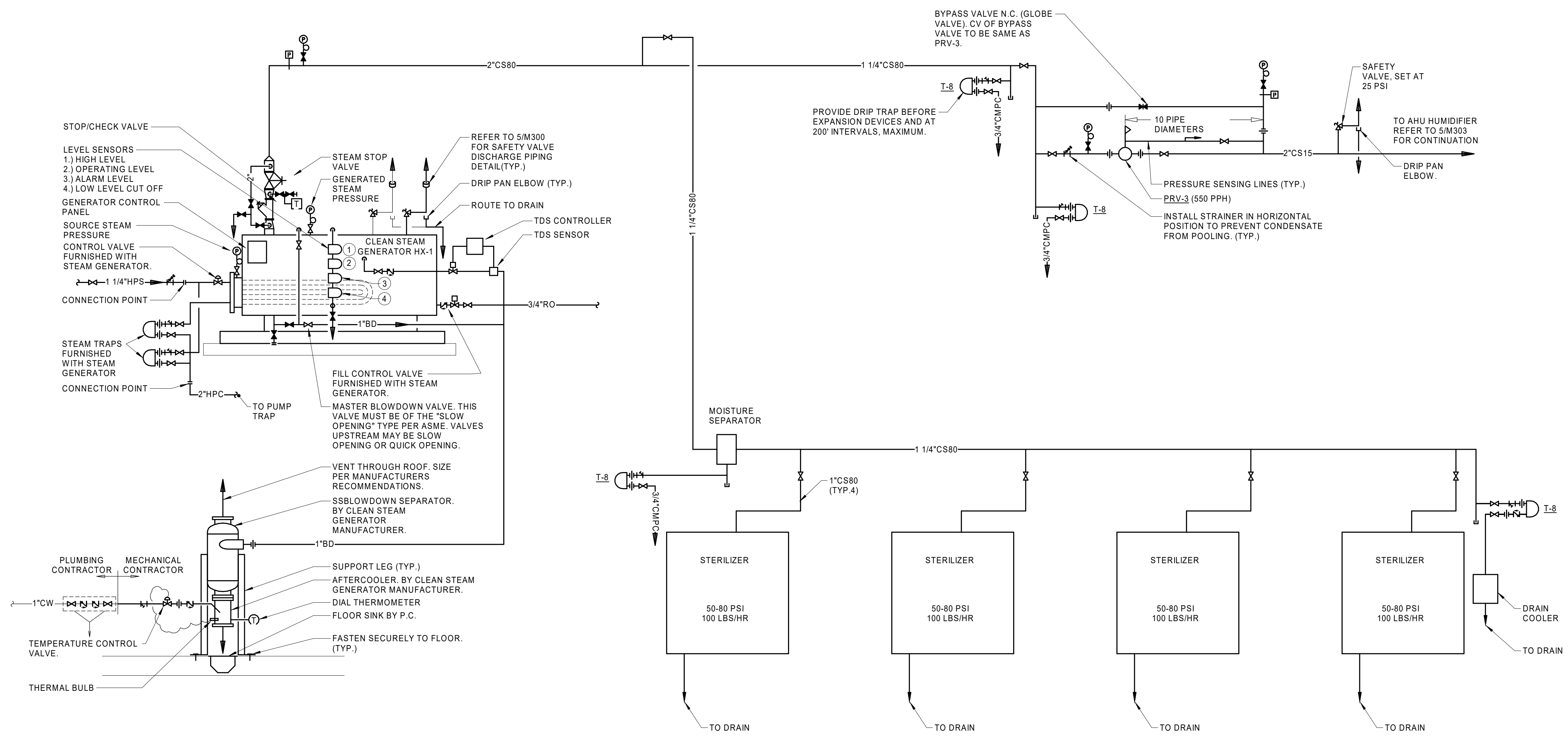
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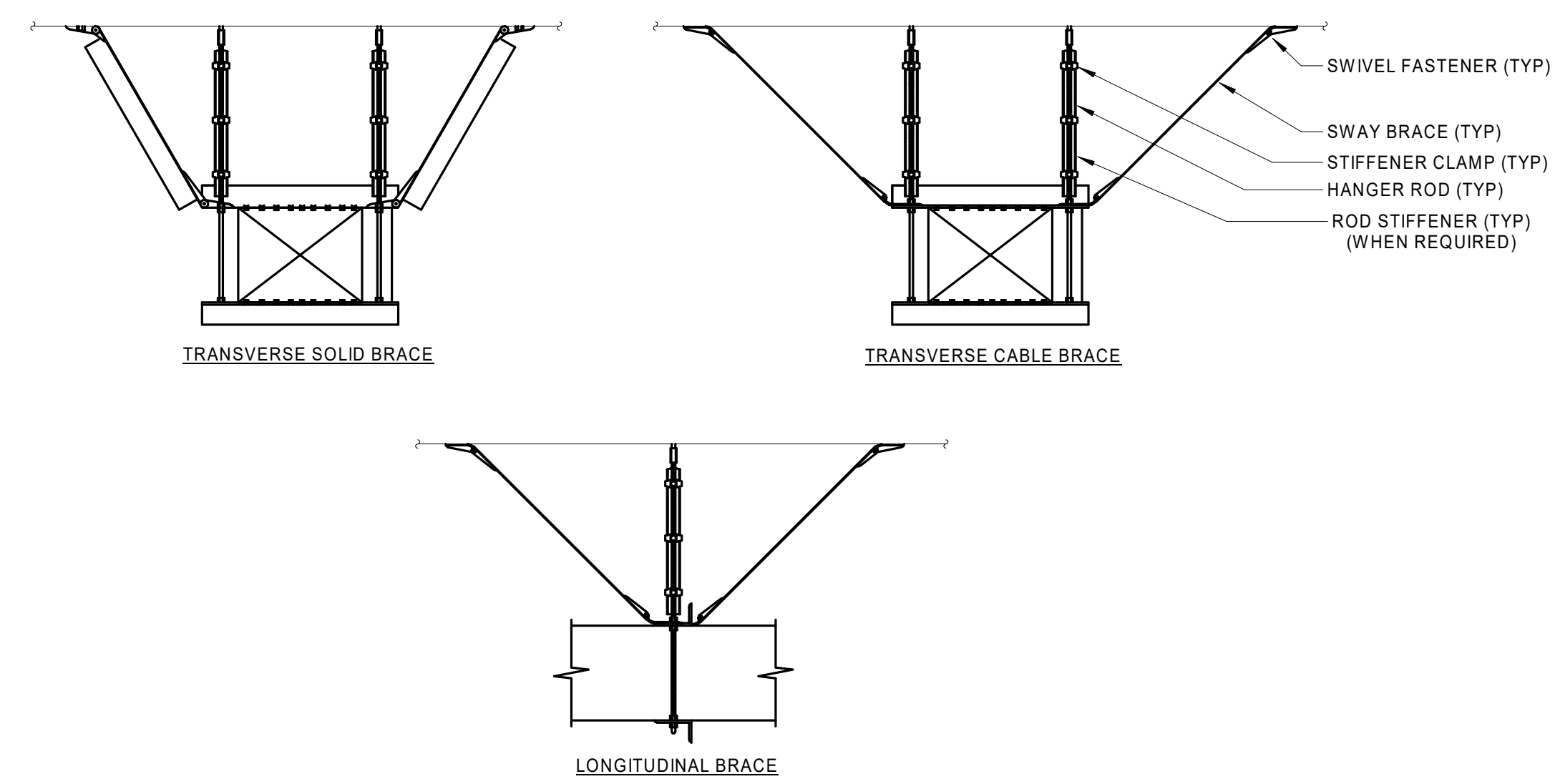
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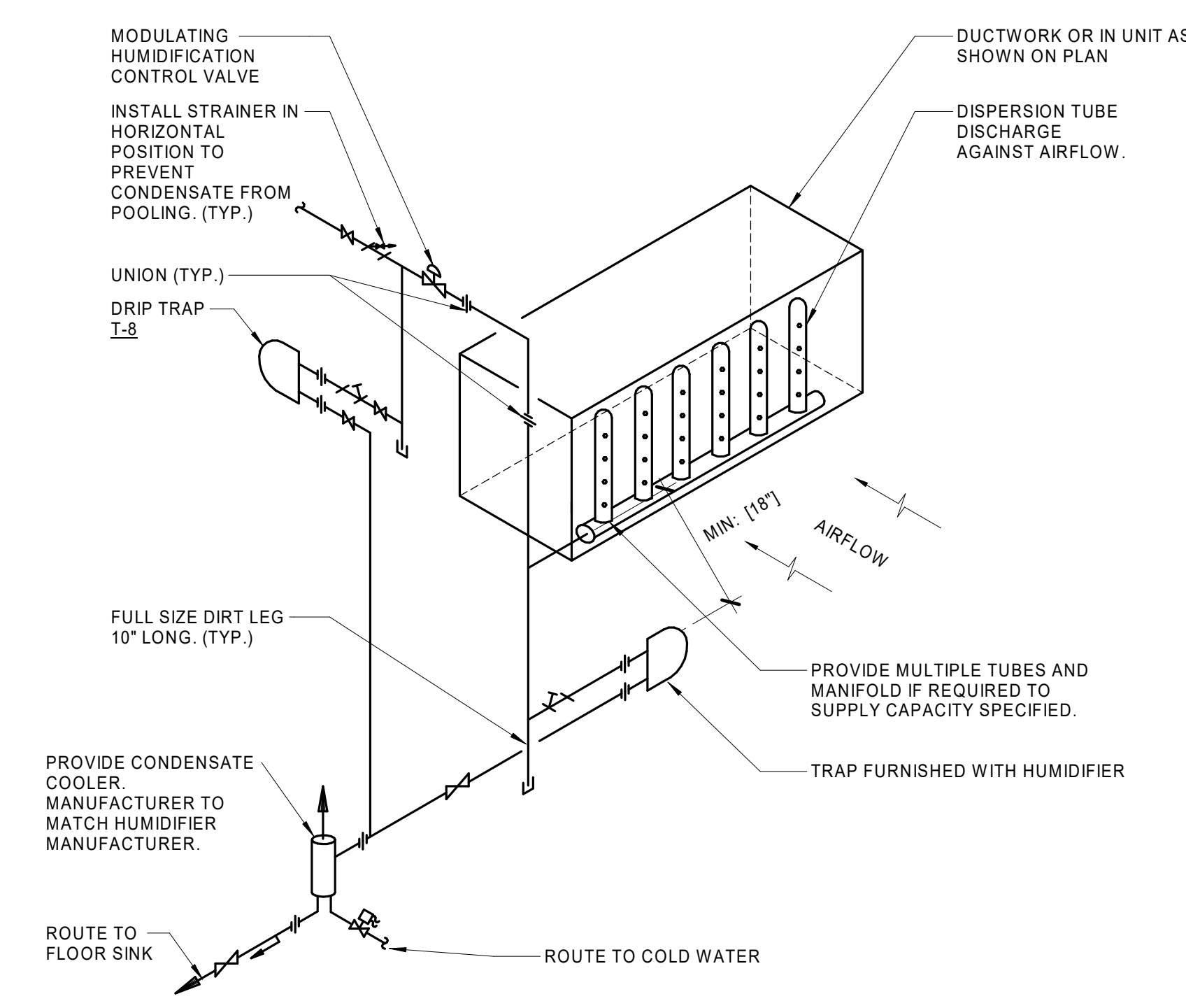
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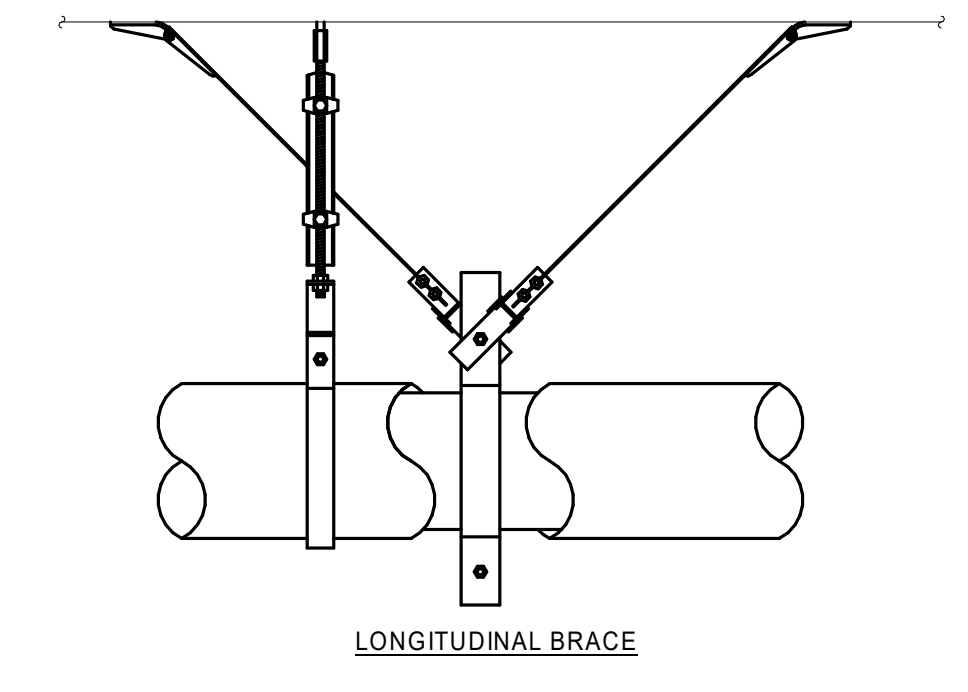
1 CLEAN STEAM FLOW DIAGRAM
NO SCALE
NOTES:
1. MAKE ALL TAPS FROM TOP OF STEAM PIPES.



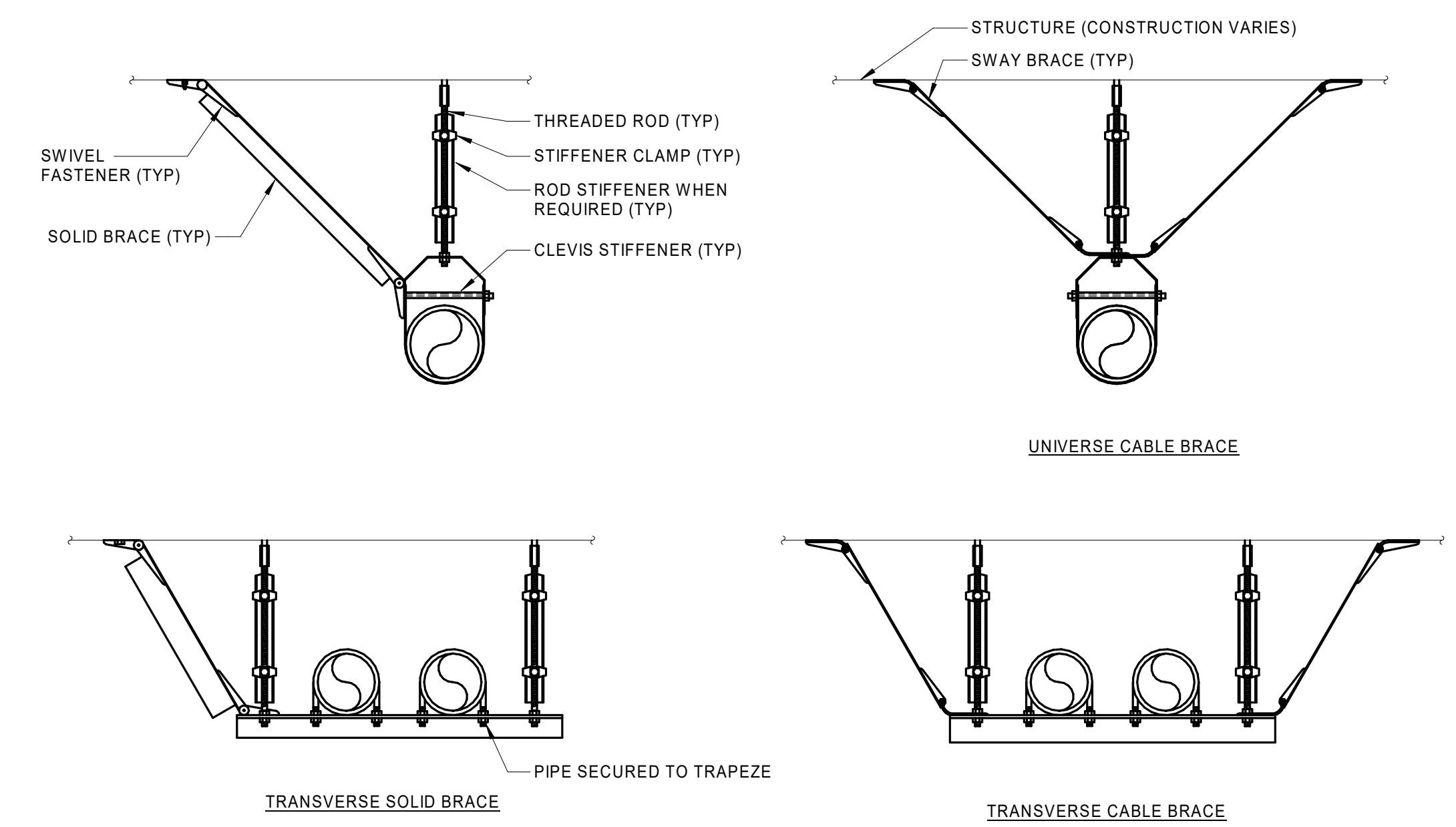
2 SEISMIC SUPPORT - DUCTWORK BRACING
NO SCALE
NOTES:
1. GENERAL REQUIREMENTS FOR SEISMIC RESTRAINT OF DUCTWORK ARE SHOWN. SPECIFIC REQUIREMENTS, INCLUDING ATTACHMENT TO BUILDING STRUCTURE, SHALL BE DETERMINED BY THE SEISMIC RESTRAINT DESIGNER. REFER TO SPECIFICATION FOR REQUIREMENTS.



5 HUMIDIFIER PIPING
12" = 1'-0"

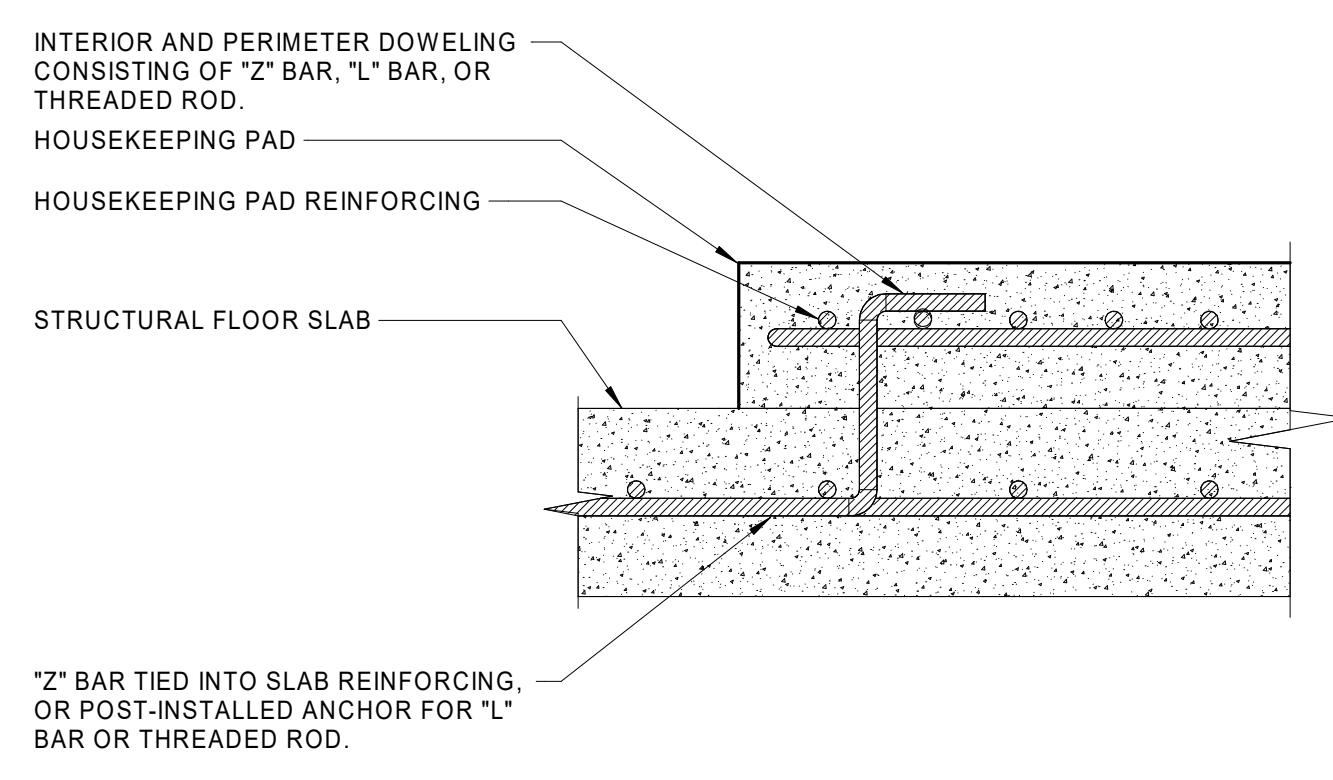


LONGITUDINAL BRACE



4 SEISMIC SUPPORT - PIPING BRACING
NO SCALE
NOTES:
1. GENERAL REQUIREMENTS FOR TRANSVERSE AND LONGITUDINAL BRACING OF PIPES ARE SHOWN. SPECIFIC REQUIREMENTS, INCLUDING ATTACHMENT TO BUILDING STRUCTURE, SHALL BE DETERMINED BY THE SEISMIC RESTRAINT DESIGNER. REFER TO SPECIFICATION FOR REQUIREMENTS.
2. REFER TO GENERAL PIPE SUPPORT DETAIL AND SPECIFICATION FOR INSULATED PIPING CONSIDERATIONS.

3 SEISMIC SUPPORT - HOUSEKEEPING PAD
NO SCALE
NOTES:
1. GENERAL REQUIREMENTS FOR SEISMIC HOUSEKEEPING PADS ARE SHOWN. SPECIFIC REQUIREMENTS, INCLUDING ATTACHMENT TO BUILDING STRUCTURE, SHALL BE DETERMINED BY THE SEISMIC RESTRAINT DESIGNER. REFER TO SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.



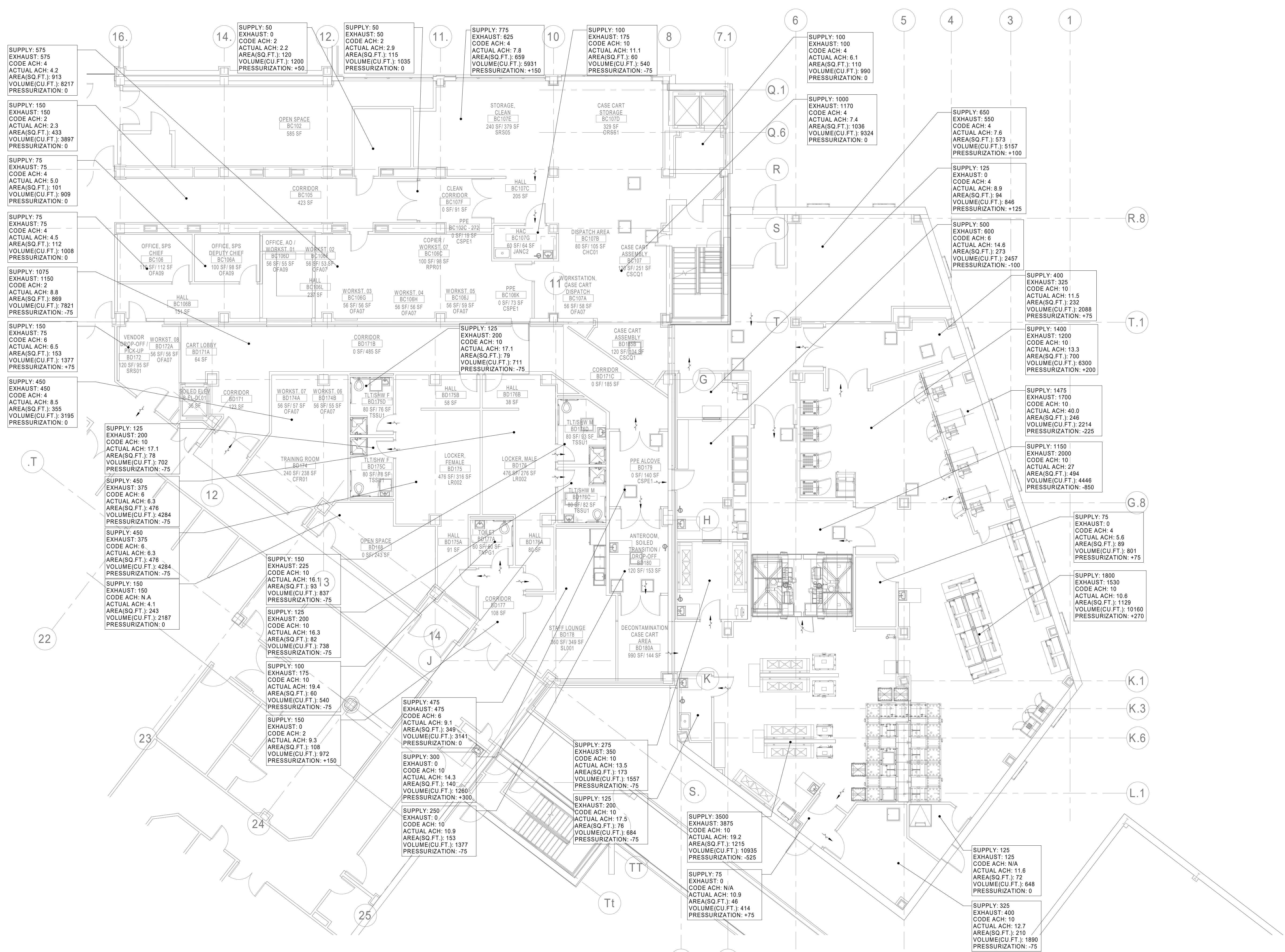
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Brook Design Firm Registration #184-007637-0014
REF. SCALE IN INCHES PROJECT #2100895.00

ARCHITECT/ENGINEER OF RECORD		STAMP		Office of Construction and Facilities Management		Drawing Title MECHANICAL DETAILS	Phase 100% CONSTRUCTION DOCUMENTS	Project Title EXPAND STERILE PROCESSING SERVICE (SPS)	Project Number 556-402
Alesia Architecture P.C. 2705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941		incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010		Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407		Approved:		Building Number 133CA	
Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7098		IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320		Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434		FULLY SPRINKLERED		Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	
Revisions:		Date:		U.S. Department of Veterans Affairs		Issue Date 11/29/2022		Checked PGP	
								Drawn NVP	
								Drawing Number M303	

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VA Form 08 - 6231



AIRFLOW MAP - BASEMENT
1/8" = 1'-0"

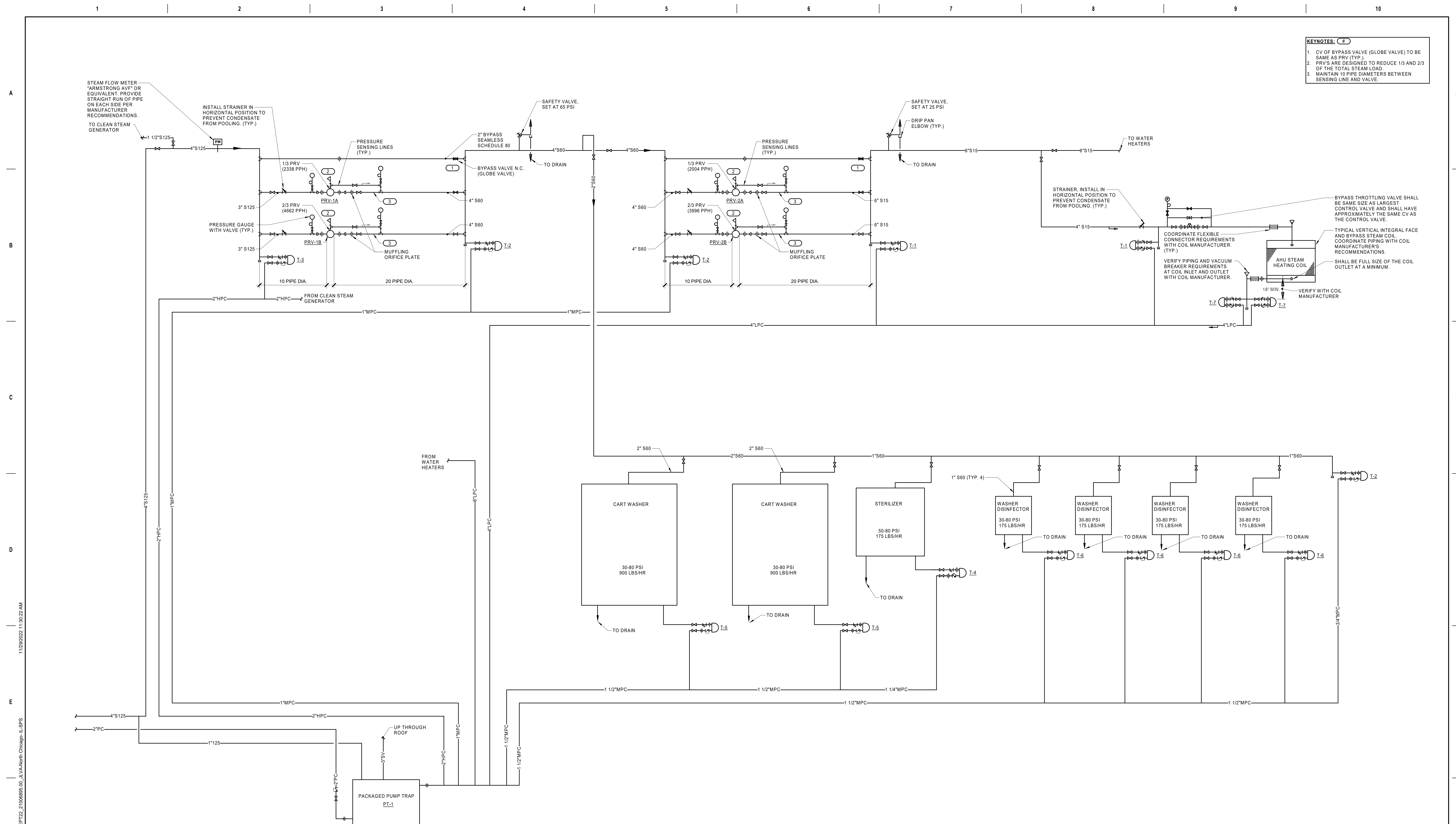
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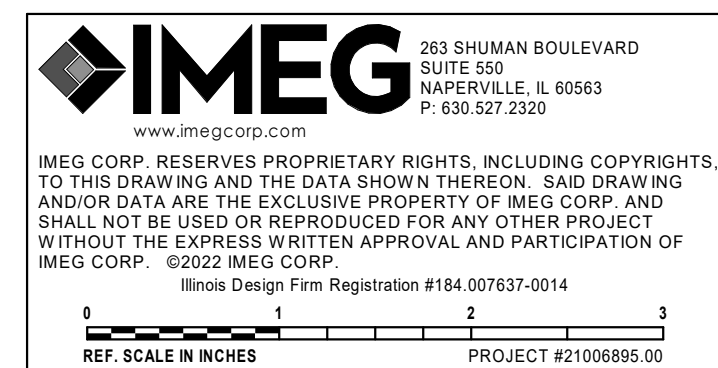
REF. SCALE IN INCHES PROJECT #2100895-05

ARCHITECT/ENGINEER OF RECORD		STAMP		Office of Construction and Facilities Management		Drawing Title AIRFLOW MAP		Phase 100% CONSTRUCTION DOCUMENTS		Project Title EXPAND STERILE PROCESSING SERVICE (SPS)		Project Number 556-402	
Ay Alesia Architecture P.C. 2705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941		incandescence life safety, inc. 3575 Ringsby Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010		Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407		Approved:		FULLY SPRINKLERED		Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064		Building Number 133CA	
S3 Shaffer Stevens & Schapp P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698		IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320		Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434		Issue Date 11/29/2022		Checked MRM		Drawn NVP		Drawing Number M305	
Revisions:		Date:		VA U.S. Department of Veterans Affairs									

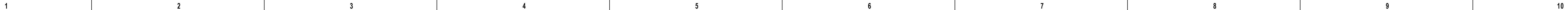
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1 STEAM FLOW DIAGRAM



		ARCHITECT/ENGINEER OF RECORD				STAMP	Office of Construction and Facilities Management	Drawing Title	Phase	Project Title	Project Number		
		 Alexia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 Incandescence life safety, inc. 3576 Renspyr Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3910	 Force Protect Security Cons. 10901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407	 11/29/2022 11/30/2023	STEAM FLOW DIAGRAM		100% CONSTRUCTION DOCUMENTS	EXPAND STERILE PROCESSING SERVICE (SPS)	556-402			
		 Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calibre Engineering, Inc. 9005 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434				FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Drawing Number			
Revisions:	Date:					VA U.S. Department of Veterans Affairs				Issue Date 11/29/2022	Checked PGP	Drawn NVP	M307



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CONTROL SYMBOL LIST			
NOT ALL SYMBOLS MAY APPLY.			
SYMBOL:	DESCRIPTION:		
	CONTROL VALVE (THREE-WAY)		
	CONTROL VALVE (TWO-WAY)		
	SOLENOID VALVE		
	CHECK VALVE		
	THERMOSTAT		
	THERMOSTAT/SENSOR WITH HEAVY DUTY ENCLOSURE		
	TEMPERATURE SENSOR (DUCT MOUNTED)		
	TEMPERATURE SENSOR WITH WELL		
	THERMOMETER WITH WELL (DIAL TYPE)		
	THERMOMETER WITH WELL (FILLED TYPE)		
	AVERAGING TEMPERATURE SENSOR		
	LOW LIMIT TEMPERATURE SWITCH		
	PROBE TEMPERATURE SENSOR		
	PRESSURE SENSOR (FURNISHED WITH BALL VALVE)		
	PRESSURE GAUGE (FURNISHED WITH BALL VALVE)		
	DIFFERENTIAL PRESSURE SENSOR		
	PRESSURE SENSOR (DUCT MOUNTED)		
	STATIC SWITCH		
	ANALOG INPUT		DIGITAL INPUT
	ANALOG OUTPUT		DIGITAL OUTPUT
	FLOW METER		HUMIDISTAT SENSOR
	FLOW SWITCH		HUMIDISTAT/SENSOR
	FLOW SENSOR		HUMIDITY SENSOR (DUCT MOUNTED)
	AIR FLOW SWITCH		FILTER
	DUCT FLOW METER		TERMINAL AIR BOX
	HUMIDIFIER		TERMINAL AIR BOX W/ REHEAT
	DUCT SMOKE DETECTOR		HEATING/ COOLING COIL
	MANUAL MOTOR STARTER WITH THERMAL OVERLOAD		ACTUATOR
	FAN		DOOR SWITCH
	MOTOR		DIFFERENTIAL PRESSURE SWITCH
	CONTACTOR		CURRENT SWITCH
	PUMP		VIBRATION SWITCH
	OCCUPANCY SENSOR		NORMAL CLOSED CONTACT
	SENSOR		NORMALLY OPEN CONTACT
			OPPOSED BLADE DAMPER
			PARALLEL BLADE DAMPER

TEMPERATURE CONTROL GENERAL NOTES:

- REFER TO EQUIPMENT SCHEDULES TO CROSS REFERENCE WHICH CONTROL DIAGRAMS APPLY TO WHICH ITEMS OF EQUIPMENT. REFER TO TERMINAL AIR BOX (TAB) SCHEDULES FOR TEMP SENSOR REQUIREMENTS FOR EACH TAB.
- EACH D.I., D.O., A.I., AND A.O. POINT SHOWN FOR ALL CONTROL DIAGRAMS SHALL BE DISCRETE FROM ALL OTHER POINTS EXCEPT AS SPECIFICALLY NOTED.
- ALL WIRING, CONTROL COMPONENTS, DEVICES AND PROGRAMMING SHOWN ON THESE CONTROL DRAWINGS SHALL BE PROVIDED BY THE TCC UNLESS SPECIFICALLY NOTED OTHERWISE.
- TEMPERATURE CONTROL CABLEING, CONDUIT, BOXES, IDENTIFICATION: REFER TO THE SPECIFICATIONS FOR A COMPLETE LIST OF REQUIREMENTS. THE FOLLOWING SCHEDULE IS PROVIDED AS A CONVENIENCE. REFER TO SECTION 19552 23 09 00 AND DIV 16/26 FOR ADDITIONAL DETAILED REQUIREMENTS.
A. CABLE/WIRE JACKET COLOR: BLUE
B. CONDUIT BOX COLOR ABOVE FINISHED CEILINGS AND UNFINISHED SPACES WITHOUT CEILINGS: BLUE
C. CONDUIT BOX COLOR IN SPACES WITH EXPOSED FINISHED STRUCTURE: MILL FINISH TO BE FIELD PAINTED
D. CABLE/WIRE INSTALLATION: IN CONDUIT
- ALL ACTUATORS SHALL BE OF THE ELECTRICAL TYPE FOR THIS PROJECT UNLESS AN ACTUATOR IS SPECIFICALLY INDICATED ON THE DRAWINGS OR SPECIFICATIONS TO BE PNEUMATIC.
- ALL MODULATING DAMPER AND VALVE ACTUATORS SHOWN WITH POSITION FEEDBACK SHALL HAVE THE VALVE POSITION DISPLAYED ON GRAPHICAL SCREEN ADJACENT TO THE DAMPER/VALVE COMMAND SIGNAL. DISPLAYED VALVE POSITION SHALL BE FROM THE FEEDBACK DEVICE/CIRCUIT (OUTPUT SIGNAL FROM THE FMCS TO THE ACTUATOR IS NOT ACCEPTABLE).
- MODULATING SIGNALS SHALL BE DISPLAYED AS % OPEN (SIGNALS DISPLAYED AS % CLOSED ARE NOT ACCEPTABLE).
- PRESSURE TRANSMITTERS WHOSE SIGNAL IS UTILIZED FOR MAINTAINING DUCT STATIC PRESSURE SHALL BE WIRED DIRECTLY TO THE DUCTROLLER TO MODULATE FAN SPEED. SIGNAL SHALL BE COMPLETELY INDEPENDENT OF THE FMCS NETWORK.
- PRESSURE TRANSMITTERS WHOSE SIGNAL IS UTILIZED FOR MAINTAINING DIFFERENTIAL PRESSURE OF ANY PUMPED WATER SYSTEM (E.G., HEATING HOT WATER, CHILLED WATER AND THE LIKE) SHALL BE WIRED DIRECTLY TO THE CONTROLLER THAT MODULATES PUMP SPEED. SIGNAL SHALL BE COMPLETELY INDEPENDENT OF THE FMCS NETWORK.
- ALL CONTROL COMPONENTS SUCH AS RELAYS, SWITCHES, CONTROLLERS, ETC. SHALL BE MOUNTED IN STEEL ENCLOSURES WITH STEEL MOUNTING BACKPLATES PER SPECIFICATION 23 09 23.
- EACH CONTROL PANEL SHALL HAVE A LAMINATED COPY OF THE APPLICABLE SEQUENCE OF OPERATION AND CONTROL DIAGRAM INDICATING THE POINTS, COMPONENTS AND OPERATION OF EQUIPMENT ASSOCIATED WITH EACH PANEL. REFER TO SECTION 23 09 23 FOR ADDITIONAL REQUIREMENTS.
- TCC SHALL WIRE THE CONTROL SIGNAL FROM THE ASSOCIATED AIR HANDLING UNIT RTU CONTROL PANEL TO CONTROL THE OPERATION OF SMOKE DAMPERS IN ACCORDANCE WITH SEQUENCE OF OPERATION. TCC SHALL PROVIDE ALL WIRING, CONDUIT, TRANSFORMERS, FUSING AND ALL OTHER ELECTRICAL COMPONENTS REQUIRED FOR COMPLETE INSTALLATION.
- TCC SHALL EXTEND CONTROL SIGNAL FROM ADDRESSABLE RELAY DEVICE SERVING EACH AIR HANDLING UNIT. REFER TO ELECTRICAL DRAWINGS FOR LOCATIONS. TCC SHALL EXTEND AND TERMINATE WIRINGS AS REQUIRED FOR EQUIPMENT SHUTDOWN.
- TCC SHALL PROVIDE POWER SUPPLIES FOR ALL 24VAC POWER REQUIREMENTS TO INCLUDE, BUT NOT LIMITED TO, APPLICATION SPECIFIC CONTROLLERS, DAMPER AND VALVE ACTUATORS, BUILDING PRESSURE SENSORS, AND OTHER CONTROL COMPONENTS AND DEVICES. REFER TO CONTROL SPECIFICATIONS FOR POWER SUPPLY REQUIREMENTS. REFER TO MECHANICAL FLOOR PLANS FOR POWER SUPPLY LOCATIONS. PROVIDE LOW VOLTAGE WIRING FROM POWER SUPPLIES TO ALL CONTROLLERS, MONITORS, COMPONENTS AND DEVICES REQUIRING 24VAC. ADDITIONAL POWER SUPPLIES NOT SHOWN AND REQUIRED FOR A COMPLETE AND OPERATIONAL SYSTEM SHALL BE PROVIDED BY THE TEMPERATURE CONTROL CONTRACTOR. THE TEMPERATURE CONTROL CONTRACTOR SHALL PROVIDE FINANCIAL PROVISIONS WITHIN THEIR BID FOR THE ELECTRICAL CONTRACTOR TO PROVIDE BRANCH POWER TO THE ADDITIONAL POWER SUPPLIES. COORDINATE THE LOCATION OF ADDITIONAL POWER SUPPLY CABINET WITH THE ELECTRICAL CONTRACTOR.
- ELEMENT LENGTHS FOR BOTH MIXED AIR TEMP SENSORS AND LOW LIMIT TEMP SWITCHES SHALL BE MINIMUM 1 LINEAR FOOT PER SQUARE FOOT OF COIL SURFACE AREA. PROVIDE MULTIPLE SENSORS AND SWITCHES AS NEEDED TO ACHIEVE REQUIRED ELEMENT LENGTHS. LOCATE RESET SWITCHES MAX. 6'-6" ABOVE ADJACENT STANDING SURFACE (E.G. ROOF, PLATFORM OR FLOOR) SO THE RESET SWITCH CAN BE CYCLED WITHOUT THE NEED FOR A LADDER.
- TO PREVENT GENERATOR OVERLOADING, TCC SHALL PROGRAM A STAGGERED START TIME FOR ALL MECHANICAL EQUIPMENT THAT IS CONTROLLED BY FMCS TO INCLUDE, BUT NOT LIMITED TO, AIR HANDLERS, PUMPS, EXHAUST FANS, AND CHILLERS. THE FIRST EQUIPMENT SHALL START 2 MINUTES (ADJ.) FROM THE TIME THE FMCS RECEIVES THE SIGNAL THAT THE TRANSFER SWITCH CHANGES TO EMERGENCY POWER SOURCE WITH ALL EQUIPMENT BEING ENERGIZED WITHIN A 20 MINUTE (ADJ.) TIME FRAME. COORDINATE ORDER OF EQUIPMENT STARTING WITH OWNER'S REPRESENTATIVE.
- CONTROL DIAGRAMS ARE SCHEMATIC IN NATURE AND DO NOT SHOW ALL REQUIRED CONTROL DEVICES AND COMPONENTS. REFER TO FLOOR PLANS, FLOW DIAGRAMS AND DETAILS FOR ADDITIONAL CONTROL DEVICES, COMPONENTS AND REQUIREMENTS NOT SHOWN ON THESE CONTROL DRAWINGS.
- TCC SHALL PROVIDE ALL CONTROL COMPONENTS AND ACCESSORIES AS REQUIRED FOR EQUIPMENT TO BE CONTROLLED AS DESCRIBED IN THE SEQUENCE OF OPERATION REGARDLESS OF WHETHER ALL CONTROL COMPONENTS OR POINTS ARE SHOWN IN THE ASSOCIATED CONTROL DIAGRAM.
- COORDINATE DDC CONTROL PANEL EMERGENCY POWER SUPPLY REQUIREMENT WITH ELECTRICAL CONTRACTOR. ALL CONTROLS ASSOCIATED WITH MECHANICAL SYSTEMS REQUIRING EMERGENCY POWER SHALL BE CONNECTED TO THE EMERGENCY POWER SYSTEM.

SEQUENCE OF OPERATION:

- FMCS TAB CONTROLLER SHALL MODULATE THE TAB DAMPER AND TAB HW REHEAT COIL CONTROL VALVE TO MAINTAIN SPACE TEMPERATURE OF 72°F (ADJ.) WITH 2°F (ADJ.) DEAD BAND BASED ON A SIGNAL FROM A WALL MOUNTED TEMPERATURE SENSOR. SEE DRAWINGS FOR TEMPERATURE SENSOR REQUIREMENTS. SPACES WITH ADJUSTABLE THERMOSTATS WILL ALLOW A +/- 3°F (ADJ.) OFFSET FROM THE DDC SETPOINT.
- AT FULL COOLING, THE TAB SHALL BE OPEN TO MAXIMUM CFM POSITION. THE REHEAT COIL CONTROL VALVE SHALL BE CLOSED.
- UPON A FALL IN SPACE TEMPERATURE, THE TAB SHALL MODULATE CLOSED UNTIL SPACE SETPOINT IS MAINTAINED, OR UNTIL IT REACHES ITS MINIMUM SCHEDULED CFM POSITION PER THE TAB SCHEDULE. THE REHEAT COIL CONTROL VALVE SHALL BE CLOSED.
- UPON A FURTHER FALL IN SPACE TEMPERATURE, THE REHEAT COIL CONTROL VALVE SHALL MODULATE OPEN TO MAINTAIN SPACE SETPOINT UNTIL THE SUPPLY AIR TEMPERATURE IS 20°F ABOVE ROOM TEMPERATURE SETPOINT.
- UPON A FURTHER FALL IN SPACE TEMPERATURE, TAB SHALL OPEN TO MAINTAIN SETPOINT UNTIL TAB AIRFLOW REACHES ITS MAXIMUM HEATING SETTING. THE REHEAT COIL CONTROL VALVE SHALL CONTINUE TO MODULATE OPEN TO MAINTAIN MAXIMUM DELTA T LISTED ABOVE.
- THE FMCS SHALL UTILIZE OUTPUT FROM ALL TERMINAL AIR BOX POSITIONS TO RESET THE SUPPLY DUCT DIFFERENTIAL STATIC PRESSURE.
- WHEN FLOATING CV'S ARE USED, FMCS SHALL PERFORM AN AUTO-ZERO FUNCTION EVERY NIGHT DURING UNOCCUPIED TIMES. THE FMCS SHALL STAGGER AUTO-ZERO SEQUENCES SO THAT ALL VALVES DO NOT SIMULTANEOUSLY CLOSE.

EXHAUST/RETURN TAB SEQUENCE OF OPERATION:

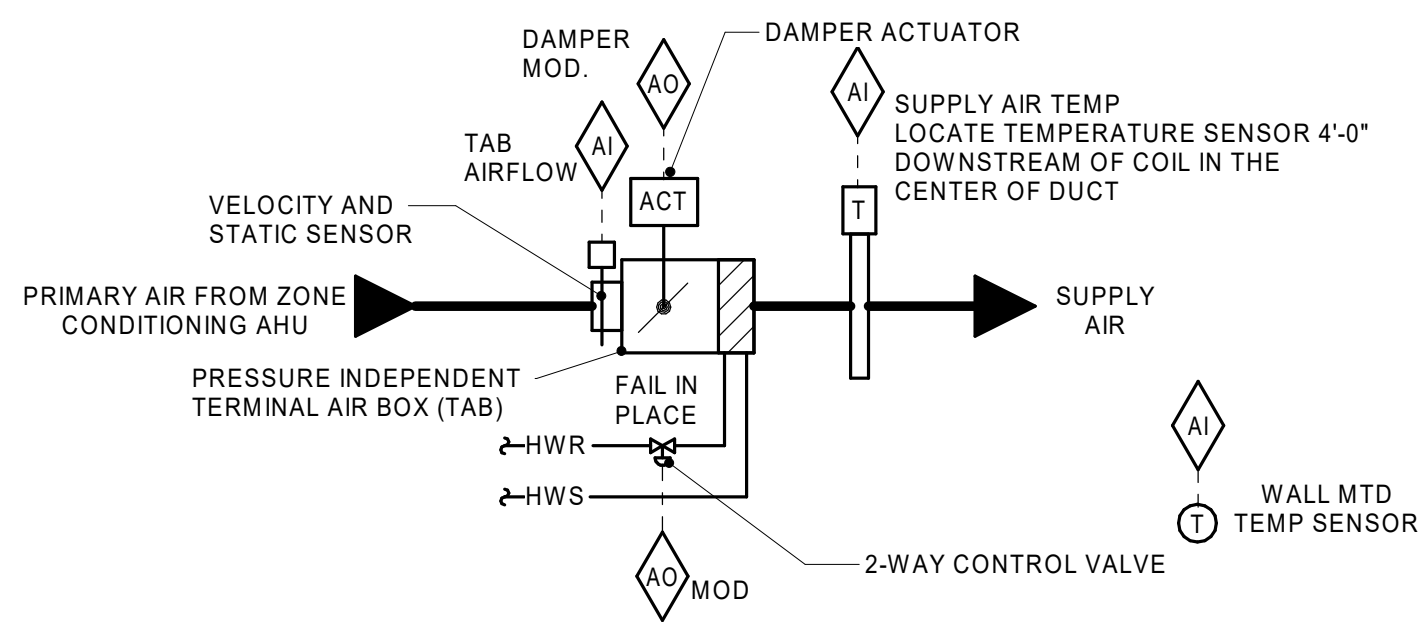
FMCS TAB CONTROLLER (OR AIR FLOW STATION/MOTOR OPERATED DAMPER COMBINATION) SHALL MODULATE THE TAB DAMPER TO MAINTAIN A CONSTANT VOLUME OFFSET. INITIAL CFM OFFSET SHALL BE THE DIFFERENCE BETWEEN THE TAB MAXIMUM VALUES ON THE DRAWINGS BUT SHOULD BE ADJUSTED BY THE TCC AND BALANCING CONTRACTOR TO ENSURE PROPER AIR FLOW DIRECTION. OFFSET CFM SHALL BE ADJUSTABLE THROUGH THE FMCS OPERATOR INTERFACE.

ALARMS, INTERLOCKS & SAFETIES:

SEND AN ALARM TO THE FMCS OPERATOR INTERFACE IF THE SPACE TEMPERATURE IS MORE THAN 10°F (ADJ.) ABOVE OR BELOW SETPOINT.

TAB CONTROL W/HOT WATER REHEAT AND CFM OFFSET - TAB-A

NO SCALE



SEQUENCE OF OPERATION:

- FMCS TAB CONTROLLER SHALL MODULATE THE TAB DAMPER AND TAB HW REHEAT COIL CONTROL VALVE TO MAINTAIN SPACE TEMPERATURE OF 72°F (ADJ.) WITH 5°F (ADJ.) DEAD BAND BASED ON A SIGNAL FROM A WALL MOUNTED TEMPERATURE SENSOR. SEE DRAWINGS FOR TEMPERATURE SENSOR REQUIREMENTS. SPACES WITH ADJUSTABLE THERMOSTATS WILL ALLOW A +/- 3°F (ADJ.) OFFSET FROM THE DDC SETPOINT.
- AT FULL COOLING, THE TAB SHALL BE OPEN TO MAXIMUM CFM POSITION. THE REHEAT COIL CONTROL VALVE SHALL BE CLOSED.
- UPON A FALL IN SPACE TEMPERATURE, THE TAB SHALL MODULATE CLOSED UNTIL SPACE SETPOINT IS MAINTAINED, OR UNTIL IT REACHES ITS MINIMUM SCHEDULED CFM POSITION PER THE TAB SCHEDULE. THE REHEAT COIL CONTROL VALVE SHALL BE CLOSED.
- UPON A FURTHER FALL IN SPACE TEMPERATURE, THE REHEAT COIL CONTROL VALVE SHALL MODULATE OPEN TO MAINTAIN SPACE SETPOINT UNTIL THE SUPPLY AIR TEMPERATURE IS 20°F ABOVE ROOM TEMPERATURE SETPOINT.
- UPON A FURTHER FALL IN SPACE TEMPERATURE, TAB SHALL OPEN TO MAINTAIN SETPOINT UNTIL TAB AIRFLOW REACHES ITS MAXIMUM HEATING SETTING. THE REHEAT COIL CONTROL VALVE SHALL CONTINUE TO MODULATE OPEN TO MAINTAIN MAXIMUM DELTA T LISTED ABOVE.
- THE FMCS SHALL UTILIZE OUTPUT FROM ALL TERMINAL AIR BOX POSITIONS TO RESET THE SUPPLY DUCT DIFFERENTIAL STATIC PRESSURE.
- WHEN FLOATING CV'S ARE USED, FMCS SHALL PERFORM AN AUTO-ZERO FUNCTION EVERY NIGHT DURING UNOCCUPIED TIMES. THE FMCS SHALL STAGGER AUTO-ZERO SEQUENCES SO THAT ALL VALVES DO NOT SIMULTANEOUSLY CLOSE.

ALARMS, INTERLOCKS & SAFETIES:

SEND AN ALARM TO THE FMCS OPERATOR INTERFACE IF THE SPACE TEMPERATURE IS MORE THAN 10°F (ADJ.) ABOVE OR BELOW SETPOINT.

TAB CONTROL W/ HOT WATER REHEAT - TAB-D

NO SCALE

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STAMP

Office of Construction and Facilities Management

U.S. Department of Veterans Affairs

Drawing Title

MECHANICAL DIAGRAMS

Approved:

Phase

100% CONSTRUCTION DOCUMENTS

FULLY SPRINKLERED

Project Title

EXPAND STERILE PROCESSING SERVICE (SPS)

Project Number

556-402

Building Number

133CA

Drawing Number

M401

Location

Captain James A. Lovell Federal Healthcare Center
3001 Green Bay Road, North Chicago, IL 60064

Issue Date

11/29/2022

Checked

MRM

Drawn

NVP

SEQUENCE OF OPERATION:

- FMCS TAB CONTROLLER SHALL MODULATE THE TAB DAMPER AND TAB HW REHEAT COIL CONTROL VALVE TO MAINTAIN SPACE TEMPERATURE OF 66°F (ADJ.) WITH 5°F (ADJ.) DEAD BAND BASED ON A SIGNAL FROM A WALL MOUNTED TEMPERATURE SENSOR. SEE DRAWINGS FOR TEMPERATURE SENSOR REQUIREMENTS. SPACES WITH ADJUSTABLE THERMOSTATS WILL ALLOW A +/- 3°F (ADJ.) OFFSET FROM THE DDC SETPOINT.
- AT FULL COOLING, THE TAB SHALL BE OPEN TO MAXIMUM CFM POSITION. THE REHEAT COIL CONTROL VALVE SHALL BE CLOSED.
- UPON A FALL IN SPACE TEMPERATURE, THE TAB SHALL MODULATE CLOSED UNTIL SPACE SETPOINT IS MAINTAINED, OR UNTIL IT REACHES ITS MINIMUM SCHEDULED CFM POSITION PER THE TAB SCHEDULE. THE REHEAT COIL CONTROL VALVE SHALL BE CLOSED.
- UPON A FURTHER FALL IN SPACE TEMPERATURE, THE REHEAT COIL CONTROL VALVE SHALL MODULATE OPEN TO MAINTAIN SPACE SETPOINT UNTIL THE SUPPLY AIR TEMPERATURE IS 20°F ABOVE ROOM TEMPERATURE SETPOINT.
- UPON A FURTHER FALL IN SPACE TEMPERATURE, TAB SHALL OPEN TO MAINTAIN SETPOINT UNTIL TAB AIRFLOW REACHES ITS MAXIMUM HEATING SETTING. THE REHEAT COIL CONTROL VALVE SHALL CONTINUE TO MODULATE OPEN TO MAINTAIN MAXIMUM DELTA T LISTED ABOVE.
- THE FMCS SHALL UTILIZE OUTPUT FROM ALL TERMINAL AIR BOX POSITIONS TO RESET THE SUPPLY DUCT DIFFERENTIAL STATIC PRESSURE.
- IN MINOR PROCEDURE AND G.I. ROOMS WHEN BOTH ROOM OCCUPANCY SENSORS DO NOT DETECT MOTION FOR 15 MIN (ADJ.) FMCS SHALL ENABLE UNOCCUPIED MODE AND SUPPLY TAB SHALL SEQUENCE AS DESCRIBED ABOVE. WHEN ANY ROOM OCCUPANCY SENSOR DETECTS MOTION FMCS SHALL ENABLE OCCUPIED MODE AND SUPPLY TAB SHALL SUPPLY MAX SCHEDULE AIR FLOW WHILE REHEAT COIL HW CONTROL VALVE MODULATES TO MAINTAIN SPACE SETPOINT.
- WHEN FLOATING CV'S ARE USED, FMCS SHALL PERFORM AN AUTO-ZERO FUNCTION EVERY NIGHT DURING UNOCCUPIED TIMES. THE FMCS SHALL STAGGER AUTO-ZERO SEQUENCES SO THAT ALL VALVES DO NOT SIMULTANEOUSLY CLOSE.

EXHAUST TAB SEQUENCE OF OPERATION:

FMCS SHALL MODULATE EXHAUST TAB DAMPER TO ACHIEVE THE ROOM AT MINIMUM -0.010 OR +0.010 (REFER TO DRAWINGS FOR CORRECT PRESSURE RELATIONSHIP) IN W.G. PRESSURE RELATIVE TO THE ADJACENT CORRIDOR.

ALARMS, INTERLOCKS & SAFETIES:

- SEND AN ALARM TO THE FMCS OPERATOR INTERFACE IF THE SPACE TEMPERATURE IS MORE THAN 10°F (ADJ.) ABOVE OR BELOW SETPOINT.
- DIFFERENTIAL PRESSURE MONITOR SHALL INDICATE A LOCAL ALARM IN THE EVENT THE ROOM PRESSURE IS GREATER THAN -0.010 OR +0.010 (REFER TO DRAWINGS FOR CORRECT PRESSURE RELATIONSHIP) IN W.G. RELATIVE TO THE ADJACENT CORRIDOR/ROOM FOR MORE THAN 30 SECONDS (ADJ.).

TAB CONTROL W/ HOT WATER REHEAT AND ROOM PRESSURE TAB CONTROL - TAB-C

NO SCALE

TERMINAL AIR BOX REPORT & DUCT MOUNTED HOT WATER REHEAT COIL GENERATION:

DDC FMCS SHALL BE PROGRAMMED TO GENERATE THE FOLLOWING REPORT BASED ON A MANUAL COMMAND FROM THE DDC FMCS WORKSTATION BY CLICKING ON A GRAPHICAL BUTTON. UPON INITIATING COMMAND THE DDC FMCS SHALL COMPILE A REPORT AS FOLLOWS:

TAB/COIL	AIRFLOW(CFM)	DMPR POS	VALVE POS	SUP AIR TEMP	ROOM TEMP	ROOM SETPOINT
SYMBOL	MAX/ACTUAL/MIN	(% OPEN)	(% OPEN)	(DEG. F)	(DEG. F)	(DEG. F)
***	***	85%	10%	85.2	73.6	72.0
***	***	80%	60%	75.1	71.1	72.0

WHEREAS THE SAMPLE REPORT ABOVE SHOWS ONLY A COUPLE TABS/COILS, THE FINAL PROGRAMMED REPORT SHALL LIST ALL TABS/COILS SERVED BY A AHU TO A SINGLE REPORT SHALL BE PROGRAMMED FOR EACH AHU AND FOR EACH FLOOR.

AFTER THE REPORT PRINTS OUT ALL TABS/HEATING COIL DATA, THE DDC FMCS SHALL AUTOMATICALLY TOTAL ALL THE INDIVIDUAL TAB AIRFLOW TO A SINGLE VALUE.

AFTER PRINTING THE SUM OF THE TABS/HEATING COIL AIRFLOW CFM, THE DDC FMCS SHALL THEN AUTOMATICALLY PRINT OUT THE AIR HANDLER REPORT FOR THE AHU WHICH SERVES THE TABS/HEATING COILS LISTED IN THE REPORT.

DDC FMCS SHALL ALLOW THE DDC FMCS OPERATOR TO ISSUE A SINGLE COMMAND THAT WILL AUTOMATICALLY CHANGE THE LOCAL SETPOINT FOR EACH TAB SERVED BY A AHU TO A SINGLE VALUE (E.G. A SINGLE COMMAND WILL SET ALL TABS/HEATING COILS SERVED BY AHU-A TO 80°F). A SEPARATE TABS/HEATING COIL SETPOINT OVERRIDE COMMAND SHALL BE PROGRAMMED IN THE FMCS FOR EACH AHU.

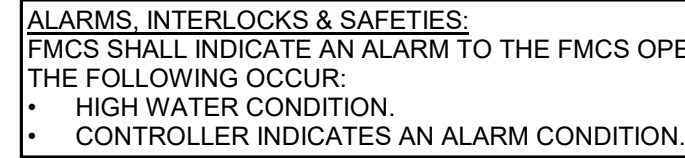
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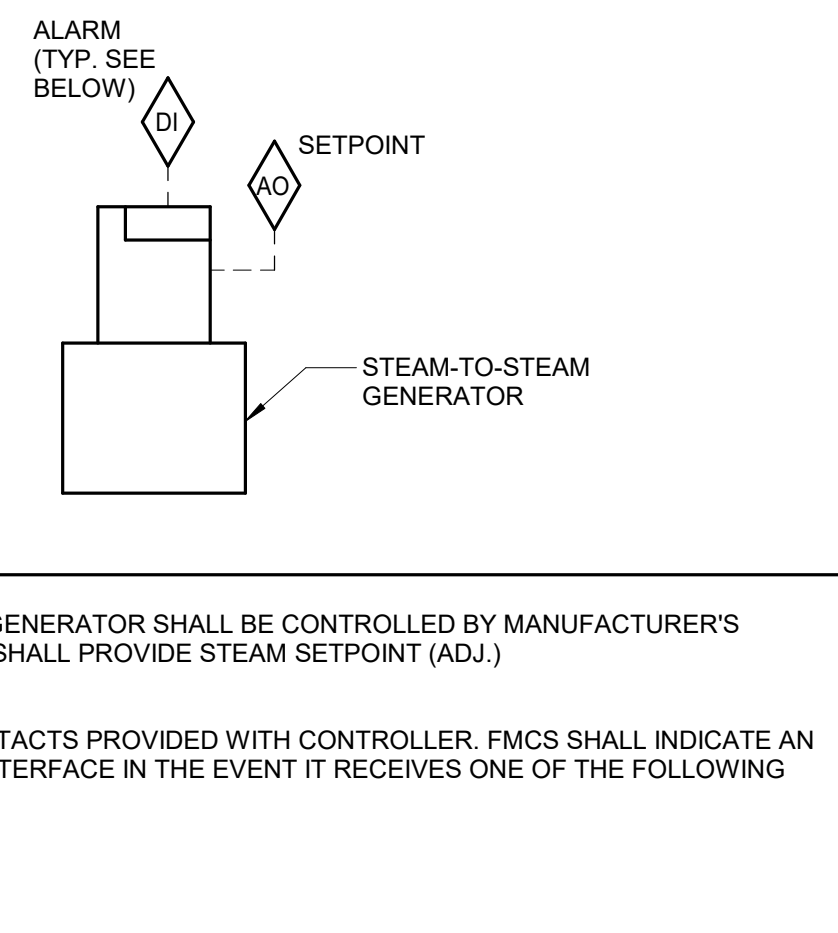
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Broad Design Firm Registration #184-007637-0014

REF. SCALE IN INCHES

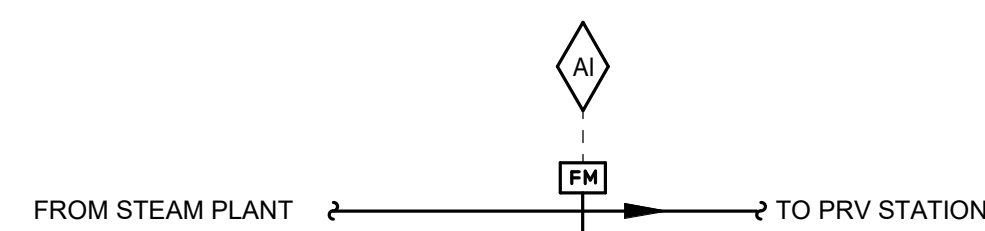
PROJECT #2100895.05



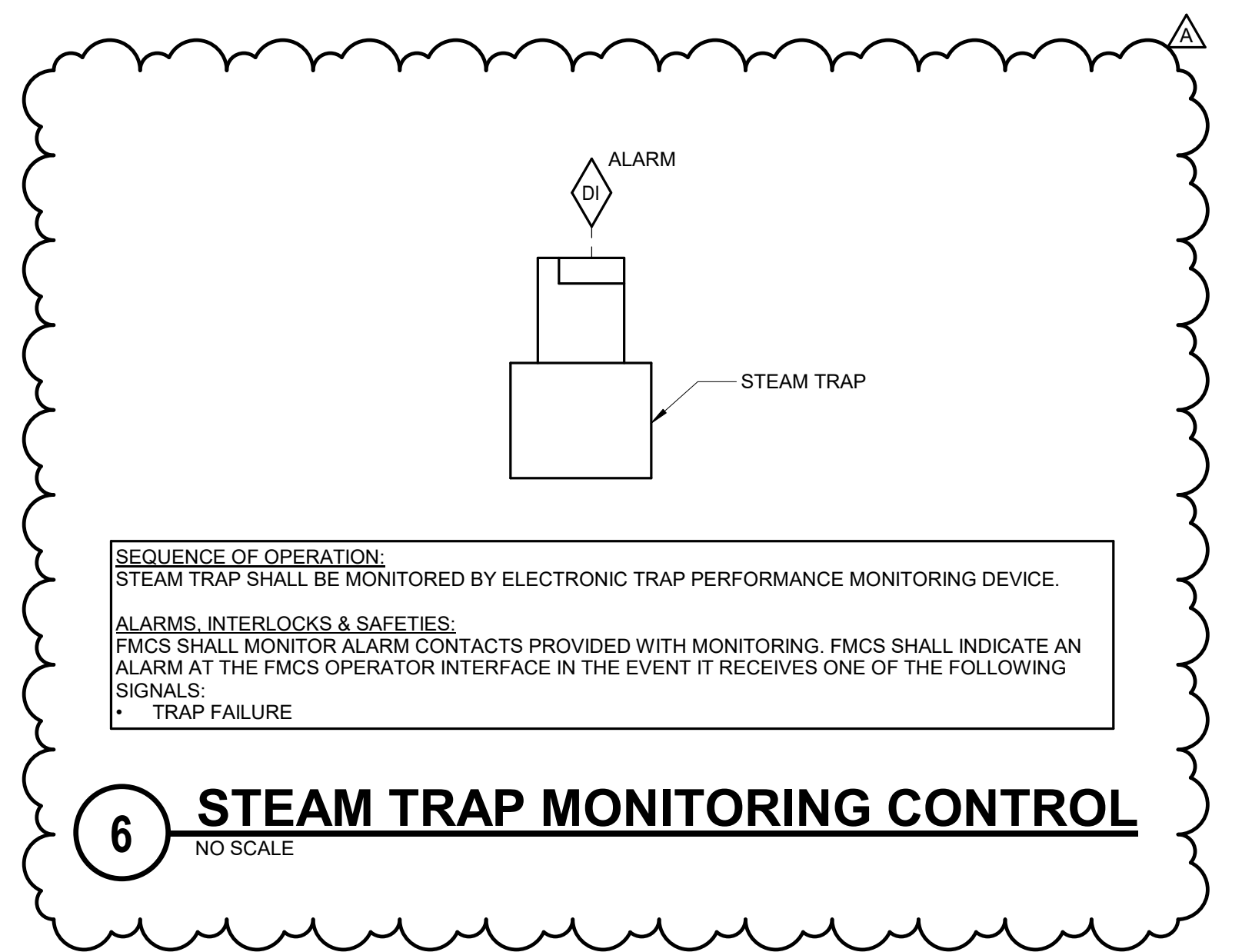
3 DI/RO WATER METERING CONTROL



5 STEAM METERING CONTROL
NO SCALE



6 STEAM TRAP MONITORING CONTROL
NO SCALE



ARCHITECT/ENGINEER OF RECORD			STAMP	Office of Construction and Facilities Management	Drawing Title MECHANICAL DIAGRAMS	Phase 100% CONSTRUCTION DOCUMENTS	Project Title EXPAND STERILE PROCESSING SERVICE (SPS)	Project Number 556-402
 Alesia Architecture P.C. 3705 North 200th Street Elkhorn, Nebraska 68022 Phone: (402) 291-6941	 incandescence life safety, inc. 3575 Ringstgy Ct Unit 408, Denver, CO 80216 Phone: (720) 722-3010	 Force Protect Force Protect Security Cons. 15901 Front Beach Road, Suite 1415 Panama City Beach, FL 32407			Approved:	FULLY SPRINKLERED	Location Captain James A. Lovell Federal Healthcare Center 3001 Green Bay Road, North Chicago, IL 60064	Drawing Number M403
 Shaffer Stevens & Schapp P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698	 IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320	 Calib e Calibre Engineering, Inc. 9090 S Ridgeline Blvd #105, Naperville, IL 60563 Phone: (303) 730-0434		VA U.S. Department of Veterans Affairs		Issue Date 11/29/2022 07/05/2023 REVISIONS	Checked MRM	Drawn NVP

11. REFER TO DIVISION 26 SPECIFICATIONS FOR DETAILS OF VFD.

1	2	3	4	5	6	7	8	9	10
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PHASE 4 BASE BID: _____
**PROVIDE ALL TERMINAL AIR
 BOXES AS SHOWN IN
 SCHEDULES.**

NOTES:

1. NEITHER RADIATED NOR DISCHARGE SOUND LEVEL SHALL EXCEED NC 35 AT 1" (5" INLET) STATIC PRESSURE WHEN TESTED PER AHRI STANDARD 885-2008 USING 5/8" 20-LB DENSITY MINERAL FIBER CEILING TILE.
2. TOTAL AIR PRESSURE DROP OF TAB AND REHEAT COIL SHALL NOT EXCEED 0.5" WC.
3. REFER TO CONTROL DRAWINGS FOR DESCRIPTION OF CONTROL TYPE.
4. SENSOR TYPE ONLY: 2- SENSORS ONLY, 2- SENSORS WITH OVERRIDE, 4- SENSORS WITH ADJUSTMENT AND OVERRIDE.
5. HEATING COIL IS BASED ON HEATING AIR FLOW. WATER PRESSURE DROP OF REHEAT COILS SHALL NOT EXCEED 5'. PROVIDE REHEAT COILS SEPARATE FROM BOXES IF REQUIRED TO MEET WATER TEMPERATURE DROP REQUIREMENTS.
6. HEAT TEMPERATURE SELECTION SHALL BE BASED ON A FIXED LEAVING AIR TEMPERATURE AND VARIABLE FLOW (GPM). PROVIDE FINAL MAXIMUM FLOW RATE (GPM) TO TEST AND BALANCE & TEMPERATURE CONTROLS CONTRACTORS.

TAG NAME	AREA SERVED	COOLING CAP. (T)	CFM		HEATING CAP. (NOTES 5, 6)				MIN. INLET SIZE (IN.)		MIN. INLET TYPE		CONTROL TYPE (NOTE 3)	SENSOR TYPE (NOTE 4)	MANUFACTURER	MODEL	REMARKS	
			HEATING MAX	MIN.	HEATING LAT	HEATING EWT	HEATING INLET	HEATING INLET	HEATING INLET	HEATING INLET								
TAB-101	CRAWL SPACE	600	600	600	460	85.0	15.0	2.9	10	-	-	-	TAB-A	2	TITUS	DESUV	NOTES 1, 2, 3	
TAB-102	DECONTAMINATION	3900	3900	3900	460	85.0	15.0	11.0	-	24	16	TAB-B	-	TAB-B	2	TITUS	DESUV	NOTES 1, 2, 3
TAB-103	WATER TREATMENT/STORAGE	625	625	625	460	85.0	15.0	2.8	-	-	-	-	-	TAB-C	2	TITUS	DESUV	NOTES 1, 2, 3
TAB-104	SCOPE PROCESSING	625	625	625	460	85.0	15.0	3.2	10	-	-	-	-	TAB-C	2	TITUS	DESUV	NOTES 1, 2, 3
TAB-105	CLEAN WORKROOM	1875	1875	1875	460	85.0	15.0	5.3	-	24	16	TAB-C	-	TAB-C	2	TITUS	DESUV	NOTES 1, 2, 3
TAB-106	COLD WALKER	1150	1150	1150	460	85.0	15.0	2.5	10	-	-	-	-	TAB-C	2	TITUS	DESUV	NOTES 1, 2, 3
TAB-107	STERILIZER AREA	2875	2875	2875	460	85.0	15.0	8.1	-	24	16	TAB-C	-	TAB-C	2	TITUS	DESUV	NOTES 1, 2, 3
TAB-108	STORAGE STERILE DURABLES	650	650	650	460	85.0	15.0	2.5	8	-	-	-	-	TAB-B	2	TITUS	DESUV	NOTES 1, 2, 3
TAB-109	UNLOADING/COOLING	400	400	400	460	95.0	15.0	1.5	8	-	-	-	-	TAB-B	2	TITUS	DESUV	NOTES 1, 2, 3
TAB-201	ELECTRIC ROOM	0	400	400	460	95.0	15.0	0.0	-	-	-	-	-	TAB-B	2	TITUS	DESUV	NOTES 1, 2, 3
TAB-202	MECH ROOM	600	200	200	460	95.0	15.0	1.1	8	-	-	-	-	TAB-B	2	TITUS	DESUV	NOTES 1, 2, 3

NOTES:
1. NEITHER RADIATED NOR DISCHARGE SOUND LEVELS SHALL EXCEED NC 35 AT 1.5" INLET STATIC PRESSURE WHEN TESTED PER AHRI STANDARD 885-2008 USING 5/8" 20-LB DENSITY MINERAL FIBER CEILING TILE.

TAG NAME	AREA SERVED	CFM		MIN. INLET SIZE (IN.) DIA	MIN. INLET SIZE (IN.)		MANUFACTURER	MODEL	REMARKS
		COOLING MAX.	MIN.		WIDTH	HEIGHT			
TAB-101E	CRAWL SPACE	600	600	8"			TITUS	DESV	NOTES 1, 2
TAB-102E	DECONTAMINATION	4475	4475		24	16	TITUS	DESV	NOTES 1, 2
TAB-104E	SCOPE PROCESSING	600	600	8"			TITUS	DESV	NOTES 1, 2
TAB-107E	STERILIZER/CLEAN WORK	4450	4450		24	16	TITUS	DESV	NOTES 1, 2
TAB-108E	STORAGE / UNLOADING COOLING	875	875	10"			TITUS	DESV	NOTES 1, 2

BALANCE TERMINAL AIR BOXES WITHIN
 BID ALTERNATE 4 SCOPE PER THESE
 SCHEDULES.

PHASE 4 BID ALTERNATE 3: _____
PROVIDE ALL TERMINAL AIR BOXES AS
SHOWN IN BASE BID SCHEDULES

BALANCE TERMINAL AIR BOXES WITHIN
 BID ALTERNATE 3 SCOPE PER THESE
 SCHEDULES. PROVIDE TAB-119E.

TAG NAME	SERVICE	CFM			HEATING COIL (NOTES 5 & 6)					MIN. INLET SIZE (IN.)	CONTROL TYPE (NOTE 3)	SENSOR TYPE (NOTE 4)	MANUFACTURER	MODEL	REMARKS
		COOLING MAX.	HEATING MAX.	MIN.	EAT 'F'	LAT 'F'	EWI 'F'	GAX. MAX.							
TAB-116	SHELL SPACE	400	400	400	46	85	150	1.2	10"	TAB-A	2	TITUS	DESV	NOTES 1, 2	
TAB-119	ANTER ROOM	775	775	757	46	85	150	2.2	10"	TAB-A	2	TITUS	DESV	NOTES 1, 2	

NOTES:

1. NEITHER RADIATED NOR DISCHARGE SOUND LEVELS SHALL EXCEED NC 35 AT 1.5" INLET STATIC PRESSURE WHEN TESTED PER AHRI STANDARD 885-2008 USING 5/8" 20-LB DENSITY MINERAL FIBER CEILING TILE.
2. TOTAL AIR PRESSURE DROP OF TAB AND REHEAT COIL SHALL NOT EXCEED 0.50" WC.

TAG NAME	AREA SERVED	CFM		MIN. INLET SIZE (N) DIA.	MIN. INLET SIZE (IN) DIA.	WIDTH	HEIGHT	MANUFACTURER	MODEL	REMARKS
		COOLING MAX.	MIN.							
TAB-116E	SHELL SPACE	400	400	12"	-	-	-	TITUS	DES/V	NOTES 1, 2
TAB-116E	ANTE ROOM	450	450	8"	-	-	-	TITUS	DES/V	NOTES 1, 2

NOTES:

1. NEITHER RADIATED NOR DISCHARGE SOUND LEVELS SHALL EXCEED NC 35 AT 1.5" INLET STATIC PRESSURE WHEN TESTED PER AHRI STANDARD 885-2008 USING 58" 20-LB DENSITY MINERAL FIBER CEILING TILE.
2. TOTAL AIR PRESSURE DROP OF TAB AND REHEAT COIL SHALL NOT EXCEED 0.50" WC.
3. REFER TO CONTROL DRAWINGS FOR DESCRIPTION OF CONTROL TYPE.
4. SENSOR TYPES: 1 - SENSOR ONLY, 2 - SENSOR WITH ADJUSTMENT, 3 - SENSOR WITH OVERRIDE, 4 - SENSOR WITH ADJUSTMENT AND OVERRIDE.
5. HEATING COIL IS BASED ON HEATING AIR FLOW. WATER PRESSURE DROP OF REHEAT COILS SHALL NOT EXCEED 5'. PROVIDE REHEAT COILS SEPARATE FROM BOXES IF REQUIRED TO MEET WATER PRESSURE DROP REQUIREMENTS.
6. HEATING COIL SELECTION SHALL BE BASED ON A FIXED LEAVING AIR TEMPERATURE AND VARIABLE FLOW (GPM). PROVIDE FINAL MAXIMUM FLOW RATE (GPM) TO TEST & BALANCE TEMPERATURE CONTROLS CONTRACTORS.

TERMINAL AIR BOX SCHEDULE - SINGLE DUCT EXHAUST - BID ALTERNATE 1

NOTES:
1. NEITHER RADIATED NOR DISCHARGE SOUND LEVELS SHALL EXCEED NC 35 AT 1.5" INLET STATIC PRESSURE WHEN TESTED PER AHRI STANDARD 885-2008 USING 5/8" 20-LB DENSITY MINERAL FIBER CEILING TILE.
2. TOTAL AIR PRESSURE DROP OF TAB AND REHEAT COIL SHALL NOT EXCEED 0.50" WC.

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Illinois Design Firm Registration #184-007837-0014

0 1 2 3
REF. SCALE IN INCHES PROJECT #21008950

ARCHITECT/ENGINEER OF RECORD			
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	Shaffer Stevens & Schaap P.C. 1070 Howard St Omaha, Nebraska 68102 Phone: (402) 455-7698		IMEG Corp. 263 Shuman Boulevard, Suite 550 Naperville, IL 60563 Phone: (630) 527-2320
			Caliber Engineering, Inc. 9090 S Ridgeline Blvd #105, Highlands Ranch, CO 80129 Phone: (303) 730-0434
			Force Protect Security Cons. 10901 Fort Beach Road, Suite 1415 Panama City Beach, FL 32407

STAMP



11/29/2022 exp 11/30/2023

Office of
Construction
and Facilities
Management

VA | U.S. Department
of Veterans Affairs

Drawing Title

MECHANICAL SCHEDULES

Approved:

Phase
100% CONSTRUCTION
DOCUMENTS

FULLY SPRINKLERED

Project Title
EXPAND STERILE PROCESSING
SERVICE (SPS)

Location
Captain James A. Lovell Federal Healthcare Center
3001 Green Bay Road, North Chicago, IL 60064

Issue Date
11/29/2022

Checked

Drawn

Project Number	556 402
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Building Number	133CA
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Drawing Number

M601